



Fig. 5D.26: Preferred method of palpation of liver.

- If resistance is felt, move the hands further down until resistance disappears.
- Exert gentle pressure and ask the patient to inspire deeply.
- The border of the liver can be felt on the tips of the fingers.
- This procedure can be repeated from lateral to medial to trace the entire edge of the liver.

3. Alternate method (Fig. 5D.27):

- Place the right hand below and parallel to the right subcostal margin.
- The liver edge will then be felt against the radial border of the index finger.



Fig. 5D.27: Alternate method of palpation of liver.

4. Hooking method of liver examination (Fig. 5D.28):

- Examiner stands at the patient's right shoulder, facing the foot end and examines the lower edge of the liver by curling the fingertips under the right costal margin.



Fig. 5D.28: Hooking method of palpation of liver.

5. Dipping method of liver palpation in ascites (Fig. 5D.29):

- Place both hands one over the other, over the area to be palpated.
- Rapidly flex your metacarpophalangeal joints, so that your fingers suddenly dip into the patient's abdomen.
- This displaces the fluid, enhancing the palpation of underlying organ.

Liver Span

- The liver span is the distance in centimeters between the upper border of the liver in the right midclavicular line, as determined by percussion (i.e. where lung resonance changes to liver dullness), and the lower border, as determined by either percussion or palpation (**Figs. 5D.30 to 5D.32**).
- The upper border of the liver is assessed using a heavy percussion technique. Light percussion is used to locate the lower edge of the liver. Light percussion is required because heavy percussion may underestimate the lower extent of the liver border.
- The normal liver span is less than 13 cm.
- In midclavicular line: Normally 6–12 cm.
- In midsternal line (left lobe): Normally 4–8 cm.
- The clinical estimate of the liver span is usually an underestimation of the actual liver size by about 2–5 cm.

Liver span	Condition seen
Increased	Hepatomegaly
Decreased	Shrunken liver as in cirrhosis
False positive for enlarged liver	• Right sided pleural effusion • Right lower lobe consolidation

Note: In conditions like emphysema of the lung, the liver may be pushed down. The edge may be palpable, leading the examiner to believe that the patient has hepatomegaly when the real problem is a hyperinflated lung. Percussion will reveal that the upper border is lower than expected.



Fig. 5D.29: Dipping method of palpation of liver.

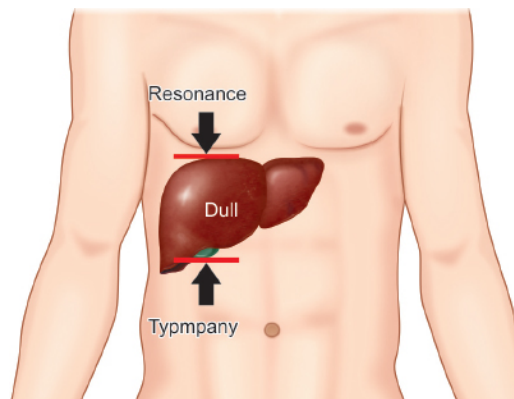


Fig. 5D.30: Liver span.

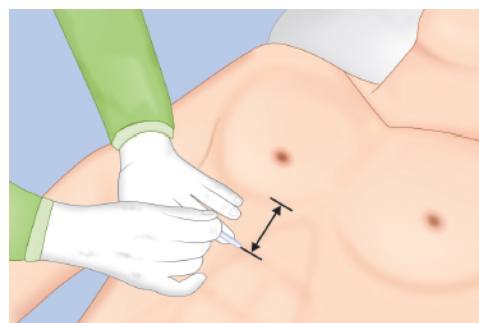


Fig. 5D.31: Percuss along the midclavicular line.

➤ **Description:**

- On palpation, a **resistance** and a **mass** is felt in the left hypochondrial and left lumbar regions. The mass is felt ____ cm below the left costal margin the **left midclavicular line**. It has

- rounded borders, has a
- characteristic notch,
- moves with respiration,
- non ballotable,
- not bimanually palpable,
- Pulsatility
- upper pole cannot be felt
- non-tender
- Consistency:
 - ◆ Soft- suggests congestion
 - ◆ Firm- suggests work hypertrophy
 - ◆ Hard- suggests

- **Grading**

- o Mild: 1-4cm
- o Moderate: 4-8cm
- o Massive: >8cm

Notes

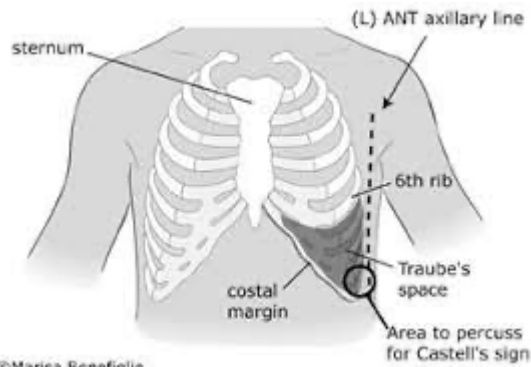
1. Why does spleen enlarge downwards and inwards? Because the **left colic flexure and phrenicocolic ligament** prevent direct downward enlargement
2. **Causes of splenomegaly** (Harrison headings)
 - a. **Work hypertrophy**: Haemolytic anaemias, Early Sickle Cell Anaemia, PNH, Thalassemia major
 - b. **Immune mediated**:
 - i. VIRAL : EBV , HIV
 - ii. BACTERIAL : TB , SCRUB , I.E , LEPTOSPIROSIS , ENTERIC FEVER
 - iii. PARASITIC : malaria , tropical splenomegaly , histoplasmosis
 - c. **Autoimmune causes**: Rheumatoid Arthritis (Felty's syndrome), SLE
 - d. **Congestive causes**: Portal hypertension, CCF, Budd Chiari , EHPVO
 - e. **Infiltrative causes**: Gaucher's, Amyloidosis, Leukemia, Lymphoma

- Good CHAMPS got no HDL**

- Percussion - hips and knee extended → so that abdomen wall becomes taut - better percussion**

-
- Right lateral decubitus position
- midaxillary line
- 8 cm
- xiphoid process
- (L) costal margin
- midpoint
- Pos Mid Ap line*
- Percuss along this L^r line
- lower Mid Ant Costal Margin

- o Traube's



- Shifting dullness- **1.5L** fluid- (Harrison's/19e/p286 & Sethuraman)
 - o Technique-
 - Start in the epigastrium and come down to rule out a distended urinary bladder
 - Percuss in the midline because bowel loops will be present and percussion conventionally moves from resonance to dullness
 - while percussing laterally - **go all the way till flank end to rule out localised dullness** (ovarian mass)
 - Wait for **15- 20 sec** for fluid to shift and bowel loops to come up
 - once resonant over side again go **back towards midline until dullness** is reached . this shows that the **DULLNESS HAS SHIFTED**
 - o Free fluid +nt but Shifting dullness -nt:
 - Small ascites
 - Loculated ascites
 - Ovarian cyst
 - o Horseshoe dullness- in minimal ascites, the dullness is confined to both the flanks and hypogastrium
 - o **Unilateral shifting dullness-** seen in splenic rupture (Ballance's sign). The blood is present in the left quadrants, so only **left sided shifting** dullness can be elicited.
- Puddle sign (not usually done)- 120mL
 - o Ask the patient to lie **prone for 5 minutes**
 - o Patient goes into **knee-elbow position** so that the middle abdomen is the most dependent
 - o Percuss the most dependent part of the abdomen for dullness
 - o Percuss the nearby non-dependent parts for resonance
 - o Change the patient's position and repercuss the area that was previously dull

Auscultation

- Bowel sounds
 - o Using diaphragm: Minimum duration to auscultate before concluding bowel sounds are absent- several minutes (**2 minutes**)
 - o Mechanism of production-??
 - o Normal frequency: Every **5-10 seconds**, but frequency varies (MacLeod)
 - o Causes of increased frequency of bowel sounds
 - Small bowel obstruction
 - Diarrhoea
 - Inflammatory bowel disease

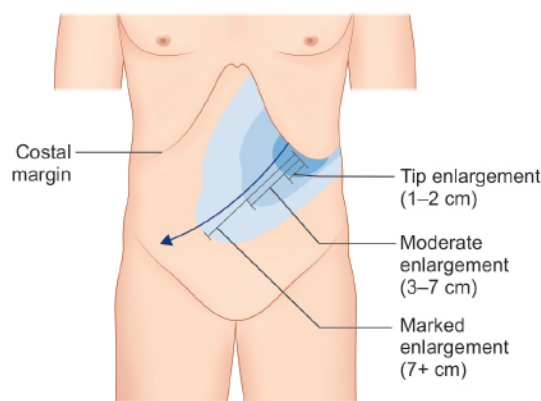


Fig. 5D.38: Grading of splenomegaly.

Hackett's grading system for palpable splenomegaly (Fig. 5D.39):

Grade	Description
Grade 0	Normal impalpable spleen
Grade 1	Spleen palpable only in deep inspiration
Grade 2	Spleen palpable on midclavicular line half way between umbilicus and costal margin
Grade 3	Spleen expands towards the umbilicus
Grade 4	Spleen goes past the umbilicus
Grade 5	Spleen expands towards pubic symphysis

Inspection:

Fullness may be seen emerging from left upper quadrant extending diagonally towards the right lower quadrant (RLQ).

Palpation:

Following methods of palpation have been discussed:

1. Classical method
2. Bimanual method
 - a. In supine position
 - b. In right lateral position
3. Hooking method
 - a. In supine position
 - b. In right lateral position
4. Middleton's maneuver
5. Dipping method.

Classical method (Fig. 5D.40):

- Patient in supine position, examine with single hand (right).
- Place the hand in the RLQ in RIF and move diagonally towards left upper quadrant.
- Hand should be firmly placed one the abdominal wall.
- Keep the hand steady during inspiration and feel for the splenic edge as it descends with each inspiration.

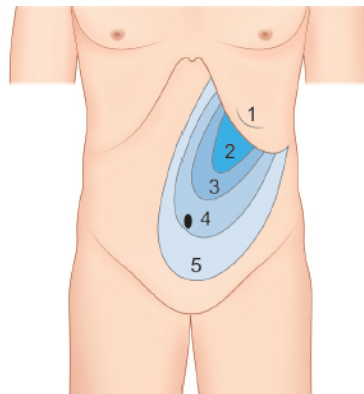


Fig. 5D.39: Hackett's grading system for palpable splenomegaly.



Fig. 5D.40: Demonstration of classical method of spleen palpation.

- If edge is not felt move the hand diagonally towards LUQ by 1 cm during expiration.
- Repeat the procedure.
- Tip of the fingers are used to feel the splenic tip.

Bimanual (supine position) (Fig. 5D.41):

- Place palm of left hand over the left lowermost rib cage posterolaterally, restricting the expansion of left lower ribs on inspiration.
- While applying firm pressure with the left hand, ask the patient to take deep inspiration.
- Insinuate the right hand beneath the left costal margin and feel for the splenic edge.

Bimanual (right lateral position):

- Done with patient lying in right lateral position with the left hip and knee flexed.
- Rest of maneuver is similar to above.



Fig. 5D.41: Demonstration of bimanual method (supine position) of spleen palpation.

Hooking method (supine position) (Fig. 5D.42):

- The physician hooks his fingers beneath the left costal margin as the patient inspires.



Fig. 5D.42: Demonstration of hooking method (supine position) of spleen palpation.

- For better appreciability, patient is asked to lie down on his left fist just inferior to his left scapula (**Middleton's maneuver**) (**Figs. 5D.43A and B**)
- From above, spleen may be continually palpable with two hands arching below the left costal margin while patient is asked to take deep breath in/out slowly.

Hooking maneuver (right lateral position):

- Examiner stands on left side facing towards the foot end
- With one hand hook the left lower costal margin and with other hand, give a counter-pressure from the posterolateral aspect.
- Now ask the patient to take a deep inspiration and feel for the tip of the spleen, by hooking the fingers.

Dipping method:

- It is done in marked ascites
- Similar to dipping method of liver (as described below under the palpation of liver).



Fig. 5D.44: Percussing the lowest left intercostal space in anterior axillary line—Castell's method of splenic percussion.

- Dullness on full inspiration indicates possible splenic enlargement (a positive Castell's sign).
- Most sensitive of all clinical signs with sensitivity 82% and specificity 83%.

	Full inspiration	Full expiration
Normal	Resonant	Resonant
Mild splenomegaly*	Dull	Resonant
Moderate/severe splenomegaly	Dull	Dull

*Percussion sign is considered positive, when a change in percussion note is observed between full expiration and full inspiration.

2. Percussion of Traube's (semilunar) space

- It is a semilunar space in the left anterior chest bounded by:
 - » Above by 6th rib
 - » Below by left costal margin
 - » Laterally by midaxillary line.
- With patient supine, percuss inferior to lung resonance from medial to lateral (**Figs. 5D.45 and 5D.47**) (as described by **Barkun**). Normally, a tympanic note heard due to gastric air bubble.

Obliteration of Traube's space	• Massive splenomegaly • Left-sided pleural effusion • Pericardial effusion • Enlarged left lobe of the liver • Full stomach or fundic mass
Upward shift of Traube's space	• Left diaphragmatic paralysis • Left lower lobe collapse or fibrosis



Fig. 5D.45: Percussion of Traube's space.



Fig. 5D.46: Percussing the posterior axillary line in right lateral position (Nixon's method).

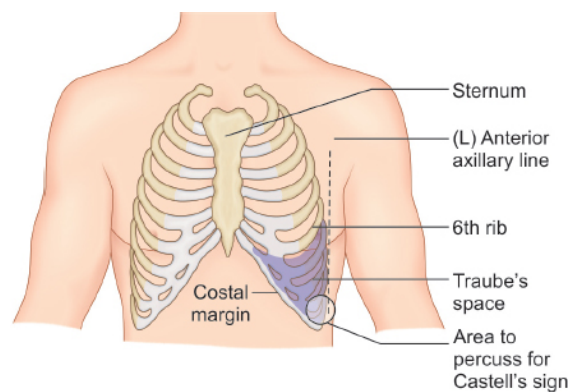


Fig. 5D.47: Landmarks of Traube's space and Castell's sign.

3. Percussion by Nixon's method

- Patient is first placed in the right lateral decubitus position.

- o Causes of absent bowel sounds
 - **Paralytic ileus**
 - Mesenteric ischaemia
 - **Peritonitis**
 - Pancreatitis
- Cruveilhier baumgarten sign- continuous venous hum over the portal vein in portal HTN
- Hepatic bruit- **Bell** over the right hypochondrium on the R MCL
- Splenic bruit - Bell over the left hypochondrium on the L MCL

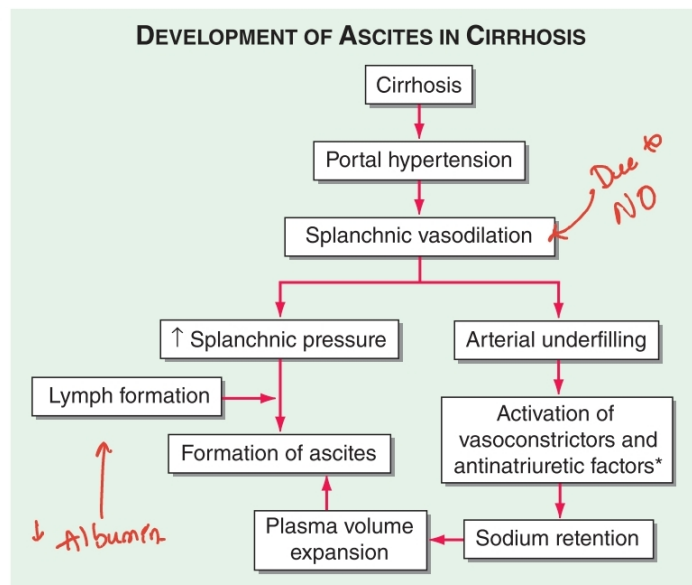
Hepatic Encephalopathy

1. **How?**
 - a. Toxic substances increase in blood bc
 - i. Abnormal Liver cant **detoxify** the substances
 - ii. More portosystemic **shunting**
 - iii. Increased **BBB** permeability
2. **What** are these toxic substances? **AGAMS**
 - a. Ammonia
 - b. GABA
 - c. AA
 - d. Mercaptans
 - e. SCFA
3. What **ppts** HE?
 - a. GI Bleeding
 - b. High protein intake
 - c. Vomiting, diarrhea, Increased use of diuretics → Elx disturbances like HypoK and HypoN
 - d. Large volume paracentesis
 - e. Constipation
 - f. TIPSS
 - g. Uremia
 - h. Viral/Alch hepatitis
 - i. SBP
 - j. Infections
4. **CF:**
 - a. Consciousness and behavior:
 - i. Hypersomnia → Inversion of sleep rhythm
 - ii. Apathy and slow response → Confusion, disorientation → Coma
 - b. Personality:
 - i. Childish and irritable
 - ii. Defecation and micturition in inappropriate places
 - c. Fluctuating neurological signs:
 - i. Slurred speech
 - ii. Constructional Apraxia
 - iii. Hypertonia, hyperreflexia, B/L extensor plantars
 - iv. Hyperventilation and Hyperpyrexia
 - d. Asterixis flapping tremor.

5. **Ix:**
 - a. Blood: NH₃ increased (>1mcg/mL)
 - b. CSF: high glutamine, normal prot and cells.
 - c. EEG: Delta waves - symm, high voltage and slow (2-5/sec)
 - d. VEP abn
 - e. LFT
6. **Rx**
 - a. Treat and remove ppt causes.
 - b. No need to restrict proteins, take normally 1-1.5g/kg/d.
 - c. Rifaximin to reduce Intestinal bacterial NH₃ production 550mg BD
 - d. **Syr. lactulose 15-30mL** until patient passes **2-3 soft stools/day**
 - i. Acidic pH from Lactulose → Ammonium ion from NH₃.
 - ii. Directly reduces bacterial metabolism
 - e. Alternatives to lactulose
 - i. Lactitol (better palatable)
 - ii. PEG
 - f. Other modalities:
 - i. Hemoperfusion to remove toxins.
 - ii. Ultimately liver transplant.

Ascites

Ascites causes



* **West Haven — clean AASLD/EASL table, then HE precipitant mechanisms,**
then abdomen additions from Niharika's notes, on the pages that follow.

WEST HAVEN CRITERIA — AASLD / EASL VERSION

Harrison's 21e does not tabulate the West Haven criteria — it describes hepatic encephalopathy in prose and tables only Child-Pugh and MELD. This is the AASLD/EASL 2014 version, which is the one to quote: it adds the ISHEN covert/overt split and, crucially, operative criteria that tell you what actually to test at the bedside.

Grade (WHC)	ISHEN	Clinical description	Operative criterion — what to test
Unimpaired	—	No encephalopathy, no history of HE	Tested and proved normal
Minimal	Covert	Psychometric or neuropsychological abnormality of tests of psychomotor speed or executive function, or neurophysiological abnormality — without any clinical change in mental state	Abnormal psychometric or neuropsychological test with no clinical manifestation . No universal cut-off; local norms needed
Grade I	Covert	Trivial lack of awareness · euphoria or anxiety · shortened attention span · impaired addition or subtraction · altered sleep rhythm	Oriented in time and space , but shows cognitive or behavioural decay relative to his or her own baseline, on examination or per the caregiver. <i>Findings usually not reproducible</i>
Grade II	OVERT	Lethargy or apathy · disorientation for time · obvious personality change · inappropriate behaviour · dyspraxia · ASTERIXIS	Disoriented for time — wrong on ≥3 of: day of the month, day of the week, month, season, year. <i>Variable but somewhat reproducible</i>
Grade III	OVERT	Somnolence to semi-stupor · responsive to stimuli · confused · gross disorientation · bizarre behaviour	Disoriented also for space — wrong on ≥3 of: country, state or region, city, place, floor or ward. <i>Somewhat reproducible</i>
Grade IV	OVERT	Coma	Does not respond even to painful stimuli . <i>Reproducible</i>

The shaded rows are overt HE. The single most important line in the table: overt HE begins at grade II, and the two findings that define the boundary are disorientation and asterixis. Below that, the patient looks clinically normal and you need psychometric testing — you cannot grade covert HE at the bedside.

THE FOUR-AXIS CLASSIFICATION — EVERY CASE GETS ALL FOUR

Axis	Categories
1. Type (underlying cause)	A — Acute liver failure · B — portosystemic Bypass/shunt with no intrinsic liver disease · C — Cirrhosis (<i>mnemonic: A, B, C = Acute, Bypass, Cirrhosis</i>)
2. Severity	West Haven grade (table above), plus GCS if grade III-IV
3. Time course	Episodic · Recurrent (≥2 episodes within 6 months) · Persistent (behavioural change always present, with relapses of overt HE)
4. Precipitant	Spontaneous (no precipitant identified) vs Precipitated — and if precipitated, <u>name it</u>

So a complete answer sounds like: 'Hepatic encephalopathy, type C, grade II, recurrent, precipitated by spontaneous bacterial peritonitis.' That single sentence is what separates a good answer from a grade.

POINTS THE AASLD GUIDELINE MAKES THAT ARE EASY MARKS

- **Ammonia is not a staging or prognostic test.** A raised level adds nothing to diagnosis, grading or prognosis in chronic liver disease. But a **normal** ammonia in a patient you think has HE should make you reconsider the diagnosis.
- **Precipitants — look for them in every case:** infection (SBP is the classic), GI bleed, electrolyte disturbance (hypokalaemia, hyponatraemia), constipation, dehydration or over-diuresis, sedatives, renal failure, dietary protein load, and TIPS or a spontaneous portosystemic shunt.
- **Treat every episode of overt HE**, spontaneous or precipitated. Lactulose is first line; **rifaximin is add-on**, not monotherapy.
- **Secondary prophylaxis after the first episode of overt HE** — lactulose, with rifaximin added after a second episode. This is the point most people forget.
- **Do not restrict protein.** Recommended intake is 1.2-1.5 g/kg/day; restriction worsens sarcopenia and outcomes.
- Grade III-IV needs **ICU-level care** and airway protection.

BEDSIDE TESTS FOR COVERT / MINIMAL HE

- **Psychometric:** PHES battery (the reference standard), number connection test A and B (Reitan's), digit symbol test, block design test, line tracing test.
- **Neurophysiological:** EEG (slowing, triphasic waves - non-specific), critical flicker frequency, inhibitory control test.
- **Asterixis is not a sign of covert HE** - its appearance means the patient has already crossed into grade II. It is also not specific to liver disease: uraemia, CO₂ retention and drug toxicity all produce it.

HEPATIC ENCEPHALOPATHY — WHY EACH PRECIPITANT WORKS

The list above (a-j) is correct but bare. Examiners rarely stop at naming a precipitant — they ask why. Every mechanism below funnels into one of four final pathways: ① more nitrogen delivered to gut bacteria, ② more ammonia absorbed or produced, ③ less ammonia cleared (worse liver function or more shunting), or ④ a brain made more vulnerable to a given ammonia level.

Precipitant	Mechanism — the answer to “why?”	Path way
GI bleeding (variceal, ulcer)	Blood is a protein meal delivered straight into the gut — roughly 100 g of protein per litre of blood. Colonic bacteria deaminate it to ammonia. Hypovolaemia from the bleed simultaneously drops hepatic and renal perfusion, and transfusion adds a further nitrogen load. This is the commonest precipitant to quote.	① + ③
High protein intake	More substrate reaching colonic urease-producing bacteria → more ammonia. Note this is why the old teaching was protein restriction — which is now wrong : chronic restriction causes sarcopenia, and muscle is the main extrahepatic site of ammonia disposal (glutamine synthetase). Losing muscle makes HE worse in the long run.	①
Vomiting, diarrhoea, over-diuresis → hypokalaemia, hyponatraemia, alkalosis	Hypokalaemia drives K ⁺ out of cells in exchange for H ⁺ → intracellular acidosis in the renal tubular cell → increased renal ammoniogenesis . It also causes a metabolic alkalosis , and alkalosis shifts the NH ₄ ⁺ ⇌ NH ₃ equilibrium towards NH₃ — the un-ionised, lipid-soluble form that crosses the blood-brain barrier . Hyponatraemia independently worsens HE by causing astrocyte swelling, compounding the osmotic effect of glutamine accumulation. Volume depletion adds prerenal azotaemia.	② + ④
Large-volume paracentesis	Post-paracentesis circulatory dysfunction — removing fluid without albumin cover unmasks the underlying splanchnic vasodilatation, effective arterial blood volume falls, renin-angiotensin and sympathetic activation follow, and renal and hepatic perfusion drop. Give 6-8 g albumin per litre removed when >5 L is drained.	③
Constipation	Prolonged transit = longer bacterial contact time and a longer absorptive window for ammonia generated in the colon. This is also the mechanism in reverse for lactulose, which works partly by being a cathartic — hence the endpoint of 2-3 soft stools a day .	②
TIPSS / spontaneous portosystemic shunt	An anatomical bypass of the hepatic sinusoids . Ammonia-rich portal blood reaches the systemic circulation without ever meeting a hepatocyte or the reticuloendothelial system. HE occurs in roughly 30-50% after TIPSS; a large spontaneous shunt does the same thing and can be embolised.	③
Uraemia / renal failure	Urea diffuses back into the gut lumen , where bacterial urease splits it into ammonia and CO ₂ — so renal failure paradoxically raises the gut ammonia load. The failing kidney also stops excreting ammonia. Think hepatorenal syndrome in this context.	② + ③
Recent binge / alcoholic or viral hepatitis	An acute-on-chronic insult . Superimposed hepatocyte necrosis acutely reduces the already-marginal residual capacity for ureagenesis, so a patient who was compensated decompensates. Alcohol adds its own direct cerebral depressant effect and often brings thiamine deficiency — always consider Wernicke’s alongside HE in a binge drinker.	③ + ④
SBP and other infections	Infection is the single commonest precipitant in practice . Systemic inflammation acts synergistically with ammonia: cytokines (TNF-α, IL-6) increase blood-brain barrier permeability and prime astrocytes, so the <i>same</i> ammonia level produces deeper encephalopathy. Sepsis also drives catabolism, releasing endogenous nitrogen, and worsens hepatic perfusion. Tap the ascites in every encephalopathic cirrhotic .	① + ④

PRECIPITANTS MISSING FROM THE LIST ABOVE — ADD THESE

- **Sedatives — benzodiazepines and opioids.** The commonest iatrogenic precipitant. Endogenous benzodiazepine-like ligands already act on the **GABA_A-benzodiazepine receptor complex** in HE; exogenous drug is additive. Reduced hepatic clearance prolongs the effect. Flumazenil produces a transient improvement — which is the evidence for the GABA hypothesis.
- **Dehydration and hypovolaemia** from any cause — same final pathway as over-diuresis.
- **Hypoglycaemia and hypoxia** — the failing liver cannot maintain gluconeogenesis; both reduce the brain's tolerance of a given ammonia level.
- **Portal vein thrombosis** or a new **hepatocellular carcinoma** — a sudden unexplained decompensation in a stable cirrhotic should trigger imaging and an AFP.
- **Non-compliance with lactulose** — ask about it explicitly; it is the most easily reversed cause and the one patients under-report.
- **Zinc deficiency** — zinc is a cofactor for ornithine transcarbamylase in the urea cycle; deficiency is common in cirrhosis and worth replacing.

Bringing it together in one line for the viva: "Any event that raises the nitrogen load, increases ammonia absorption, further impairs an already marginal liver, or lowers the cerebral threshold, will precipitate encephalopathy — and in practice infection and GI bleeding are the two to exclude first."

ABDOMEN — ADDITIONS FROM NIHARIKA'S NOTES

Only points not already covered in these notes are reproduced. Everything on this page is tagged to its source.

HISTORY AND GENERAL EXAMINATION

- **Poor nutrition — Dr Kadhiraavan's checklist.** He is particular about these four: **skin flakes (depigmentation) · lustreless hair · beefy tongue · angular cheilitis.** Present them together as a set rather than scattered through the general examination. [Niharika's notes]
- **Inflammatory diarrhoea** — look for **blood and mucus, fever, and abdominal pain.** In any diarrhoea, always ask the **relation to food.** [Niharika's notes]
- On examination think of two things locally: **(1) local peritonitis, (2) a mass.** [Niharika's notes]
- **Fetor hepaticus** is due to **dimethyl sulphide** as well as methyl mercaptan. [Niharika's notes]
- **Palmar erythema spares the centre of the palm** — it involves the thenar and hypothenar eminences. [Niharika's notes]
- **Bile salt deposition causes BRADYCARDIA,** not tachycardia — in obstructive jaundice expect a slow pulse. [Niharika's notes]
- **Obstructed bowel sounds are high-pitched and tinkling.** [Niharika's notes]
- **Ascites precox — always rule out constrictive pericarditis,** because it is treatable. Ascites out of proportion to the peripheral oedema, in a high-SAAG high-protein (≥ 2.5 g/dL) fluid, is the giveaway. [Niharika's notes]

PORTAL HYPERTENSION — ANATOMICAL CLASSIFICATION

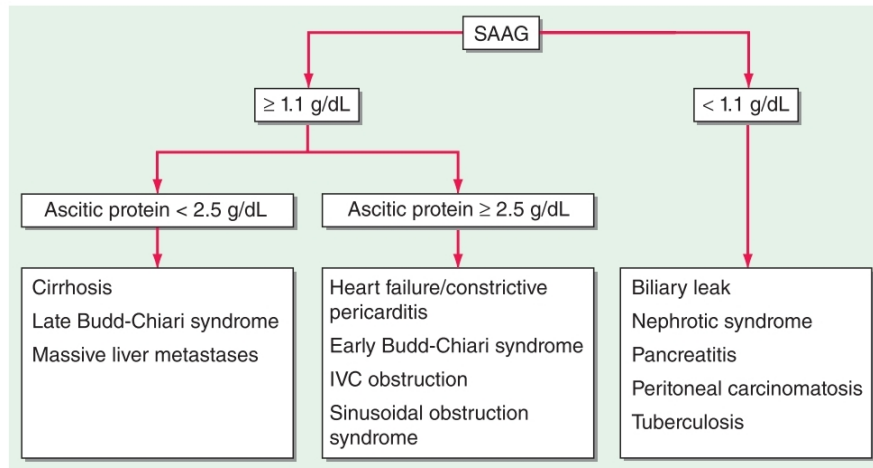
Learn it by where the block sits relative to the sinusoid — this is the structure the figure in Niharika's copy is labelled with. [Niharika's notes]

Level	Causes
Pre-hepatic	Portal vein thrombosis (PVT) · splenic vein thrombosis · massive splenomegaly with increased splenic inflow
Intrahepatic — presinusoidal	Schistosomiasis · non-cirrhotic portal fibrosis (NCPF) · congenital hepatic fibrosis · sarcoidosis
Intrahepatic — sinusoidal	Cirrhosis (any cause) · acute alcoholic hepatitis
Intrahepatic — postsinusoidal	Veno-occlusive disease (VOD) / sinusoidal obstruction syndrome
Post-hepatic	IVC tumour, thrombus or web · large hepatic vein obstruction (Budd-Chiari) · small hepatic vein disease (VOD) · constrictive pericarditis, severe TR

- **Why the level matters clinically:** presinusoidal block gives portal hypertension with **preserved liver function** — variceal bleeding in a patient who is otherwise well, with a normal albumin and no encephalopathy. Sinusoidal block gives portal hypertension *plus* liver cell failure. That single distinction separates NCPF and EHPVO from cirrhosis at the bedside.

SCORES — TWO POINTS OF EMPHASIS

- **Child-Pugh:** learn the **score-2 column** — the middle column is what gets asked, because the 1 and 3 columns are guessable. **A = 5-6** (good survival), **B = 7-9** (moderately impaired), **C = ≥ 10** (high mortality). [Niharika's notes]
- **Maddrey discriminant function = $4.6 \times (\text{PT test} - \text{PT control}) + \text{bilirubin}$; > 32 = poor outcome,** and the threshold for considering steroids in alcoholic hepatitis. [Niharika's notes]
- **MELD = Model for End Stage Liver Disease. Higher MELD → higher on the transplant list.** [Niharika's notes]
- **DSM-V alcohol use disorder** = problems in **2 of 11 life areas** clustering in the same 12-month period. DSM = Diagnostic and Statistical Manual of Mental Disorders. [Niharika's notes]
- **1 unit = 10 mL of pure ethanol.** Teachers differ on the standard drink — $\approx 7-8$ g is quoted, some teach 8 g or 10 g. (See the alcohol page in this document for the full comparison.) [Niharika's notes]



- o High SAAG → Most likely Portal HTN
 - An ascitic protein level of ≥ 2.5 g/dL indicates that the hepatic sinusoids are normal and are allowing passage of protein into the ascites, as occurs in cardiac ascites, early Budd-Chiari syndrome, or sinusoidal obstruction syndrome
 - An ascitic protein level < 2.5 g/dL indicates that the hepatic sinusoids have been damaged and scarred and no longer allow passage of protein, as occurs with cirrhosis, late Budd-Chiari syndrome, or massive liver metastases.

Ascites precoax: Ascites appearing before pedal oedema

- Liver cirrhosis
- TB peritonitis
- **Constrictive pericarditis**** - **ALWAYS RULE OUT** - cause it is treatable {pericardial stripping} unlike cirrhosis of liver causing ascites(reason- one hepatic vein ends close to the IVC-right atrial junction which is **intrapericardial**, so increased pressure is reflected in hepatic veins)

Ascites- workup

1. Blood:
 - a. CBC- neutrophilia indicates infection. Also BUSE LFT
 - b. Serum albumin- decreased in cirrhosis, nephrotic syndrome, protein losing enteropathy - also to check for **SAAG**
 - c. **Cholesterol level** - increase in **nephrotic** , decrease in **cirrhosis**
 - d. LFT
 - e. RFT for HRS
2. Urine examination-
 - a. Nephrotic: Albuminuria, PCR > 0.3 , 24h $> 3g$
 - b. Nephritic: casts (RBC and WBC)
3. Diagnostic tap
 - a. gross appearance
 - i. Clear, straw/light green - Cirrhosis, CHF, Nephrotic
 - ii. **Hemorrhagic: Malignancy, TB, Pancreatitis**
 - iii. Cloud turbid: SBP
 - iv. **Deep green/Dark brown:** Biliary leak
 - v. Milky white/Chylous: Lymphatic obstruction ($> 200mg/dL$ Triglycerides)
 - vi. **Black: Pancreatic necrosis, Malignant melanoma.**

- b. biochemistry -
 - i. protein , (<2.5 and >2.5)
 - ii. specific gravity:
 - iii. glucose(Low/ $<50\text{mg/dL}$ in Malignancy, TB and peritonitis)
 - iv. two special tests
 1. **Adenosine deAminase >40** - TB ,
 2. **amylase($>1000\text{mg/dl}$)** - pancreatitis as suspicion
 - c. Microscopy
 - i. PMN $>250/\text{mm}^3$ in SBP
 - ii. Lymphocytes in TB and malignancy
 - iii. Gm and ZN staining for infections, Cells for Malignancy.
 - d. Culture and PCR.
 [TB peritonitis on tap - via mesenteric lymphnode or miliary tb (hematogenous)
 - i. Straw coloured/haemorrhagic
 - ii. Specific gravity >1.016 (normal is 1.010)
 - iii. Protein $>2.5\text{g/dL}$
 - iv. SAAG $\leq 1.1\text{g/dL}$
 - v. ADA >40 - adenosine de aminase
 - vi. smears are not diagnostic much - laparoscopy with biopsy is done
 - vii. For CNS TB, skeletal TB, disseminated TB, extension of 12 – 24 weeks is added to the continuation]
4. USG
- a. Most sensitive, even 30mL of ascitic fluid can also be detected
 - b. Presence of any intra abdominal mass or lymph nodes- TB/lymphoma
 - c. Splenic enlargement
 - d. Liver enlargement, consistency
 - i. **heterogenous echoes in cirrhosis}** ,
 - ii. **caudate lobe** enlargement - budd chiari (cause of direct separate drainage into ivc hence spared)
 - e. Portal vein- with doppler . **Phtn - $> 1\text{cm}$ or 13 mm in size**
5. Chest X Ray if suspecting pleural effusion

Ascites management-

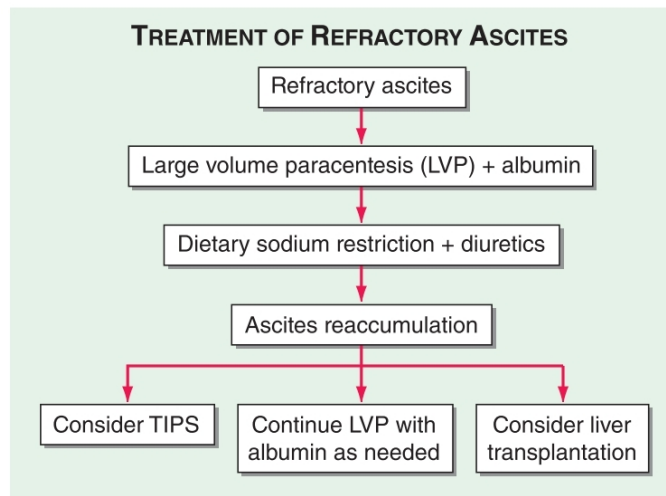
1. **Goal:** Not more than 0.5kg per day to be lost. (not more than 1kg per day in % co-existent pedal edema)
2. Daily charting of: Weight, I/O, Abdominal girth. 2x/wk: BUSE and RFT.
3. Bed rest → More renal perfusion → diuresis.
4. Start with no salt → 4-6g/d (=No added salt)
5. Normal fluids. If HypoNa, restrict to 1-1.5L/day.
6. Diuretics
 - a. K Sparing Diuretics: **Spironolactone start 25mg QD** → Weekly step up → **Max 400mg/day**. → Add Furosemide(**max 120-160mg**)/Torsemide/Thiazide.
→ N/R = **Refractory Ascites**
 - b. STOP Diuretics if:
 - i. SEVERE HypoNa of $<120\text{ mEq}$
 - ii. Progressive renal failure
 - iii. Worsening of HE.

Refractory ascites: no effect even after **Na restriction** and **high dose medical rx**

Causes:

1. non compliance
2. alcohol binge
3. functional renal failure
4. infection
5. SBP
6. nsaid

- rx -first medical (**midodrine** and **clonidine**) → N/R → LARGE VOL PARACENTESIS + sod retention + albumin (or) **TIPSS** (but no effect on mortality)



•
treat underlying condition -

- tb (att) - nephrotic syndrome etc
- if hrs hps pps - require liver transplant
- if **hepatic encephalopathy** - give **lactulose** until patient passes **2-3 soft stools/day** and also **rifaximin**- gut cleanser - 550mmg bd
- endoscopy and banding - varices
- if portal pressure persistently high and risk of variceal bleed - **TIPSS** but risk of HE

Paracentesis- technique and complications - harrison

Technique

- for tap - 20 -30 ml removed
- The left lower quadrant is preferred because of the greater **depth** of ascites and the **thinner** abdominal wall.
- Paracentesis is a **safe procedure** even in patients with **coagulopathy**; complications, including abdominal wall hematomas, hypotension, hepatorenal syndrome, and infection, are infrequent.
- large volume paracentesis - 5-15 litres removed in 24 hrs - (usually only 0.5 - 1 litres removed) and pt is given HAS(Human albumin solution) **6-8g/litre** of ascitic fluid (harrison)

Complications

- Hypoproteinemia- replace with HAS
- sudden cardiorespiratory shock - due to volume depletion
- ppts H.E - portocaval shunts open up and nitrogenous material reach the brain bypassing liver
- Peritonitis
- Trauma and bleed , perforation of hollow viscus (*Dx : presence of multiple organisms on culture without Neutrophilia*)

Complications of ascites

1. Pleural effusion-hepatic hydrothorax { normally we expect hydrothorax in liver disease because of low protein then why give a separate name ? - cause there can just be isolated hydrothorax in absence of ascites in cirrhosis - hepatic hydrothorax} , **more common right side - indicates rent and direct flow of ascitic fluid**
2. SBP - increased risk in **low ascitic protein (<1.5g/dl)**
mortality upto 50% , hence ALWAYS DO CELL COUNT AND RULE OUT
 - Spontaneous bp - called because it occurs spontaneously without any cause - like tap , perforation etc - usually gram negative organism. **E Coli** , klebsiella > strep and Ecocci. Said to be due to Translocation of Gut bacteria against edematous bowel wall + Reduced Immunity (Local and systemic)
 - 2° BP: if more than two organisms are grown in culture
 - DX : > 250 neutrophil/mm³, also pH<7.3.
 - rx :
 - Empirical:
 - o books: CEFOTAXIME (3rd gen) 2g IV q8h x 5d
 - o JIPMER: CEFTRIAXONE 2g i.v od x 5 days
 - o Alternatives: Amoxiclav, ciproflox and oflox.
 - prophylaxis Rx i/c/o : NORFLOX 400 mg /d or once a week
 - h/o SBP with cirrhosis
 - ascitic protein < 1g/dl
 - active GI bleed

cell count(Neutrophilia)	cell count +	cell count -	cell count -
+nt			
culture +nt	culture -	culture +	culture -
RX	RX	DONT	DONT

3. Hernia and divarication of recti
4. Hepatorenal syndrome - **renal vasoconstriction** and **systemic arterial vasodilation**.
 - **criteria**
 - - **s creat > 1.5**
 - - **cirrhosis with ascites**
 - - **no improvement with diuretic or albumin**
 - - **no shock**
 - - **no h/o nephrotoxic drugs**
 - - **no kidney disease**

- two types -
 - **type 1** rapid progression (<48 hrs) - rapid elevation in creatinine levels
 - **type 2** chronic or insidious progression - slow elevation and better outcome
 - med rx - midodrine(alpha agonist) , **octreotide** (basically splanchnic vasoconstriction) and albumin
- 5. hepatopulmonary syndrome - NO -MEDIATED AV SHUNTS - PLATYPNEA and ORTHODEOXIA (**atleast 5 % drop** in saturation on sitting up)
- 6. portopulmonary syndrome - ENDOTHELIN MEDIATED - Rx : sildenafil(pde5) , bosentan (endothelin antagonist) - definitive is TRANSPLANTATION
- 7. Renal failure

Splenomegaly

CML:

- t(9;22)(philadelphia chromosome)
- median age - 60 yrs
- Etio: no familial associations; RFactor: ionising radiation (5-10 yrs after exposure & dose related)

Pathophysiology:

- ABL present on ch 9 ; BCR is present on ch 22
- Balanced reciprocal translocation of ch 9 and 22 leads to Philadelphia chromosome and BCR-ABL which is a HYBRID ONCOGENE (BCR-210) → Constitutional Tyr kinase activity→excessive prolifer and reduced apoptosis of cml cells
- Variations - minor bcr -190 kDa - worse outcome (*Forty favors*: +40kDa)
- micro bcr- 230 kDa - better outcome
- Other pathways are also involved like: MAP kinases, STAT PATHWAY, PI3K, MYC
- Three stages:
- chronic stable phase - less than 10% blasts
- Accelerated phase - blasts between 15-29%
- blast phase - >30% blasts. transforms into all(30%) ,aml(70%)
- Tackled by Tyrosine kinase inhibitors → bind to BCR-ABL1 kinase domain preventing downstream pathways → 'stabilize' the genome , prevent additional mutations and hence preventing transformation to blast phase
- **Untreated cml per se causes genomic instability - accumulation of more mutations - blast crisis**

Symptoms

- Asymptomatic (picked up on routine screening)
- If symptomatic:
 - ◆ Splenomegaly (early satiety , LH pain and dragging sensation)
 - ◆ Anemia (malaise , easy fatigability)
 - ◆ less common- thrombotic and vaso-occlusive events , bleeding diathesis
 - ◆ Patient comes with bone pain, fever, weight loss - suspect blast transformation

Signs

- Massive spleen> hepatomegaly> lymphadenopathy
- Raised basophil- pruritus
- sternal tenderness, skin and subcut rash - blast transformation

Investigations:

- Blood:
 - Hemogram- leukocytosis
 - P.S- More of myelocytes metamyelocytes , neutrophils and basophils
 - Blasts and thrombocytopenia - progression
- Biochemical -
 - Low LAP score
 - High uric acid, LDH
- Marrow -
 - Hypercellular
 - Myeloid hyperplasia (M:E = 20:1)
 - Blasts - indicate acceleration
- Cytogenetics
 - Document ph chromosome - FISH , PCR
 - Masked ph - translocation between ch 9 and ch other than 22
 - Clonal evolution - Additional ch abnormalities (eg - trisomy 8, double ph) - bad prognosis

SOMETIMES FISH AND PCR CAN YIELD FALSE RESULTS.. SO ALWAYS CORRELATE WITH MARROW FINDINGS

Stage wise management recommendations

1. **Chronic phase:** Imatinib → if fails → Change TKI → Allogenic Stem cell transplant
2. **Accelerated Phase:** Hydroxyurea + TKI → Once converted to Chronic → SCT
3. **myeloid blast crisis:** TKI + Chemo ± SCT
4. **lymphoid blast crisis:** treat like ALL

Drug descriptions

1. Tyrosine Kinase Inhibitors (TKI)

first gen - imatinib

sec gen - dasatinib , nilotinib

third gen - ponatinib

Doses -

imatinib - 400 mg daily

dasatinib - 100mg daily (first line), 140 mg (salvage)

Side effects -

DO NOT GIVE TKI IN PREGNANCY

imatinib - wt gain , fluid retention , nausea, diarrhea

dasatinib - Myelosuppression , pleural effusion, thrombocytopenia

ponatinib - hypertension

2. Allogeneic stem cell transplantation (SCT)

second line

curative modality

Whom to consider sct - clonal evolution , failure of salvage therapy

mortality in subseq decades post sct is due to complications than due to cml

relapse (gvhd, organ dysfunction , steroid side effects)

3. Miscellaneous

Other agents used before tki - hydroxyurea(2g/day), busulfan , interferon

alpha

Note:

Cytogenetics on tki therapy - monitor response at 1 yr

partial response - **35%** decrease in positive ph positive metaphases

complete response - no ph positive metaphases detected. **More than 35% decrease?**

10 yr survival rate is now 90%

2. CLL:

Mature looking lymphocytes accumulate in Blood, BM, Spleen and LN.

Male>>Female. 65yo Peak.

Types:

1. Due to B cell expansion, 95% cases.
2. Due to T cell expansion

CF:

1. 1/4th of them asymptomatic
2. Gen LN and HSM
3. Slowly Hemolysis + BM infiltration → Anemia
4. Recurrent infections: S.pneumonia, Staph and H. influenza
5. Rarely bleeding

Complications:

1. Prolymphocytic leukemia (>55% of prolymphocytes)
2. Large cell lymphoma/Richter's syndrome (**More aggressive type**)

Investigations:

1. Blood:
 - a. CBC: Anemia, 50k-2L WBC, $\pm$ Low platelets (due to AI Tcytopenia)
 - b. PS: >95% of all the cells are mature-appearing lymphocytes + Varied amounts of prolymphocytes
 - c. DCT +ve
 - d. Low Folate
2. BM: Hypercellular with small and medium sized lymphocytes
3. LN Bx: Well differentiated small non cleaved lymphocytes

Dx criteria

- ☐ 5×10^9 Lymphocytes
- ☐ >2mons
- ☐ $\leq 55\%$ Atypical/immature lymphocytes
- ☐ Low density of IgM/IgD with κ or λ light chains
- ☐ CD19, CD20, CD23 +ve representing B cells
- ☐ CD5 surface antigen

Staging:

	<u>Binet staging</u>
A (>10y survival)	No anemia. No Tcytopenia
B	<3 areas of lymphoid enlargements
C (<2y)	> 3 areas (OR) <10g/dL Hb and/or <1L/mm ³ Plat

<u>Revised Rai staging</u>	<u>Lymphocytes</u>	<u>LN + HSM</u>	<u>Anemia + Tcytopenia</u>
Low risk (>10y)	Same cutoff as Dx criteria	-	-
Medium		LN $\pm$ HSM	-
High(<2y)		$\pm$	<11 Hb / <1L Plat

Rx

Binet/Rai staging	Rx
A/Low risk	Reassure.
B/intermediate	Treat only symptomatic ones: 1. <u>Symptomatic + Medically Fit</u>: (CRF)CPx + Rituximab + Fludarabine ± Human hyaluronidase. 2. <u>Symptomatic + Medically Unfit</u>: Bendamustine/Chlorambucil 3. Local RadTx for Troublesome LN.
C/High risk	Same as B + Alemtuzumab and SCT CHOP (Cpx, Doxorubicin, Vincristine and Predn) in refractory <u>Other associated symptomatic Mx</u> : PRBC for anemia Predn for hemolytic anemia and Tcytopenia Splenectomy for Symptomatic splenomegaly

<u>Hodgkin Lymphoma</u>	<u>Non Hodgkin Lymphoma</u>
Bimodal age distribution	Older age
Contiguous lymph node involvement	Non contiguous lymph node involvement
"B Symptoms"- fever, night sweats, significant weight loss	No B symptoms
Pel- Ebstein pattern of fever- relapsing type	
	Extranodal involvement- splenomegaly common
Alcohol test : On consumption of alcohol, increased pain and lymphadenopathy	Mediastinal, abdominal, Waldeyer's ring commonly involved

<u>Disorder</u>	<u>Haemogram+PS findings</u>
CML	Anaemia, Thrombocytosis, Basophilia, Myelocytes in peripheral smear
Lymphoma	Anaemia, Eosinophilia (because RS cells release IL-4, 5)
CLL	Smudge cells/basket cells (artefact) Bone marrow lymphocytosis, CD5, 23 positive
Hairy cell Leukaemia	Pancytopenia, TRAP positive hairy cells
AIHA	Spherocytes may be seen
Hyperreactive malarial splenomegaly	Peripheral smear usually negative for malaria
Primary Myelofibrosis	Immature WBCs, nucleated RBCs, teardrop shaped RBCs (Dacryocytes), pancytopenia in later stages, Bone marrow is dry tap with calreticulin mutations (diagnostic)
Polycythemia Vera	Increased Hb, Increased Hct, Decreased serum erythropoietin, Thrombocytosis

CHRONIC MYELOID LEUKAEMIA

- **Peripheral smear — the “college girl appearance”** (the full spectrum of granulocyte maturation on one film). Add **thrombocytosis** to the findings. [Niharika's notes]
- **Marrow: dwarf / micromegakaryocytes, sea-blue histiocytes and pseudo-Gaucher cells** — the last two reflect the high cell turnover overwhelming macrophage lipid handling. [Niharika's notes]
- **Weight loss suggests transformation to blast phase.** The **accelerated phase is 10-30% blasts** — know the criteria. [Niharika's notes]
- **Imatinib also causes hypopigmentation and periorbital oedema** — both are commonly asked and commonly forgotten. [Niharika's notes]
- **T315I mutation → resistance to all first- and second-generation TKIs → stem cell transplant** (or ponatinib, the only TKI active against it). [Niharika's notes]

CHRONIC LYMPHOCYTIC LEUKAEMIA

- **Threshold: 5000 lymphocytes/ μ L** ($5 \times 10^9/L$) of clonal B cells. [Niharika's notes]
- **Binet staging: A** — fewer than 3 nodal areas; **B** — 3 or more areas, without anaemia or thrombocytopenia; **C** — anaemia or thrombocytopenia present. [Niharika's notes]
- **Indications to treat or to splenectomise: AIHA not responding to steroids · Richter transformation · massive splenomegaly · marrow failure.** [Niharika's notes]

BONE MARROW

- **Causes of a dry tap: myelofibrosis · hairy cell leukaemia · aplastic anaemia** (add packed marrow from any infiltrative process). [Niharika's notes]

Polycythemia vera

- Hyperviscosity symptoms
 - o Tinnitus, vertigo, headache, TIA
 - o Blurring of vision
 - o Mucosal bleeding
- Aquagenic pruritus due to histamine release
- Gastric ulcer due to histamine release
- Erythromelalgia- burning pain in the hands and feet accompanied by pallor/cyanosis of the limb
- Thrombosis and haemorrhage
- Paradoxical cyanosis
- Diagnostic criteria (WHO 2016)
 - o Hb >16.5 in males, >16 in females (previously 18.5 and 16.5)
 - o OR Hct >49% in men, 48% in women
 - o Bone marrow biopsy showing trilineage hyperplasia with pleiomorphic and mature megakaryocytes
 - o JAK2 mutations
- Treatment
 - o Weekly Phlebotomy to maintain haematocrit <45%
 - o Aspirin to prevent thrombosis
 - o Hydroxyurea- cytoreductive therapy
 - o Ruxolitinib- JAK inhibitor

Hairy cell Leukaemia

- Chronic B cell lymphoproliferative disorder of old age
- Activating mutation in BRAF
- Differential count reveals more lymphocytes
- Hairy cells
 - o Atypical lymphocytes
 - o Fine cytoplasmic processes resembling hairs
 - o TRAP positivity- Tartrate resistant acid phosphatase
 - o CD19, 20 positive- Pan B cell markers
 - o **CD25 positive- specific marker**
- Bone marrow biopsy
 - o Fried egg appearance of cells
 - o Honeycomb appearance of marrow due to infiltration of leukemic cells
- Treatment- Cladribine. Symptomatic treatment. Steroids and Myelotoxic drugs are C/I.

Hyperreactive malarial splenomegaly (Formerly Tropical Splenomegaly)

- **No fever**, Peripheral smear **negative** for malaria
- **IgM** anti malarial antibodies **raised**
- **Liver** biopsy is done to show sinusoidal lymphocytosis/ Kupffer cell hyperplasia
- Treatment- long term Chloroquine (Proguanil- Dr. Hamide)
- Criteria***
 - o Major criteria
 - Hailing from endemic area - africa , india
 - Massive splenomegaly **>10cm**
 - Elevated serum IgM
 - Clinical and immunological response to antimalarials
 - Regression of splenomegaly by **40% 6 months** after starting **therapy**
 - High antibody levels of Plasmodium species

- o Minor criteria
 - Hepatic sinusoidal lymphocytosis
 - Hypersplenism
 - Lymphocyte proliferation
 - Familial occurrence

Chronic fever d/d (for months)

- TB
- Lymphoma
- IE
- Brucellosis
- CMV syndromes
- Drug fever

Hypersplenism

- In the presence of **splenomegaly**,
- **Decrease in one or more cellular elements of the blood**
- With a normal or hyperplastic bone marrow
- Which responds to splenectomy (although it is not the standard of care)
- Causes- any cause of splenomegaly, but more often in **congestive** aetiologies
- Mechanism
 - o Sequestration of cells
 - o Immune mediated destruction
 - o Haemodilution due to increased plasma volume
- Management
- When to do splenectomy?

Permutations and Combinations

1. Hepatosplenomegaly

- **Infective:** Disseminated TB, Infectious mononucleosis, Infective endocarditis, Chronic malaria, Infective hepatitis, Enteric fever, Kala-azar (don't tell consultants)
- **Haematological:** Haemolytic anaemia, CML, Myelofibrosis, PCV, HL, NHL, HCL
- **Congestive:** CCF, Constrictive pericarditis, Cirrhosis with portal hypertension, Budd chiari
- **Storage disorders:** Gaucher, Amyloidosis, Glycogen storage disorders
- **Autoimmune-** SLE, Felty

2. Hepatosplenomegaly with lymphadenopathy

- Lymphoma
- Leukaemia (ALL, CLL)
- Disseminated TB
- Infectious mononucleosis
- AIDS
- CML with blast crisis

3. Splenomegaly with icterus

- Cirrhosis
- Acute malaria
- Haemolytic anaemia
- Lymphoma
- Acute viral hep
- Miliary TB

4. **Fever with splenomegaly**: Malaria, Leptospirosis, Scrub typhus, Infective Endocarditis, Dengue, Kala azar
5. **Acute Fever with thrombocytopenia**: Dengue, Malaria, Leptospirosis, Scrub typhus. In Dengue, because thrombocytopenia is immune mediated, the fever will be first, then thrombocytopenia. In the others, fever and thrombocytopenia will occur on the same day
6. **Fever with lymphadenopathy**: Leptospirosis, Scrub Typhus, Viral infections. Long duration- lymphoma

When to suspect what?

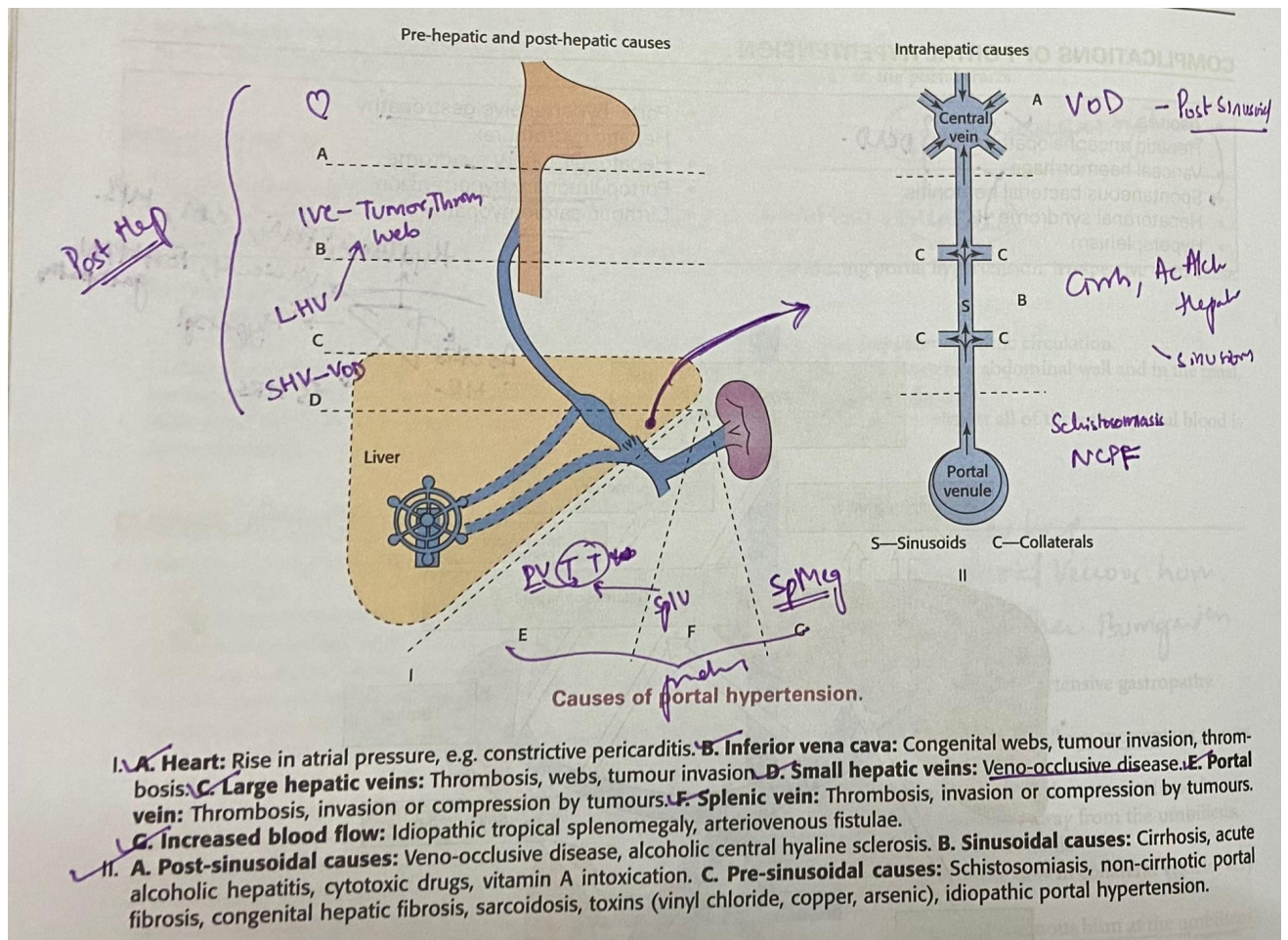
- Age groups of malignancies
 - o CML: 50-60yo (UpToDate)
 - o CLL: 60-70yo > 30-40yo (UpToDate)
 - o Lymphoma: HL is bimodal: 20-35yo and 50-65yo; NHL is

Portal Hypertension

definition - portal hepatic venous pressure gradient > 5 mm Hg (gradient in pressure between portal vein and IVC) - invasive method - HARRISON

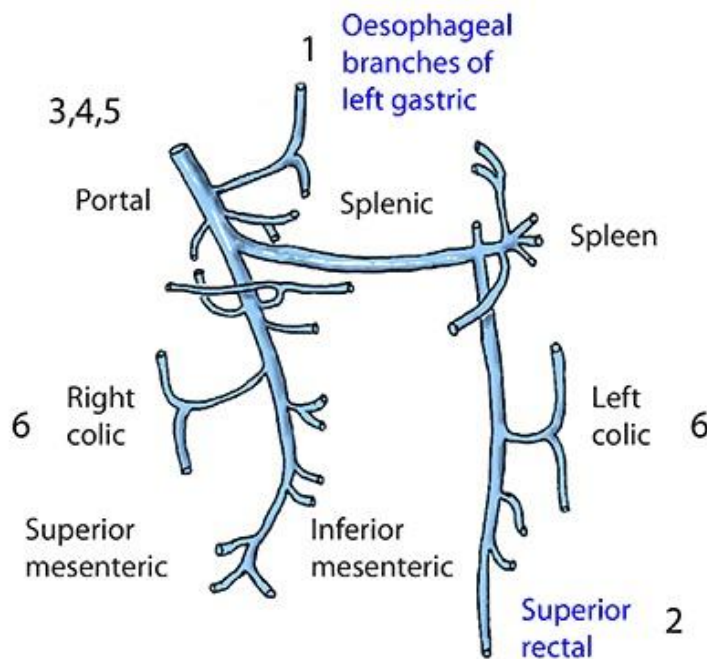
- Aetiology

TABLE 337-3 Classification of Portal Hypertension	
Prehepatic	
	Portal vein thrombosis
	Splenic vein thrombosis
	Massive splenomegaly (Banti's syndrome)
Hepatic	
	Presinusoidal
	Schistosomiasis
	Congenital hepatic fibrosis
	Sinusoidal
	Cirrhosis—many causes
	Alcoholic hepatitis
	Postsinusoidal
	Hepatic sinusoidal obstruction (venoocclusive syndrome)
Posthepatic	
	Budd-Chiari syndrome
	Inferior vena caval webs
	Cardiac causes
	Restrictive cardiomyopathy
	Constrictive pericarditis
	Severe congestive heart failure



- Normal portal vein pressure: 5-8 mmHg
- Portal hypertension: >12 mmHg
- Diagnosis
 - o Non-invasive techniques
 - USG- Findings that may be present in portal HTN include
 - Ascites
 - Splenomegaly
 - Nodular liver
 - Portal flow mean velocity <12 cm/second
 - Portal vein diameter >12mm

PORTOSYSTEMIC ANASTOMOSES



1 Lower oesophagus

Portal: Oesophageal branches of left gastric veins
Systemic: Azygos veins

4 Bare area of liver

Portal: Hepatic/portal veins
Systemic: Inferior phrenic veins

2 Upper anal canal

Portal: Superior rectal vein
Systemic: Middle/inferior rectal veins

5 Patent ductus venosus (rare)

Portal: Left branch of portal vein
Systemic: Inferior vena cava

3 Umbilical

Portal: Veins of ligamentum teres
Systemic: Superior/inferior epigastric veins

6 Retroperitoneal

Portal: Colonic veins
Systemic: Body wall veins

- Treatment

- o Endoscopy to screen for varices. EVBL if varices are present
- o Nonselective beta blockers- Propranolol/Nadolol
- o Treat complications as they arise
- o Surgical- TIPSS

Acute hepatic failure

- Aetiology- Drugs (paracetamol, ATT, Halothane), Viral hep, Cryptogenic, Wilson's, shock, budd chiari, leptospirosis
- Dr. Hamide
 - Acute liver failure- Encephalopathy and coagulopathy that occurs within **26 weeks** of initial liver injury in the absence of pre-existing liver disease
 - Fulminant hepatic failure- 4 to 8 weeks
 - Treatable causes of Acute liver failure
 - Paracetamol toxicity
 - Severe malaria
 - Leptospirosis
 - ?Wilson's
- Fulminant hepatic failure- features of hepatic encephalopathy and INR >1.5 developing within 8 days
- Hyperacute-
 - Jaundice to encephalopathy <7 days
 - Cerebral oedema common
 - Common causes- viral, paracetamol
- Acute
 - Jaundice to encephalopathy 8-28 days
 - Cerebral oedema common
 - Causes- Cryptogenic, drugs
- Subacute
 - Jaundice to encephalopathy 29 days to 12 weeks
 - Cerebral oedema uncommon
 - Causes- Cryptogenic, drugs

Maddrey score

4.6x (PT test-PT control)+ Serum bilirubin in mg/dL

Used in alcoholic hepatitis

Score **>32** implies poor outcome

CHILD PUGH TURCOTT SCORE

FACTOR	1	2	3
A lbumin		2.8 - 3.5 g/dl	
B ilirubin		2-3 mg/dl	
C oagulopathy - <i>PT</i>		4-6 sec prolongation	
D istension - <i>ascites</i>		controlled with diuretics	
E ncephalopathy		1-2	

CHILD A : 5-6 ; B : 7-9 ; C : >10

LIVER TRANSPLANT

- Indications
 - Acute liver failure : Highest priority
 - Cirrhosis with complications
 - liver neoplasm
 - metabolic condition due to liver disease
 - familial amyloid polyneuropathy,
 - primary hyperoxaluria
 - cystic fibrosis
- cirrhosis pts are typically candidates if MELD score is ≥ 15 (MELD-Na score). but evaluation started once MELD ≥ 10
- **MELD SCORE** - min : 6 & max : 40
 - $MELD = 3.8 \cdot \log_e(\text{serum bilirubin [mg/dL]}) + 11.2 \cdot \log_e(\text{INR}) + 9.6 \cdot \log_e(\text{serum creatinine [mg/dL]}) + 6.4$
 - $MELD\text{-}Na = MELD + 1.32 \cdot (137 - Na) - [0.033 \cdot MELD \cdot (137 - Na)]$
 - Na value < 125 mmol/L are set to 125, and values > 137 mmol/L are set to 137.
 - S.Na is a reflection of the vasodilatory state in cirrhosis and predicts waitlist mortality independent of the MELD score
 - There is a linear increase in mortality by 5 percent for each mmol decrease in serum sodium between 125 and 140 mmol/L
- MELD standard **exception** points (these are not accounted in normal MELD scoring but are indication for transplantation)
 - hepatocellular carcinoma(HCC),
 - hepatopulmonary syndrome, and
 - portopulmonary hypertension.
- Contraindications
 - cardiopulmonary disease - that cant be corrected
 - Malignancy (other than liver) - not meeting cure criteria
 - metastatic HCC
 - intrahepatic cholangiocarcinoma,
 - persistent nonadherence with medical care
 - Uncontrolled sepsis
 - Acute liver failure with a sustained intracranial pressure >50 mmHg or a cerebral perfusion pressure <40 mmHg

- **KINGS COLLEGE CRITERIA - for ACUTE LIVER FAILURE**

- I/C/O **ACETAMINOPHEN** induced
 - An **arterial pH < 7.30**, irrespective of grade of encephalopathy
 - or
 - Grade III or IV encephalopathy with both a **PT** greater than **100** seconds & s.CREATININE concentration > **3.4 mg/dL**

- For **other** causes of ALF , referral for liver transplantation is indicated for:
 - A PT greater than 100 seconds, irrespective of the grade of encephalopathy
 - or
 - Any three of the following:
 - Age < 10 or > 40 years
 - **Unfavorable** disease **etiology**, such as non-A, non-B viral hepatitis, idiosyncratic drug reactions, Wilson disease
 - Duration of jaundice before development of encephalopathy greater than 7 days
 - PT > 50 sec
 - S.Bil > 18 mg/dL

Alcohol and disease-

read more from Harrison's/19e/p2723

- CAGE questionnaire
 - Cut down- Have you ever felt that you should cut down on your alcohol intake?
 - Annoyed- Have you ever felt annoyed when people criticise your drinking habits?
 - Guilty- Have you ever felt guilty about your drinking habits?
 - Eye opener- Do you need a drink first thing in the morning?
 - Inference
 - One "yes" response- indicative of an alcohol use problem
 - More than one- strong indication of alcohol abuse or dependence
- FAST questionnaire- fast alcohol screening test questionnaire
 - How often do you have 8 or more drinks (men)/6 or more drinks (women) on one occasion?
 - How often during the last year have you been unable to remember what happened the night before because you were drinking?
 - How often during the last year have you failed to do what was normally expected of you because of your drinking?
 - Has a relative/friend/doctor been concerned about your drinking and suggested you to cut down?
- Robbins
 - Daily ingestion of 80gm ethanol generates significant risk for severe hepatic injury
 - Daily ingestion of 160g or more for 10-20 years is more consistently associated with severe injury
 - Only 10-15% of alcoholics develop cirrhosis
- Hazardous drinking is the regular consumption of 3 units per day for men (24g pure ethanol) and 2 units per day for women (14g pure ethanol)- MacLeod
- Calculating units of alcohol

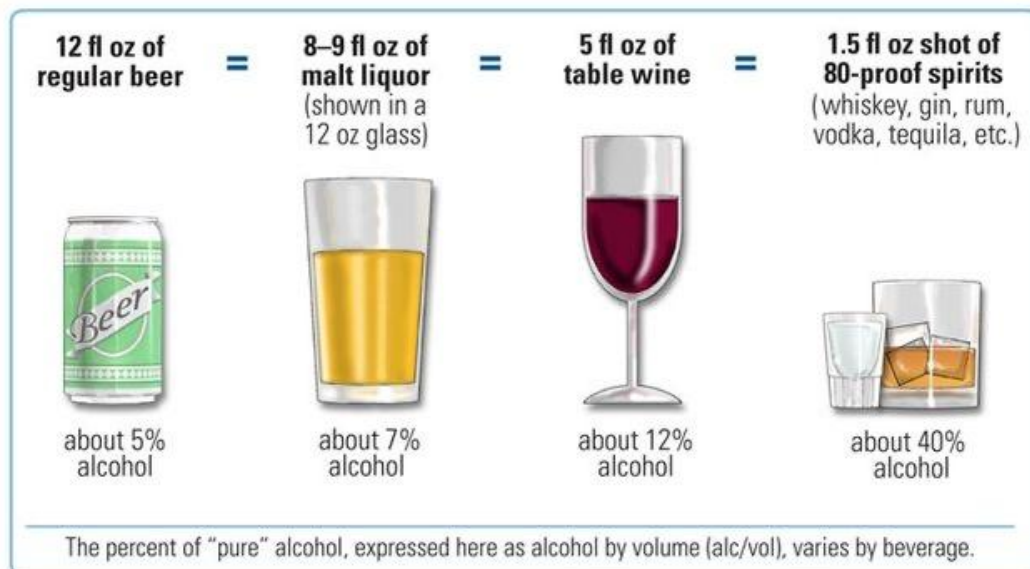
widmark formula

BAC in g = BAR -> body weight x alcohol conc x R (constant male=0.68, female=0.55)

Alcohol Content Chart

	Type of alcohol	Average alcohol %	Main Ingredients	Origin
Distilled Alcohol	Shōchū	12 ~ 45	Sweet potato, rice, wheat, etc....	Mostly produced in Japan
	Whisky	40 ~ 60	Barley	Originated in England
	Brandy	40 ~ 60	Grapes	Originated in France
	Vodka	40 ~ 60	Barley, Wheat & Corn	Originated in Russia
	Tequila	35 ~ 40	Blue Agave	Originated in Mexico
	Pao-Chu	30 ~ 60	Rice & beans	Chinese Shochu
Brewed Alcohol	Saké	14 ~ 18	Rice	Mostly produced in Japan
	Wine	12 ~ 16	Grapes	Produced in all over the world.
	Beer	4 ~ 15	Barley	Produced in all over the world.
	Shōkōshū	15 ~ 55	Sticky rice, wheat	Produced and originated in Shōkō China

- o 1 unit= 1 small glass of wine, ½ pint of beer, 1 shot (30ml) of spirits (rum/vodka/whiskey/gin)-> standard drink
- o 1 unit= 25ml of 40% ethanol= 10ml pure ethanol
- o X% proof= X units of alcohol per litre



- System wise effects of alcohol**
 - CVS-
 - Acutely- decreases myocardial contractility and causes peripheral vasodilation, thus mildly decreasing BP and increasing CO.
 - Chronic consumption causes a dose dependent increase in BP which returns to normal within weeks of abstinence
 - Increased risk of coronary artery disease because of increased LDL
 - Increased risk of (dilated) cardiomyopathy. Mechanism-
 - Holiday heart syndrome- Atrial or ventricular arrhythmias, especially paroxysmal tachycardias, can occur temporarily after heavy drinking in individuals who do not have heart disease
 - CNS
 - Blackout- temporary episode of anterograde amnesia
 - Disturbed sleep
 - Relaxes the muscles of the pharynx, so can exacerbate obstructive sleep apnoea
 - Peripheral neuropathy
 - Cerebellar degeneration or atrophy
 - Wernicke's and Korsakoff syndromes (read from theory)
 - Cognitive problems
 - GIT
 - Esophageal varices
 - Acute pancreatitis
 - Liver
- Alcohol use disorder definition (DSM-V) Repeated alcohol related difficulties in at least 2 of 11 life areas that cluster together in the same period. (Ref
- Fulminant hepatic failure- encephalopathy within 4 weeks of onset of jaundice
- *****Melaena**- Slimy, smelly, sticky dark stools (Dr. Hamide) (dark, smelly, tarry)
 - Drug induced- iron
- Compensated vs. Decompensated cirrhosis
 - Compensated cirrhosis- cirrhosis without any complications secondary to hepatic dysfunction or hepatic parenchymal architecture
 - Features suggestive of compensated cirrhosis- shrunken liver with splenomegaly in an otherwise normal individual
 - Decompensated cirrhosis- Symptoms and signs suggestive or complications secondary to hepatic dysfunction or change in hepatic parenchymal architecture (ie portal HTN)
 - Ascites, GI bleed, HE are suggestive of DCLD
- Poncet disease- TB affecting joints
- Hypersplenism criteria
 - Moderate to massive splenomegaly
 - Pancytopenia
 - Normal to hyperactive bone marrow
 - Pancytopenia responds to splenectomy

ALCOHOL — STANDARD DRINK, BINGE DRINKING, AND THE CIRRHOTIC DOSE

The page above defines a unit but stops there. These are the three numbers you will be asked for: what one drink contains, what counts as a binge, and how much alcohol over how long actually produces cirrhosis.

1. HOW MUCH ALCOHOL IS IN A DRINK

Measure	Ethanol content	Equivalent to
1 UK unit	8 g (10 mL) of ethanol	½ pint of ordinary beer · 1 small (25 mL) measure of spirits · 1 small glass of wine
1 US standard drink	14 g of ethanol	360 mL beer (5%) · 150 mL wine (12%) · 45 mL spirits (40%)
Quick bedside formula	grams = volume (mL) × strength (%) ABV) × 0.789 ÷ 100	A 750 mL bottle of 12% wine ≈ 71 g ≈ 9 UK units

2. BINGE DRINKING — THE DEFINITIONS

- **NIAAA (the one to quote):** a pattern of drinking that raises the blood alcohol concentration to **≥0.08 g/dL** — typically **≥5 drinks in men** or **≥4 drinks in women within about 2 hours**.
- **WHO / UK: ≥6 units for women, ≥8 units for men** on a single occasion — that is, double the daily limit in one sitting.
- **Heavy episodic drinking** is the WHO term for binge drinking occurring at least monthly.
- **Why it matters clinically:** binge patterns cause **acute alcohol-associated hepatitis**, arrhythmia ('holiday heart' — usually AF), acute pancreatitis, hypoglycaemia, aspiration and trauma — and binge drinking on a background of steady heavy intake accelerates progression to cirrhosis independently of the total dose.

3. THE CIRRHOTIC DOSE — THRESHOLD AND DURATION

	Men	Women
Threshold above which risk of alcohol-associated liver disease rises	>30 g/day (≈ 4 units/day)	>20 g/day (≈ 2-3 units/day)
Dose classically associated with cirrhosis	60-80 g/day for >10 years	20-40 g/day for >10 years
Safe threshold	There is none for liver disease — risk is continuous	Women are more susceptible at every dose

- **Duration matters as much as dose.** Roughly **10 years** of sustained drinking above threshold is the usual figure quoted; cirrhosis is rare below that however heavy the intake.
- **Only 10-20% of heavy drinkers develop cirrhosis.** This is the most commonly asked follow-up. Cofactors that decide who does: **female sex** (lower gastric alcohol dehydrogenase, higher blood levels per unit body weight), **obesity and metabolic syndrome**, **hepatitis B or C co-infection**, **genetic polymorphisms** (PNPLA3, TM6SF2), **drinking outside meals**, and **daily rather than intermittent drinking**.
- **The spectrum is sequential but overlapping:** steatosis (>90% of heavy drinkers, fully reversible in 4-6 weeks of abstinence) → alcohol-associated hepatitis (10-35%) → fibrosis → cirrhosis (10-20%) → hepatocellular carcinoma (1-2% per year once cirrhotic).

4. SCREENING TOOLS — KNOW ONE PROPERLY

- **CAGE** — Cut down, Annoyed by criticism, Guilty, Eye-opener. **≥2 of 4 is a positive screen.** Quick, but poor at detecting hazardous drinking that has not yet caused dependence.
- **AUDIT** (10 items, WHO) — the better instrument. **≥8** indicates hazardous drinking; **≥20** suggests dependence. **AUDIT-C** is the 3-item consumption-only short form.
- **Biochemistry: AST:ALT ratio >2** is the classic pointer (both usually <300 IU/L), with raised **GGT** and **macrocytosis**. **Carbohydrate-deficient transferrin** is the most specific marker of sustained heavy intake.

JIPMER Chemotherapy Protocols

- AML- 7+3 protocol for induction
 - 24 hour continuous infusion for 7 days of Cytarabine (Ara-C) 100 mg/m²/day, PLUS
 - 3 days of Daunorubicin 60 mg/m² over 1 hr
 - Check bone marrow after 28 days
 - If induction is successful, then consolidation is started with 3 cycles of Ara-C at a higher dose for a shorter duration
 - If consolidation is successful then follow up regularly because the chance of recurrence is high
 - If induction fails, then re-induction/bone marrow transplant/newer drugs like Midostaurin can be added
- CLL- FCR regimen
 - Fludarabine, Cyclophosphamide, Rituximab
 - Bendamustine based regimens are currently first line but FCR regimens are gaining in popularity
- HL- ABVD 8 cycles
 - Adriamycin, Bleomycin, Vinblastine, Dacarbazine
- NHL- 6 cycles of R-CHOP
 - Rituximab 350 mg/m², Cyclophosphamide 750 mg/m², Doxorubicin (Hydroxydaunorubicin) 50 mg/m², Vincristine (Oncovorin) 1.4 mg/m², Prednisolone 100 mg for 5 days
 - Repeat bone marrow and imaging. If in remission then follow up. If not in remission then transplant
- Multiple myeloma- CyBorD regimen 2 times each per cycle x 4 cycles
 - Cyclophosphamide 300 mg/m², Bortezomib 1.4 mg/m², Dexamethasone 40 mg
 - ± Autologous Bone marrow transplant
 - Lenalidomide (Thalidomide analogue) based regimens are used in other centres

UPDATE — MULTIPLE MYELOMA CHEMOTHERAPY

The protocol page above lists CyBorD (cyclophosphamide-bortezomib-dexamethasone) with lenalidomide 'used in other centres'. That reflects practice around 2018–20. The field has moved from triplets to quadruplets, and CyBorD is now essentially a regimen of convenience — used where lenalidomide or an anti-CD38 antibody is unaffordable or unavailable, or in renal failure while awaiting recovery.

CURRENT FIRST-LINE (NEWLY DIAGNOSED MYELOMA)

Patient	Regimen	Evidence
Transplant-eligible	Dara-VRd (daratumumab + bortezomib + lenalidomide + dexamethasone) induction → ASCT → consolidation → lenalidomide maintenance. Isa-VRd is the equivalent isatuximab-based option.	PERSEUS (phase 3), GRIFFIN (phase 2). ~4-yr PFS around 85%.
Transplant-ineligible / deferred	D-VRd if fit (FDA-approved for this group in Jan 2026), or D-Rd (daratumumab + lenalidomide + dex) continuous. Isa-VRd is an alternative.	CEPHEUS (D-VRd), MAIA (D-Rd), IMROZ (Isa-VRd)
Frail / resource-limited	Start with a triplet (VRd, or CyBorD where lenalidomide is not feasible) and add the fourth drug once tolerance is shown — the 'when in doubt start with three, add the fourth' approach.	Pragmatic; widely used in Indian practice on cost grounds

WHAT TO SAY IF ASKED WHY CyBorD IS STILL USED

- **Renal impairment at presentation** — bortezomib needs no dose adjustment and works fast; lenalidomide is renally cleared and needs dose reduction. CyBorD or VCd is a reasonable bridge until creatinine recovers, then switch to a lenalidomide-containing regimen.
- **Cost and access** — daratumumab and lenalidomide are the expensive components. This is the honest answer in a government-hospital setting and examiners accept it.
- **Cyclophosphamide is not part of any current preferred first-line regimen** in guideline terms — the backbone is proteasome inhibitor + IMiD + anti-CD38 antibody + steroid.

THE FOUR-DRUG BACKBONE — LEARN IT AS CLASSES

Class	Drug	Key toxicity to quote
Proteasome inhibitor	Bortezomib (carfilzomib, ixazomib)	Peripheral neuropathy (much less with weekly subcutaneous dosing), thrombocytopenia, herpes zoster — give acyclovir prophylaxis
IMiD	Lenalidomide (thalidomide, pomalidomide)	Venous thromboembolism — needs aspirin or LMWH prophylaxis; myelosuppression; teratogenicity
Anti-CD38 monoclonal	Daratumumab (isatuximab)	Infusion reactions, hypogammaglobulinaemia and infection; interferes with blood cross-matching (anti-CD38 binds red cell CD38) and with serum M-protein measurement (IgG kappa)
Steroid	Dexamethasone	Hyperglycaemia, insomnia, infection, myopathy

SUPPORTIVE CARE AND RELAPSE — WORTH ONE LINE EACH

- **Bisphosphonate** (zoledronic acid) or denosumab for all patients with bone disease — monthly, with dental clearance first (osteonecrosis of the jaw).
- **MRD negativity** is now the key response endpoint — deeper than complete response, and the primary endpoint on which CEPHEUS was approved.
- **Maintenance**: lenalidomide until progression is the minimum standard after ASCT; dual maintenance (add a PI or anti-CD38) for high-risk cytogenetics.
- **Relapse**: BCMA-directed **CAR-T** (ciltacabtagene autoleucel, idecabtagene vicleucel) and **bispecific antibodies** (teclistamab, elranatamab; talquetamab targets GPRC5D) have transformed relapsed disease. Not available at most Indian centres, but naming them shows currency.

For the exam, the safe formulation is: 'The current standard of care for newly diagnosed myeloma is a quadruplet — an anti-CD38 antibody with a proteasome inhibitor, an IMiD and dexamethasone — followed by autologous transplant in eligible patients and lenalidomide maintenance. In our setting CyBorD is used where lenalidomide or daratumumab is not accessible, and in renal failure.'

Approach to an abdominal mass

- Determine if it is parietal or intra-abdominal
 - Perform head raising test or leg raising test (if it is above or below the umbilicus respectively)
 - Parietal masses will become more prominent
- Determine if the intra-abdominal mass is intraperitoneal or retroperitoneal
- Retroperitoneal organs (**SAD PUCKER**)- Suprarenal gland, Aorta and IVC, Duodenum (2nd to 4th parts), Pancreas (**except tail**), Ureter, Colon (descending and ascending), Kidneys, Esophagus (lower 2/3), Rectum

<u>Intra-peritoneal</u>	<u>Retroperitoneal</u>
Head raising- disappears or becomes smaller	Head raising- disappears or becomes smaller
Becomes more prominent in knee-elbow position	Does not fall forward in the knee-elbow position
Moves with respiration	Does not move with respiration
Has intrinsic mobility	No intrinsic movement
Dull on percussion	Resonant on percussion because of the intestinal loops in front
Can cross the midline	Usually do not cross the midline

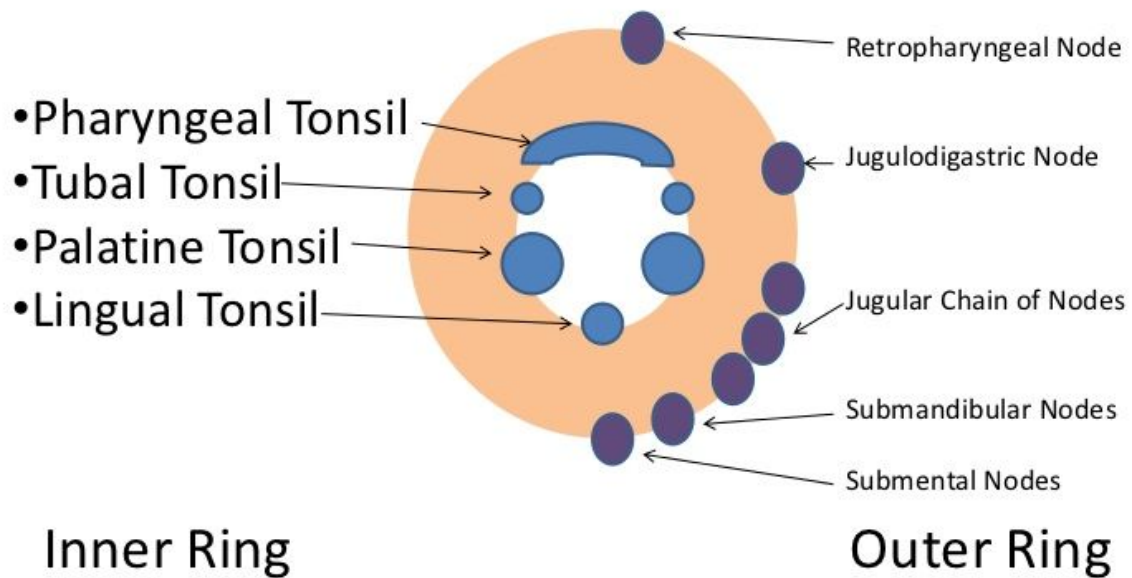
Gynae masses

<u>Ovarian mass</u>	<u>Ascites</u>
Flanks are resonant (because the ovarian mass pushes the bowels laterally)	Ascites is resonant in the midline because the bowels float in the middle of the abdomen
Shifting dullness absent	Shifting dullness present
Ruler test - A flat ruler is placed on the abdomen just above the ASIS and pressed firmly backwards. In ovarian mass, the pulsations of the aorta are transmitted to the fingers through the ruler	Negative
Umbilicus can be displaced upwards	Umbilicus can be displaced downwards

Table I : Differentiating Features of NCPF, EHPVO, Cirrhosis

S. No.	Parameter	NCPF	EHPVO	Cirrhosis
1	Mean age	28	10	40
2	Sex M:F	2-3:1	2-3:1	4-5:1
3	GI bleeding	Infrequent well tolerated	Infrequent well tolerated	Frequent and recurrent
4	Encephalopathy	Rare	Rare	Common
5	Jaundice	Rare	Rare	Common
6	Ascitis	Rare	Rare	Common
7	Spider Nevi	No	No	Yes
8	Splenomegaly	>7cm	< 7 cm	< 5 cm
9	Hepatomegaly	Firm, smooth 1-2 cm	Uncommon	Firm, irregular 3cm
10	LFT	Normal	Normal	Deranged
11	Liver			
	a. Gross	Normal/rarely irregular	Normal	Shrunken, nodular
	b. Microscopic	Normal/portal fibrosis	Normal	Necrosis and regeneration
12	Site of block	Intrahepatic presinusoidal	Extrahepatic presinusoidal	Intrahepatic, sinusoidal and post sinusoidal
13	Pressure studies			
	a. Intrasplenic pressure	High	High	High
	b. Portal vein pressure	High	Normal to raised	High
	c. WHVP	Normal –	Normal	High
	d. (wedged hepatic	raised		
	e. venous pres- sure)			
14	Long term survival (5 years)	90-95%	90-95%	50-75%

Waldeyer's Lymphatic Rings



Stroke- History, Examination, Management, Discussion

Faculty wise points

- Dr. Hamide- Integrate the risk factors into the history
- Dr. Vivekanandan- Include DM and HTN in HOPI,

CVA

- stroke is used to identify all forms of CVA , stroke is a better term than CVA
- diseases under CVA
 - cerebral thrombosis
 - cerebral embolism
 - cerebral h/h
 - SAH
 - HTN encephalopathy
 - other inobvious stroke
 - MS
 - todods palsy
 - cerebellar h/h or infarction
 - venous sinus thrombosis
 - sub dural or epidural hematoma

History

Demographic details

Dextrousness (not needed)- Right/Left handed-

How to determine? Roll a paper and ask the patient to look through it. If a person is right handed, they will use their right eye and vice versa. Or ask the patient to kick a ball.

Significance? Right handed people mostly have left hemispheric dominance (language centre located in left hemisphere) and vice versa.

Education status- for MMSE purpose- if a patient is illiterate, he won't be able to go through MMSE.

Source of history: If not from the patient, state the source of history. Say "**Reliability is fair**"

Chief Complaint- Weakness in the R/L UL/LL/both for the last ____ days

HOPI:

Patient was apparently asymptomatic ____ days ago when he developed **weakness** in <limb>.

Classic history- embolic stroke- while doing normal work; thrombotic- on waking up from sleep; haemorrhagic- after strenuous work

Points in motor history - balavinam

- At the onset of weakness, - **balavinama neratthil**.
 - Sudden onset:
 - Vascular (refer table)
 - Infections (hrs-days progression)
 - Trauma
 - Demyelination
 - Gradual onset usually indicates IC SOL as possible etiology
 - Associated with:
 - was there loss of consciousness - **suya nenaivu**. Cortical localisation; Brain stem + RAS involvement.
 - seizure - **valippu** (**mainly for venous stroke , cause arterial no supply no fire** Also Todd's palsy? but usually its monoplegia and rarely hemiplegia)- yes and also localises cortex no ?
 - speech deficit - **pesa mudiyuma appo?** Cortical and side of the lobe localisation? yes
 - deviation of angle of mouth? - **vaayi konattil orupakkama poidcha?** For internal capsule localisation.
 - **Any headache ? - if yes describe socrates.**
 - severe headache accompanying paralysis
 - SAH - sudden excruciating occipital headache
 - cerebral h/h
 - HTN encephalopathy
 - headache preceding the paralysis for months
 - SOL
 - subdural hematoma
 - mild headache
 - cerebral thrombosis
 - embolism
 - Ppt-ed by:
 - Exertion: Embolic > hge
 - Emotion/Excitement: hemorrhage
 - On waking up, morning: Thrombotic
 - Fever: Cerebral abscess, encephalitis
 - Seizure: Hge/Thrombosis, Todd's
 - Persistent vs intermittent (intermittent in Demyelinating diseases: Waxing and waning in MS; HypoK Periodic paralysis; Also recurrent TIA?)
 - **apdiye irka? - idaipatta irka?**
 - Rigid/flaccid weakness
 - **adhu katdinama/ irukkam irka?**
 - **loose ah irka**
 - Degree of weakness- paresis vs. plegia (Plegia points more towards lesion at IC and below.)
 - **suttama mudiyala ?**
 - **kojama nagarva mudiyuma?**
 - Progression: Static/dynamic weakness (Depends on which type of vascular etiology; Waxing and waning in MS)
 - **apdiye irka?**
 - **adhu meimpattadha?**
 - **adhu mosamakiyyadha?**
 - Evolution of stroke:
 - Stepwise: Stuttering stroke/repeated TIA/Thrombotic origin
 - Completed stroke: Max at onset, completed in 6h.
 - Stroke in evolution: Increasing over hours to days.

<u>Feature</u>	<u>Thrombus</u> (overall MC etiology even in HTN)	<u>Embolus</u>	<u>Haemorrhagic</u>
Onset and Progression	Can be maximal at onset(mins) (or) progress over minutes to hours.	Maximal at onset(secs) Rapid improvement may occur(d/t lysis of emboli)	Sudden onset with progression over minutes to hours. May be catastrophic.
Features of raised ICP	May develop later	Rare.	Develops early in large bleed. ± Neck rigidity as well.
Classical history includes	Mid-elderly p/h/o TIA ± HTN.	Young female k/c/o Valvular HD	>40yo k/c/o HTN

- o If progressive, how much time did it take to progress from UL to LL? What was the patient doing at the time? How was the work affected?
 - o Was the patient able to walk when transported to the hospital? How many people took the patient? How (stretcher/wheelchair)? *To assess the severity of weakness*
 - o Intermittent weakness- electrolyte imbalance like hypokalemic periodic paralysis.
 - o Remission-exacerbation pattern of weakness- MS
 - o Progressive motor weakness- malignancy
 - o Recovering motor weakness- infection
 - o Slowly progressive motor weakness- degeneration
- Proximal (Usually myopathies. Neuropathy with proximal involvement: GBS)
 - o - kullikyumbodhu thala ku mela thanni ootre mudiya
 - o shirt / tshirt poda mudiya sondama?
 - o sondhama saapada mudiya
 - o keele ukkaramudiya toilet ku pogambodhu?
 - o padi mela yerambodhu kastama irka?
 - o ukkarandhu eranga mudiya?
 - distal: more s/o stroke. (Peripheral Neuropathies, associated with paresthesias)
 - o shirt button poda mudiya?
 - o pen pudika mudiya
 - o saapad kalakka mudiya?
 - o seippu poda mudiya?
 - o nadakambodhu serippu kaal vittu poguda?
 - neck?/ breathing difficulty
 - o thala thooka mudiya? bed le irundhodu
 - o moochu prachanai irka?
 - Involving or sparing the trunk (*Cerebellar involvement*)
 - o bed la porala / illena thiripi padaka mudiya ?
 - o padukayyil irundhodu yaendhiraka mudiya?
 - Diurnal variation *Myasthenia Gravis*
 - o indha balavinam kalile ratrile marupatdu irka?
 - Pyramidal vs extrapyramidal - extrapyramidal will have only stiffness but no weakness
 - Abnormal movements (**fasc in lmn , any other extrapyramidal**) - yenthavadu asadharana asaivu irka? adhu control panna mudiya? murukkudhal (twitching)

Sensory history-

- loss of sensation of temperature while bathing. -
 - o kulikyumbodhu sooda thanni kulir tanni teriyadha?
- Loss of touch- can you feel your clothes on your body?
 - o neengal adaigalai unara mudiyuma? / thotta teridhaa?
- Tingling/numbness?
 - o koosudha? erichhal ?
 - o thala thookam bodhu yedadhu shock madiri teriyidha?
- washbasin sign (**Sensory ataxia**)
 - o mugam kalvumbodhu mayakam maadiri irka?
- Can you feel the ground with your feet?
 - o nadakambodhu tarai teridhaa
 - o saani madhiri teridhaa?

Pain

- Pain and weakness can be seen in muscle disease. Polymyositis can have pain and weakness. The weakness will be in proximal muscles.
- Radicular pain can be a harbinger of compressive myelopathy.
- Extramedullary compression has prominent radicular pain.

History suggestive of raised ICT: Headache, projectile vomiting, *Transient visual obscuration* (Dr. Vivekanandan)

Autonomic history:

- bowel
- bladder
 - o feel - cirunirppai unara mudiyudha?
 - o initiate - pogamudituda?
 - o dribbling?
 - o control? Nirutha
- postural hypotension - ukkaramdhu yaedhiricha mayakam maadiri irka
- sweating profusely? adhigama viyartatdu irka?
- palpitation - padapadappu

Cranial nerves history- Quick comment "No h/s/o other cranial nerve involvement (if VII nerve is involved)

I – alteration in smell - vasanai teridha?

II – blurring of vision/diminished vision, differentiate colours, night blindness - paarvai nalla irka?

III, IV, VI – drooping of eyelids/ double vision/ able to move eyes in all directions - kannu? rendu renda teridhaa?

V – having sensation over the face, able to chew food - mugam unarvu teridha? / saapadu mella mudiyuma?

VII – able to close the eyes and lips, deviation of angle of mouth, drooling of saliva, able to feel taste of objects - cuvaikka mudiyuma? / vaayi konattil orupakkama poidcha? / vaayalandhu umilnr varudha? endha pakkam varum? / kannu mudunga?

VIII – hard of hearing, tinnitus, vertigo - kekadhuku prachna irka? / thalai suttral madhiri irka?

IX, X – difficulty in speech, nasal twang to speech, **nasal regurgitation**, difficulty in swallowing - saapad muzhugangu kastam irka? / saapadubodhu saapad muukle varudha?

XI – move the neck in all directions, lift the shoulder - kaluttai nagarvavum mudiyuma? / tol pattai thooka mudiyuma?

XII – able to roll the tongue and push the food backwards - naak neetunga , ulle thallanga

cerebellum history

- nadakambodhu thallatam irka?
- saapad vaayi kitte editha kastama irka?
- oru pakkathil vilum pokku irka?
- narrow roads? - nera nadakamudiyuma

History suggestive of infection- TB meningitis, tuberculoma, TB vasculitis can be stroke mimics.

h/o evening rise of temperature, loss of wt, loss of appetite

h/o ear discharge- CSOM

TB and Stroke

- TB meningitis causes **endarteritis** which itself can cause stroke.
- Mycotic aneurysmal rupture causing a bleed.
- Tuberculomas are stroke mimics.

<u>Structural</u>	<u>Functional</u>
ICSOL: • Primary cerebral tumours • Metastatic cerebral tumours • Subdural haematoma • Cerebral abscess • Neurocysticercosis • Tuberculoma	<i>F. THEM</i> Focal seizures Todd's paresis- after epileptic seizure Hypoglycaemia Encephalitis Migrainous aura- with or without headache Meniere's disease or any vestibular disorder
Demyelination Peripheral Nerve lesions Toxoplasmosis	

<u>Feature</u>	<u>Stroke</u>	<u>Stroke mimic</u>
Symptom onset	Sudden (minutes)	Often slower onset
Symptom progression	Rapidly reaches maximum severity	Often gradual onset
Severity of deficit	Unequivocal	May be variable/uncertain
Pattern of deficit	Well defined pattern	May be non specific
Loss of consciousness	Less common	More common

CVS History- A fib. Ask all the cardinal symptoms- chest pain, dyspnoea, palpitations, syncope, pedal oedema

Drug history

Look out for:

- Oral contraceptive pills
- Digoxin for A fib- a tablet which is given only 5 or 6 days in the week (to avoid toxicity)
- Warfarin- haemorrhagic stroke- tablet given in the evening because PT/INR is done in the morning lol

Past history

- Past history of weakness- look out for any previous TIAs or strokes
- Vascular
 - History s/o Rheumatic or congenital heart diseases in Cardioembolic suspicion
 - Claudication pain and PVD in Thrombotic etiology
- Infection: TB, Fever and infection symptoms for Abscess, meningitis.
- Trauma: SDH

***** Elicit the history points that were risk factors for the stroke (Dr. Hamide)**

Summary (Sample)

An elderly diabetic and hypertensive man presented with sudden onset weakness of his right upper limb and lower limb which gradually progressed and was associated with deviation of angle of mouth to the left side. (Integrate the risk factors also)

Complete hemiplegia : with UMN facial palsy + hemiplegia

Incomplete hemiplegia : hemiplegia without UMN facial palsy

Crossed hemiplegia : ipsilateral LMN type of cranial nerve palsy with contralateral hemiplegia

EXAMINATION

General examination

1. Position: decubitus or supine position
 - a. affected side
 - i. UL : flexed , adducted , semi pronated ; lying helplessly
 - ii. LL : extended , adducted , plantarflexed
 - b. normal side : describe
2. Facies : angle of mouth deviated on attempted talking
3. PR- Irregularly irregular pulse in A.fib , maybe vessel wall thickening
4. BP-
 - a. Cushing's reflex, risk factor for stroke- Cushing's triad of **increased blood pressure, irregular breathing, and bradycardia**
 - b. *better to check to in unaffected limb - to avoid low BP recording d/t vasomotor paralysis in affected limb*
5. RR- Irregular respiration in Cushing's triad
6. Temperature : use AXILLA , dont take oral temp in facial palsy
7. PICCLE
 - a. Pallor : B12 def
 - b. Icterus
 - c. Cyanosis
 - d. Clubbing
 - e. Lymphadenopathy : disseminated TB
 - f. Oedema : maybe present on affected side d/t prolonged immobility (decreased venous and lymphatic flow)

NEUROCUTANEOUS MARKERS

1. TUBEROUS SCLEROSIS



adenoma sebaceum,



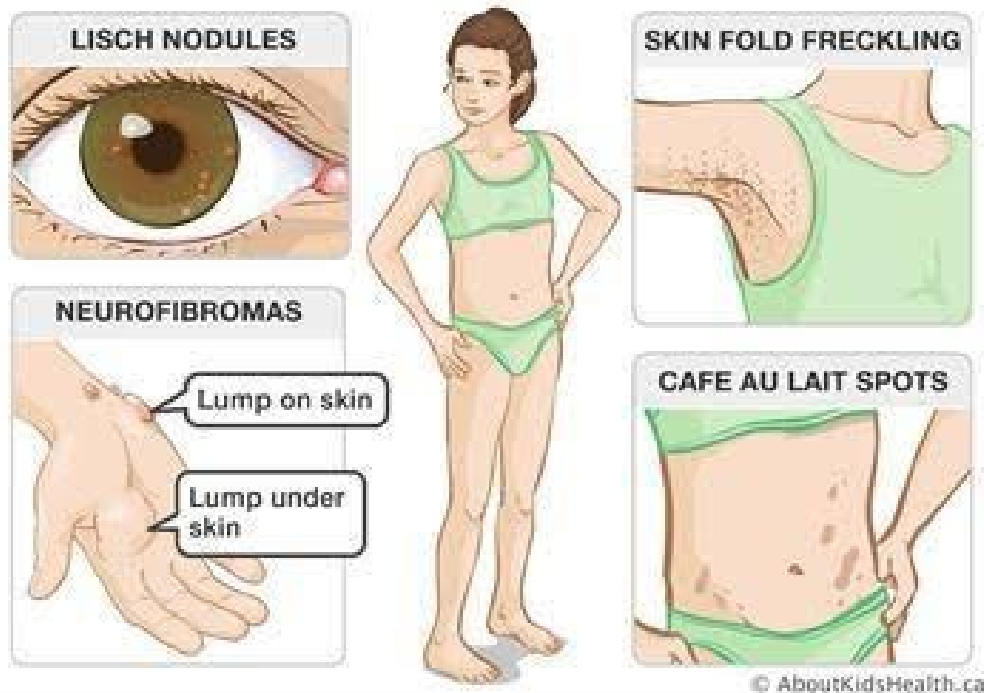
shagreen patches (raised erythematous on back) ,



ash leaf macules (non raised leaflike hypopigmented)

2. NEUROFIBROMATOSIS :

cafe au lait spots,
neurofibromas,
periungual fibromas,
axillary freckling



3. VON-HIPPEL LINDAU SYNDROME : conjunctival hemangioma

4. STURGE WEBER SYNDROME : port wine stain(capillary hemangioma)



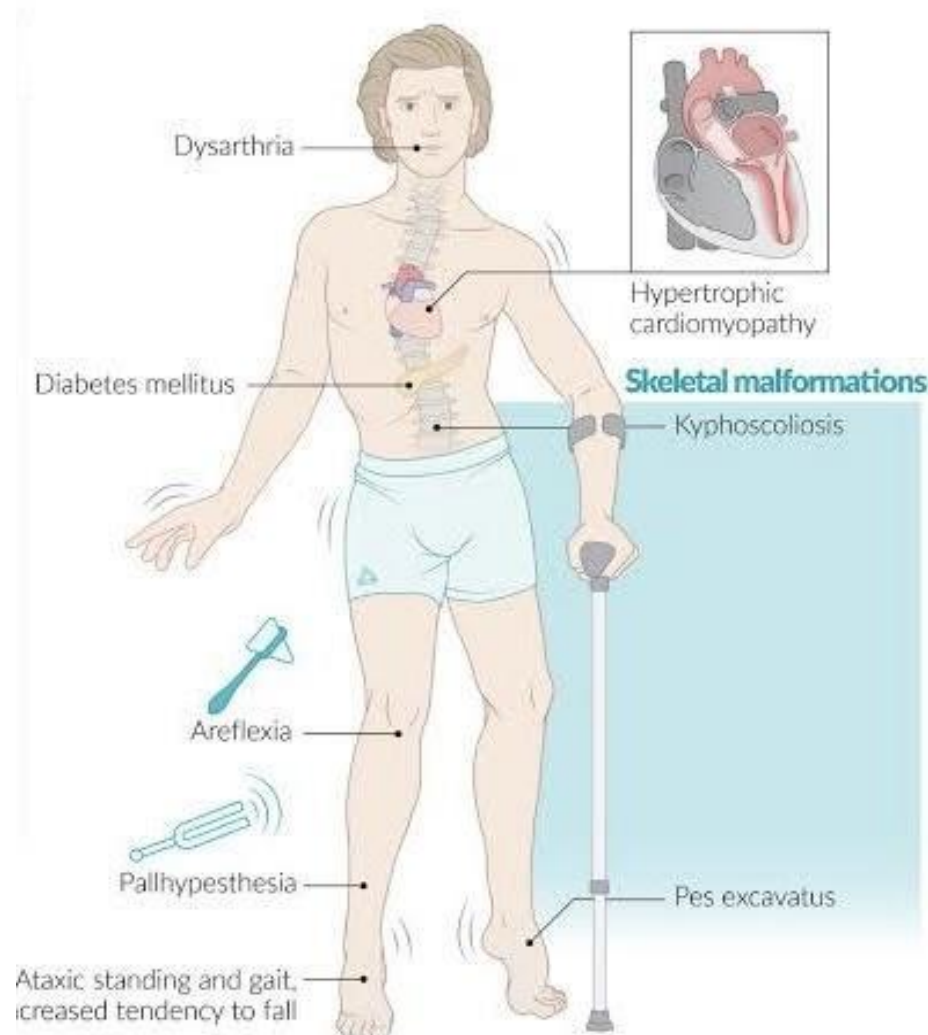
5. ATAXIA TELANGIECTASIA : telangiectasias, especially on the conjunctiva



6. SPINA BIFIDA : tuft of hair



7. FRIEDREICH'S ATAXIA : pes cavus



Focussed general examination for stroke in the young

- Arcus juvenilis
 - o Arcus senilis starts superiorly and no iris is visible between limbus and arcus
 - o Arcus juvenilis starts inferiorly and iris is visible between limbus and arcus (Lucid interval of Vogt)

- **Xanthoma**- larger, tumour like xanthelesma



- Xanthelesmata
- Frank's sign- Earlobe creases
- Signs of rheumatic activity
- Signs of infective endocarditis

Neck Examination

Carotid bruit: With Diaphragm of steth

Thyroid enlargement- may obliterate carotid pulse, may itself cause a bruit which can be mistaken for carotid bruit

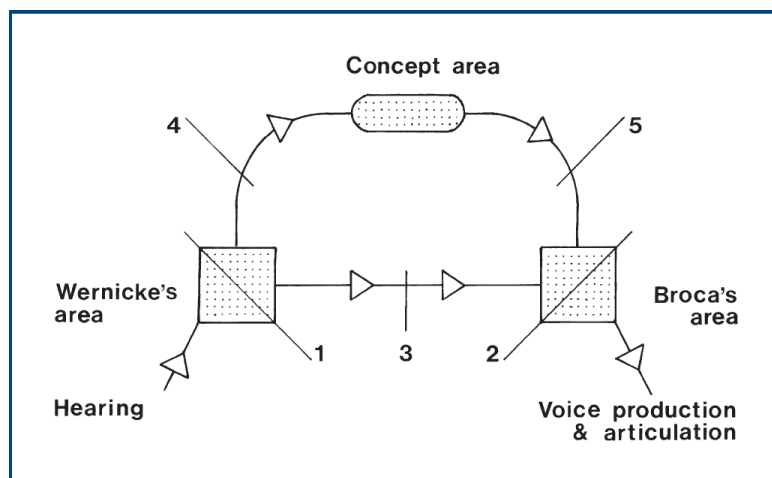
Neurological Examination

Follow this order:

- Higher mental function
- Cranial nerves
- Motor system (includes all of these components. Every one of them must be examined in a motor system short case)
 - o Bulk
 - o Tone
 - o Power
 - o Superficial reflexes
 - o Deep tendon reflexes
 - o Coordination
 - o Gait and Stance
 - o Involuntary movements
- Sensory system
- Signs of meningeal irritation

Higher mental function: Comment- "Conscious, oriented to time, place, person. Memory, speech are normal"

- Consciousness
- Orientation
- Memory
 - o Short term: Sequence of objects; 5/7 digits backwards
 - o Recent: About last meal; Duration of illness
 - o Long term: Birthday/Wedding day.
- Speech and language: By giving oral and written commands.
 1. **Dysphasia**- Abnormalities in speech production and/or understanding. It may also involve other language symptoms like reading/writing unlike the previous two.
 2. **Aphasias**



- Pathway of transmission

- The speech areas are located in the dominant hemisphere
- Wernicke's area (area 22) → sensory / receptive area
- concept area → meaning is understood
- Broca's area (area 44,45) → speech output is generated
- Wernicke's area also is connected to Broca's area directly by the arcuate fasciculus.
- **Wernicke's aphasia** (1)-
 - Poor comprehension.
 - Fluent but often meaningless speech (as it cannot be internally checked). No repetition
- **Broca's aphasia** (2) -
 - Preserved comprehension.
 - Non-fluent speech. No repetition.
- **Conductive aphasia** (3)-
 - Loss of repetition with preserved comprehension and output
- **Transcortical sensory** aphasia (4)-
 - Poor comprehension, fluent but meaningless speech but intact repetition
- **Transcortical motor** aphasia (5)-
 - Preserved comprehension, non-fluent speech but repetition intact.
- 3. **Dysphonia** : loss of volume caused by laryngeal disorders (laryngitis / vagal disorder)
- 4. **Dysarthria** : difficulty in articulating speech due to a motor deficit
 - With abnormal rhythm of speech

	<u>Spastic dysarthria</u>	<u>Extrapyramidal dysarthria</u>	<u>Cerebellar dysarthria</u>
features	1. slurred , slowed and laboured 2. barely opens mouth	1. monotonous , without rhythm , suddenly start & stop	1. slurred as if drunk 2. sometimes scanning speech
causes	1. pseudobulbar palsy 2. MND	1. parkinsonism	1. alcohol tox 2. MS 3. cerebellar lesion

- With normal rhythm
 - LMN
 - Palatal- X nerve
 - Tongue- XII nerve
 - Facial- VII nerve
 - Myasthenic
 - Muscle fatigability

- Intelligence: Simple calculations, Draw house
- **Apraxia** – difficulty with motor planning to perform purposeful tasks, provided the command is understood
- **Agnosia**- Inability to process sensory information
- **Causes of Coma:**
 - Vascular:
 - Infections: Meningitis, Encephalitis, Cerebral Malaria
 - Trauma: Cerebral contusion/concussion; SDH/EDH
 - Metabolic: DM, Uremic Encephalopathy and other Toxic Metabolic encephalopathies, Hypo/HyperCa

MINI MENTAL STATE EXAMINATION (MMSE)

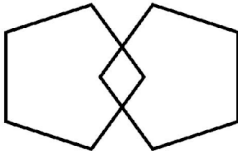
Name:

DOB:

Hospital Number:

One point for each answer

DATE:

ORIENTATION Year Season Month Date Time Country Town District Hospital Ward/Floor	/ 5	/ 5	/ 5
REGISTRATION Examiner names three objects (e.g. apple, table, penny) and asks the patient to repeat (1 point for each correct. THEN the patient learns the 3 names repeating until correct).	/ 3	/ 3	/ 3
ATTENTION AND CALCULATION Subtract 7 from 100, then repeat from result. Continue five times: 100, 93, 86, 79, 65. (Alternative: spell "WORLD" backwards: DLROW).	/ 5	/ 5	/ 5
RECALL Ask for the names of the three objects learned earlier.	/ 3	/ 3	/ 3
LANGUAGE Name two objects (e.g. pen, watch). Repeat "No ifs, ands, or buts". Give a three-stage command. Score 1 for each stage. (e.g. "Place index finger of right hand on your nose and then on your left ear"). Ask the patient to read and obey a written command on a piece of paper. The written instruction is: "Close your eyes". Ask the patient to write a sentence. Score 1 if it is sensible and has a subject and a verb.	/ 2	/ 2	/ 2
COPYING: Ask the patient to copy a pair of intersecting pentagons 	/ 1	/ 1	/ 1
TOTAL:	/ 30	/ 30	/ 30

MMSE scoring

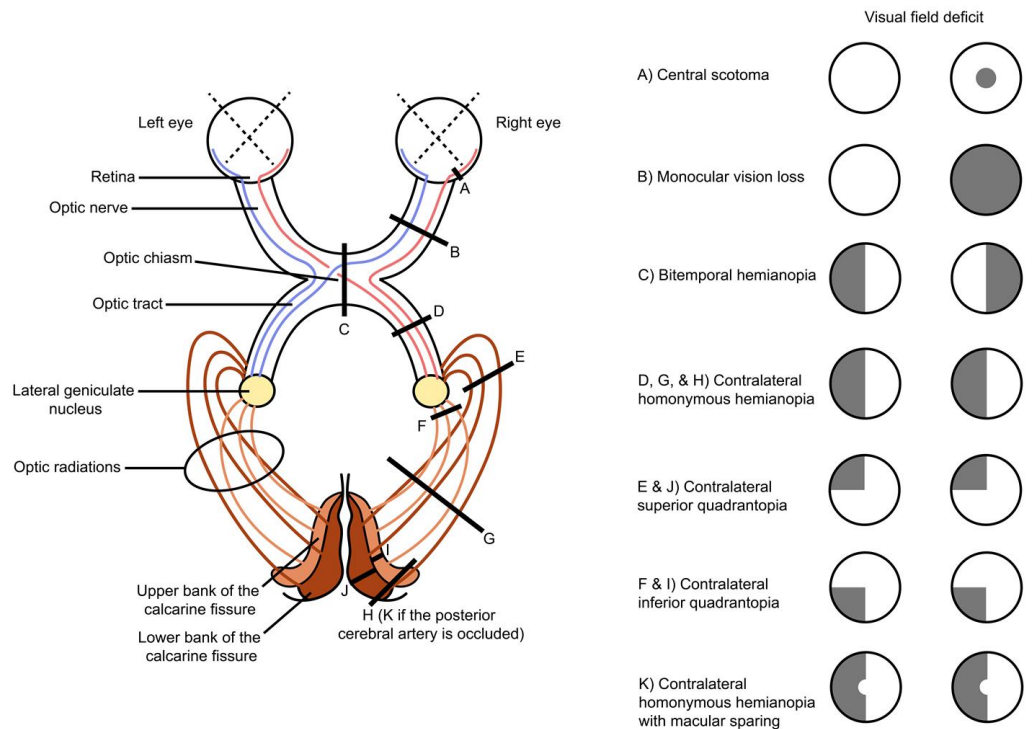
24-30: no cognitive impairment
18-23: mild cognitive impairment
0-17: severe cognitive impairment

Cranial Nerves Examination

- Olfactory-
 - Ask the patient to smell some known things like mint oil, camphor. Not pungent.
 - Check each nostril separately with a time gap of 15 min/ use different material
 - Anosmia in both nostrils
 - Common cause- common cold
 - Hyposmia can be a feature of Parkinson's(early sign of alzheimer's and other dementias)
 - Base of skull #, kallmann syndrome
 - Unilateral anosmia
 - Blocked nostril
 - Unilateral frontal lobe lesion like a meningioma or glioma- very very rare
- Optic-
 - **Visual field by confrontation (using a moving finger : better appreciated)**
 - *look for homonymous hemianopia : can be seen in internal capsule stroke*
 - Colour vision, Visual acuity and FUNDOSCOPY.
 - **Menace** reflex- in patients who do not cooperate with confrontation. The patient is asked to look straight ahead. A menacing stimulus like a hand is rapidly brought towards the patient's face from one side. Reflex blinking occurs.
 - Inference:
 - Brief anatomy:
 - **The right and left nasal hemiretina cross over.**
 - The optic tract sends fibres to the lateral geniculate nucleus in the thalamus.
 - From the thalamus, there are two projections to the visual cortex (called the optic radiation)
 - Meyer's loop through the temporal lobe to the occipital lobe
 - Baum's loop through the parietal lobe to the occipital lobe

■ Lesions and their effects

Visual Field Defects



Where is the lesion	What happens now	Cause
Optic nerve (B)	The fibres capturing images from one eye are affected, → I/L Monocular vision loss	Optic atrophy- Glaucoma, trauma,tumor,optic neuritis
Optic chiasm (C)	fibres of the nasal hemiretina cross→ bitemporal hemianopsia	Pituitary enlargement (hypertrophy/adenoma) Craniopharyngioma Meningioma of Sella turcica
Optic tract (D)	fibres of the contralateral nasal hemiretina and ipsilateral temporal hemiretina → Contralateral Homonymous HemiAnopsia	MCA stroke (ICapsule) ICSOL
Meyer's loop through the temporal lobe	Contralateral superior quadrantanopia →Pie in the sky defect	Ischemia, ICSOL at Temporal/Parietal lobe. (also check for other temporal/parietal lobe features)
Baum's loop through the parietal lobe	Contralateral inferior quadrantanopia → Pie on the floor Defect	
Occipital lobe lesions	Macular sparing bc dual blood supply (from MCA as well)	PCA strokes

- Oculomotor, Trochlear, Abducens-
 - Ptosis
 - Unilateral ptosis- IIN palsy, Horner's
 - Senile ptosis
 - Bilateral ptosis
 - Myasthenia gravis
 - Snake bite
 - Botulism
 - Bilateral Horner's syndrome
 - Nystagmus
 - How to check
 - 0.5m away from pt, pointed source object
 - Move it left, right, up and down. Keep it at that deviation for 5 sec waiting for nystagmus. Not to the lateral extremes (Physiological Directional Nystagmus!)
 - Grading: Alexander's grading of nystagmus
 - 1: +nt when looking in direction of fast comp
 - 2: when looking straight
 - 3: when looking in direction of slow comp
 - Description:
 - Nystagmoid jerks are normal at extremes of gaze
 - pendular / jerky
 - pendular : moving at same speed in both direction
 - jerky : has a fast and slow component
 - vertical : brainstem disease
 - upbeat : demyelination , stroke , wernicke's encephalopathy
 - down beat : medullary cervical Jn lesion : arnold chiari , demyelination
 - horizontal : common
 - Ataxic : nystagmus of ABducting eye > ADducting eye
 - internuclear ophthalmoplegia
 - MS , stroke
 - Multidirectional gaze evoked : nystagmus in direction of gaze , occurring in more than 1 direction
 - always central : cerebellar or vestibular
 - anticonvulsant toxicity ; MS ; alcohol
 - rotatory : usually central , can be periphery
 - Fast component direction,
 - Degree of nystagmus,
 - Probable etio
 - Gaze palsies, refer at the end.
 - Pupil size, reflexes (DL, IDL and Accomodation)**Light reflex in III nerve- Dr. Hamide**
 - Light reflex pathway
 - Optic nerve, optic chiasma, optic tract, Pretectal nucleus, bilateral Edinger Westphal nucleus, oculomotor nerve, ciliary ganglion, short ciliary nerves, sphincter pupillae, bilateral pupillary constriction

- Abnormal pupils

Type	CLINICAL FINDINGS	ANISOCORIA	LESION
Marcus-Gunn pupil	A deafferented pupil which constricts to consensual but not direct light	Absent	CN II
Hutchinson pupil	A dilated pupil that does not respond to direct or consensual light	Present	CN II
Horner's syndrome	A small pupil with associated ipsilateral ptosis and decreased facial sweating	Present	Sympathetics
Adie's tonic pupil	A dilated pupil with an impaired light response and slow constriction to near vision	Present	Parasympathetics
Argyll Robertson pupil	A small, irregular pupil that constricts to near vision but not to light	Absent	Pretectum

- EOM: **H movement**. One arm away.
- Conjugate movements of eyes (test both eyes simultaneously)

CONJUGATE EYE MOVEMENTS* (UMN PARTS OF III, IV AND VI CRANIAL NERVES)

Saccadic Movements

1. Stand in front of the patient (half to one meter apart).
2. Hold two fingers about 30° apart in the horizontal plane.
3. Ask the patient to look with both eyes.
4. Ask the patient to look at each finger alternately as rapidly as possible.

Pursuit Movements

1. Stand in front of the patient.
2. Move a finger slowly across the visual field.
3. Patient looks at the moving finger with both eyes.
4. Examine movements in the horizontal plane.
5. Examine movements in the vertical plane.

- If Diplopia, Check for Head tilt and
 - i. Head tilts in the direction of the action of the paralysed muscle as a compensatory mechanism.
 - ii. Use a definite pointed source to get good images.
 - iii. Separation of the two images is greatest in the direction of the action of the paralysed muscle.
 - iv. Of the two images, one is blurred. The image from the paralysed eye (false image) is indistinct or blurred.

○ Inference:

■ Functions EOM:

- Lateral rectus- Abduction
- Medial rectus- Adduction
- SIN- Superiors intort
- Obliques abduct
- Obliques function inverted when it comes to elevation and depression : Sup Oblique depresses and vice versa.

○ IIIN palsy-

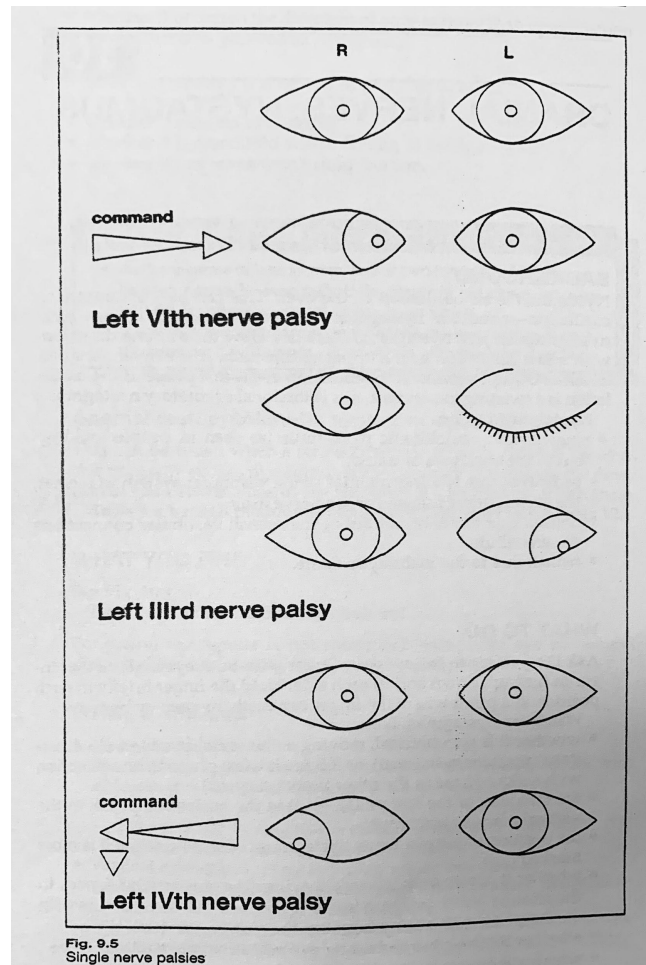
● Findings:

- Ptosis (due to paralysis of LPS)
- Down and Out
- Pupil is fixed and dilated due to paralysis of sphincter pupillae
- Accommodation is lost due to paralysis of the ciliary muscle

● Causes:

Medical/Functional/Pupil sparing IIIN Palsy	Structural/Surgical/Pupil involving IIIN Palsy
DM, HTN, Atherosclerosis, Collagen Vascular disease.	1. Aneurysms- Posterior communicating artery, Posterior cerebral artery 2. Uncal herniation 3. Cavernous sinus thrombosis
<i>vasa vasorum that are vulnerable to ischaemia do not supply the sphincter pupillae</i>	

- IVN palsy- the adducted eye does not move downwards (superior oblique dysfunction)
- VIN palsy- the adducted eye doesn't abduct fully



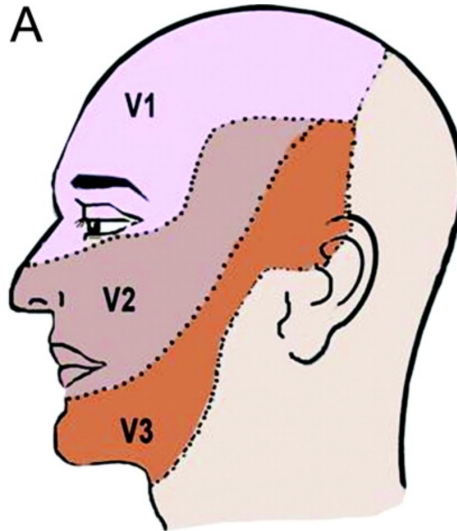
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- UMN vs LMN Gaze palsies:

UMN	LMN
Affects both eyes simultaneously	Unilateral
Normal Pupillary reflexes	Abn
Normal Brainstem doll eye movements (VOR : vestibulo ocular reflex)	Abn

- Trigeminal-
 - Three nuclei of the trigeminal nerve
 - **Principal sensory nucleus-** dorsal column **sensations** to the **face**
 - **Mesencephalic nucleus- proprioception** of the muscles of **mastication** and the periodontal membrane. *Prevents one from biting too hard.*
 - **Spinal nucleus- crude touch, pain, temperature from the ipsilateral face.** It goes as low as C2 before it crosses and ascends in the medial lemniscus

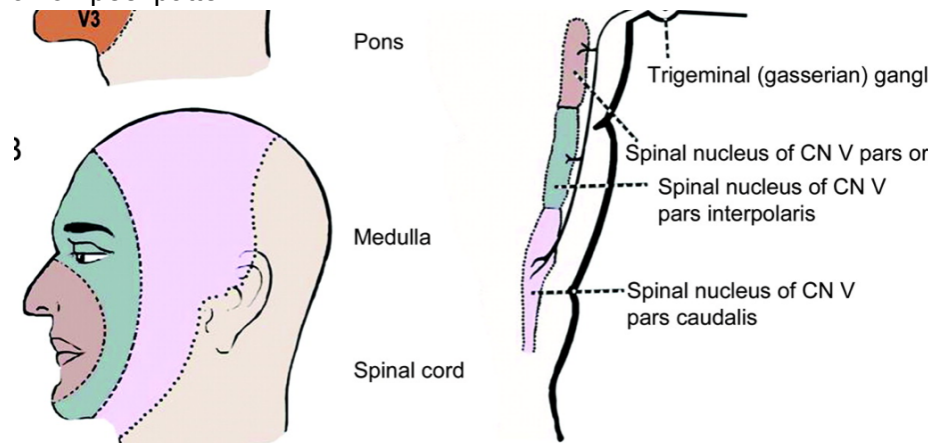
○ Sensory:

- Check touch, pain and temperature. Carried by different fibres.
- Test at: Forehead, upper part of the sides of the nose, upper



lip, chin.

- Avoid: Midline (Dual innervation) but go centrifugally; Skin over jaw angle (C2-3 innervation)
- What is onion peel sensory innervation?
 - The fibres from Spinal Nucleus of V CN supply face in onion peel pattern



○ Motor-

- Look for **symmetry** of temporal fossae and Angle of jaw.
- **ALL muscles of mastication are supplied by V3 (mandibular) and function to elevate the mandible except lateral pterygoid**
 - Ask the patient to **clench their teeth** and **palpate temporalis, palpate masseter.**
 - Ask the patient to open their **jaw against resistance.** (lateral pterygoid)

○ Reflex:

- Jaw jerk : both afferent and efferent is V
- and corneal reflex.

- Cause of V nerve lesions

Nuclear lesions : pons / medulla / upper cervical cord	Preganglionic lesions	Gasserian ganglion lesions	Post ganglionic
1. tumour 2. demyelination 3. vascular lesions 4. syringomyelia	1. tumour (meningioma, CP angle tumour) 2. meningeal irritation - associated usually with VI , VII , VIII	1. tumour 2. abscess 3. herpes zoster 4. Raeder paratrigeminal syndrome	1. cavernous sinus : thrombosis / infxn 2. Gradenigo syndrome @ petrous part of temporal bone a. otitis media b. pain @ distribution of V CN c. VI CN palsy 3. superior orbital fissure syndrome - III , IV , V , VI

- Facial Nerve-

- Look for Bell's phenomenon - seen in LMN

Test	Muscle being tested	Inference
Wrinkle forehead	Occipitofrontalis	Absent on affected side in LMN facial palsy
Close the eyes tightly and attempt to resist opening by examiner	Orbicularis oculi	Examiner can open the eyelids easily in LMN
Smile showing teeth (EEEEEE)	Levator anguli oris	Deviation of angle of mouth
Puff cheeks	Buccinator	Can easily be unpuffed in LMN or UMN
Clench teeth and make a face	Platysma	Normally longitudinal folds of skin are visible in the neck

Feature	UMN	LMN
Angle of mouth deviation	+	+
Nasolabial fold	lost	lost
Bell's phenomenon	-	+
Frontal wrinkling	+	-

- In Right sided CVA, the lesion is RIGHT UMN FACIAL PALSY which causes deviation of angle of mouth to the left. Along with left sided weakness

- Vestibulocochlear nerve-
 - Rinne- 512 Hz tuning fork on the mastoid process. Ask the patient if they can hear it. Once they can no longer hear it, move the tuning fork to the EAM
 - Weber
 - Why 512 Hz?
 - Falls in mid-speech frequency range
 - Lesser overtones
 - Lower frequencies produce more vibratory effect
 - Optimal tone decay time- higher frequencies have faster decay time
- Glossopharyngeal, Vagus- Symmetry of palatal arches and pharynx, Gag reflex
 - Gag reflex- use a tongue depressor to depress the tongue. Using a swab, touch the palate or the pharynx on one side. Note the contraction of the palate, pharynx and posterior 1/3 of the tongue (Sethuraman)
 - Affected Palatal arch is lower than the other > C/L Uvula deviation for X CN palsy
 - Palatal myoclonus- Autosomal dominant cerebellar ataxia (ADCA)
 - Exaggerated gag reflex- pseudobulbar palsy
 - How to distinguish between pure IX and pure X palsies?
 - Alagappan- In pure X nerve palsy, the patient is able to feel the tickling sensation of the throat but there is no contraction in the gag reflex
- Spinal Accessory Nerve-
 - How to:
 - elevate the shoulder against resistance,
 - Turn face to one side against resistance.
 - Inspect and palpate the **opposite SCM**.
 - Trapezius- shrug shoulders against resistance
 - Unilateral SCM paralysis causes weakness in movement to the opposite side.
 - XI nerve palsy- Jugular foramen syndrome (IX, X, XI), Lateral medullary syndrome.
- Hypoglossal- Protrude the tongue, lick teeth
 - UMN palsy of tongue- no true wasting, no fibrillation, protrusion of tongue is difficult, tongue is small and spastic
 - Jaw jerk exaggerated bilaterally- sign of pseudobulbar palsy
 - LMN tongue- wasting, fibrillation, tongue can be protruded and is **deviated to the affected side**, flaccid tongue

S O O S: V, VII, IX and XII CN respectively. S: Same side deviation; O: Opposite side deviation.

Bulbar and pseudobulbar palsy

bulbar (nuclear)

- bilateral affection of the lower cranial nerves IX-XII or their nuclei in the oblongata
- **lower** motor neuron lesion, **peripheral** palsy
- dysarthria, dysphagia, n.VII,V
- masseter reflex absent
- tongue atrophy, fasciculations
- **cause:** polyradiculoneuritis - Guillain-Barré syndrome), brainstem lesions – tumours, meningoencephalitis, motor neuron disease

pseudobulbar (supranuclear)

- bilateral lesion of cortico-bulbar tract
- **upper** motor neuron lesion, **central** palsy
- dysarthria, dysphagia, n.VII,V
- masseter reflex increased
- emotional lability (unprovoked crying or laughing)
- frontal type of the gait
- **cause:** cerebrovascular disease, arteriosclerosis (multi-infarct conditions)

Motor System Examination

Bulk

Comment about both inspection, palpation and Measurement.

- Measure at some clearly defined point wrt a bony process
- Customary points to measure bulk
 - o 18cm above patella
 - o 10cm below tibial tuberosity
 - o 10cm above and below olecranon

Tone

- Tone is defined as the resistance offered to passive movement by a resting muscle
- Check tone at the shoulder, elbow, wrist, knee, hip, ankle joints
- Remember that to check tone in one muscle, you need to do its antagonistic movement and place your hand on the muscle to be checked to feel for its tone
- Say **spasticity** instead of increased tone (then get ready for all the questions on spasticity, rigidity, paratonia etc)
- **Hypertonia-**
 - o There are three extremes. Note- antigravity muscles are flexors of the arms and extensors of the legs

Feature	Spasticity	Rigidity
Seen in	Pyramidal disease	Extrapyramidal disease
Velocity	Dependent	Independent
More in	Anti gravity muscles	Equal in both
Associated w	Clasp knife spasticity (<i>When it reaches considerable tension, there is a protective relaxation and a sudden 'clasp knife' loss of tone, usually towards completion of joint flexion/extension.</i>)	Lead pipe rigidity(Steady resistance); Cogwheel rigidity(Intermittent rigidity) - tremor superimposing on rigidity (parkinson)

- **Paratonia**: Inability to relax muscles during muscle tone assessment. Feature of frontal lobe disease. There are two types- oppositional and facilitatory *GO MF*
 Oppositional (gegenhalten)- when subjects involuntarily resist passive movements
 Facilitatory (mighalten)- subjects involuntarily assist passive movements
- **MISC - fullers**
 - **Myotonia** : slow relaxation following action
 - ask to make a fist and open suddenly → the fist open slowly
 - seen in : myotonic dystrophy
 - **Dystonia** : maintains posture at extreme of movement
 - due to co contraction of agonist and antagonist
 - **percussion myotonia** : muscle dimpling following percussion with knee hammer
 - myotonic dystrophy
- **Causes of hypotonia (Dr. Hamide)**
 1. LMN type of paralysis
 2. Spinal shock
 3. Cerebellar lesions
 4. Chorea
- **Clonus**
 - **Sudden stretching of hypertonic muscles produces reflex contraction.** → if stretch maintained during relaxation → reflex contraction occurs.
 - > 3 downward beats → abnormal
 - **Ankle clonus-**
 - flex the hip and the knee.
 - Provide support under the knee but not at the calf.
 - **Dorsiflex and evert the ankle.**



- Patellar clonus :
 - legs straight
 - hold patella & push it down suddenly and hold it there for few sec
 - rhythmic contraction can be seen
- **Well Sustained** clonus: **≥6** downward beats of the foot.
- **Ill sustained** clonus: **<6** downward beats of the foot.

- *True clonus* : *clonus + extensor plantar response*
 - True clonus ceases with flexion of the great toe.
- Pseudoclonus: Seen in anxiety, thyrotoxicosis

Power

MRC grading of power

Check power at all the joints

Comment on handgrip as **percentage**

power should be graded according to maximum power achieved no matter how brief that is maintained

0-No muscle contraction visible

1-Flicker of contraction, but no movement

2-Joint movement when effect of gravity is eliminated

3-Movement against gravity but not against resistance

4-Movement against resistance but weaker than normal

a. 4- : slight movement against resistance

b. 4 : moderate movement against resistance

c. 4+ : sub maximal movement against resistance

5- Normal power

Movement and joint	Muscle Involved	Nerve Supply	Root Value
Shoulder abduction	Deltoid	Axillary nerve	C5
Elbow flexion	Biceps brachii	Musculocutaneous nerve	C5,C6
Elbow extension	Triceps brachii	Radial nerve	(C6),C7,(C8)
Wrist Flexion	Flexor carpi ulnaris Flexor carpi radialis	Radial nerve	(C6), C7, (C8)
Finger extension	Extensor digitorum	Posterior interosseous nerve, branch of Radial	C7, (C8)
Finger flexion	FDS, FDP	Median, ulnar	C8
Finger abduction	First dorsal interosseous	Ulnar nerve	T1
Finger adduction	Second palmar interosseous	Ulnar nerve	T1
Thumb abduction	Abductor pollicis brevis	Median nerve	T1
Hip flexion	Iliopsoas	Lumbosacral plexus	L1, L2
Hip extension	Gluteus maximus	Inferior gluteal nerve	L5, S1
Knee extension	Quadriceps femoris	Femoral nerve	L3, L4
Knee flexion	Hamstrings	Sciatic nerve	L5, S1
Foot dorsiflexion	Tibialis anterior	Deep peroneal nerve	L4, L5
Foot plantarflexion	Gastrocnemius	Posterior tibial nerve	S1
Big toe extension	Extensor hallucis longus	Deep peroneal nerve	L5
Extension of the toes	Extensor digitorum brevis	Deep peroneal nerve	L5, S1

Group of Muscles	Individual Muscles	Test	Nerve supply
Thenar eminence	ABD pollicis brevis	Pen test	Median Nerve
	Flexor P Brevis		
	Opponens Pollicis	ask the patient to touch the proximal phalanx of 2 nd to 5 th digits with the tip of the thumb	
	ADD Pollicis	Book test/Froment sign	ULNAR
Hypothenar	1. Palmaris brevis 2. ABD Digiti minimi 3. FI D Minimi 4. Opp D Minimi		ULNAR
Lumbricals	4 in number	Flex MCP against resistance	lateral two by median nerve and medial two by ulnar nerve
Interossei	4 palmar(PAD) and 4 dorsal(DAB)	Card test for Palmar.	Deep branch of Ulnar nerve

Reflexes-

If low → Do jendrassik's

If ex aggregated → check for clonus

- **Superficial Reflexes - polysynaptic** : require cortex interpretation before response
 - Corneal (CN5,7)
 - Conjunctival (CN 5,7)
 - Pharyngeal (CN 9,10)
 - Abdominal (T₈-T₁₂)- MacLeod - require cortex interpretation to function
 - lost in LMN at the level : herpes zoster / surgery
 - lost in UMN lesion above the level
 - **Cremasteric (L₁)- DO NOT MISS THIS IN A MALE PT**
 - Stroke from **below upwards**
 - Can also be done by giving a stroke at the level of the lower third of the adductor canal

EXAMINATION FOR SUBTLE HEMIPARESIS [FIG. 6D(IV).58]

1. Pronator drift (Barre's sign)

- The patient stretches out both arms directly in front of him or her with palms upright (i.e. forearms supinated) and closes his or her eyes.
- This position is held for 20–30 seconds.

Normal response:

- Palm will remain flat, elbows straight and the limbs horizontal OR
- Symmetrical deviation from this position (i.e. on both the sides—dominant hand may pronate slightly more than the non-dominant hand)

Positive pronator drift: Components of pronator drift as mentioned above are seen in the weaker side (asymmetric response) which indicates a lesion in contralateral cortex

- Positive with eyes open: Motor deficit
- Positive with eyes closed: Sensory deficit (posterior column)
- Outward and upward drift: Cerebellar drift
- "Updrift" (involved arm rising overhead without patient awareness): Parietal lobe lesions (loss of position sense)
- Drift without pronation: Functional upper limb paresis (conversion disorder)

2. Forearm rolling test [Fig. 6D(iv).59]

- The patient bends each elbow and places both forearms parallel to each other.
- He or she then rotates the forearms about each other, first in one direction and then the other.
- In the abnormal response, the forearm contralateral to the lesion appears fixed while the other arm rotates around it.

3. Rapid finger tapping test

- The patient rapidly taps the thumb and index finger repeatedly at a speed of about two taps per second.
- Hemispheric lesions cause the contralateral finger and thumb to tap more slowly and with diminished amplitude.

4. Foot tapping test

- The seated patient taps one forefoot at a time for 10 seconds on the floor, as fast as possible, while the heel maintains contact with the floor.
- A discrepancy of more than five taps between the left and right foot indicates cerebral disease contralateral to the slower foot.

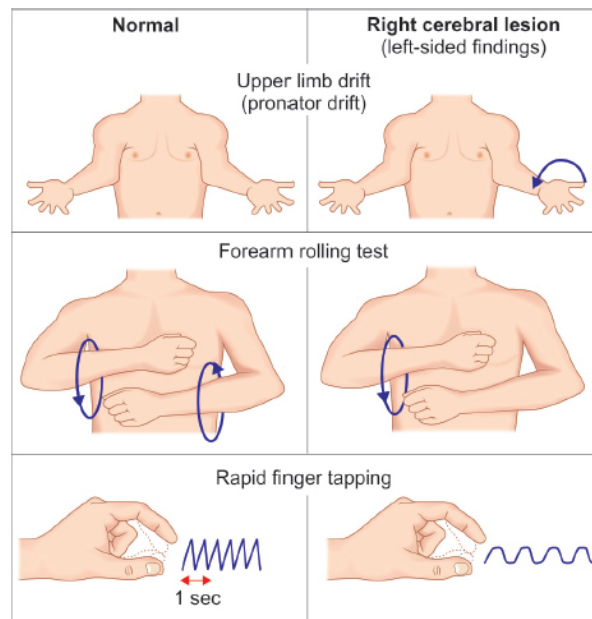


Fig. 6D(iv).58: Examination for subtle hemiparesis.



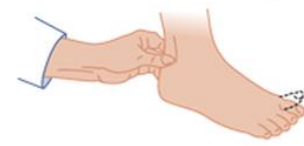
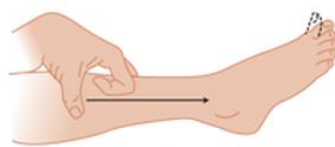
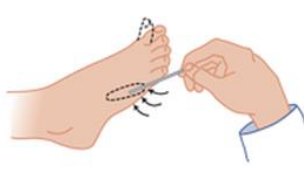
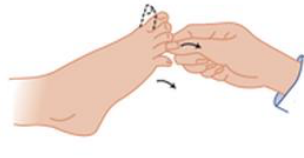
Fig. 6D(iv).59: Forearm rolling test.

- Plantar Response L₅-S₁
 - afferent : tibial nerve
 - efferent :
 - flexor response : tibial nerve
 - extensor response : peroneal nerve
 - in hemi/paraplegia (UMN) - plantar response remains extensor life long

<u>Normal plantar response</u>	<u>Extensor plantar response -</u> complete response is called brissaud's reflex	<u>How to report</u>	<u>Conditions where babinski is seen</u>
1. flexion of all toes 2. dorsiflexion @ ankle 3. flexion @ knee & hip 4. contraction of TFL	• extension @ great toe • fanning of other toes • dorsiflexion @ ankle • flexion @ knee & hip • contraction of adductor magnus - Dr. hamide • contraction of TFL - causing IR of HIP	• flexor plantar response • extensor plantar response • no response : no movement of any toes • equivocal response : when full component of extensor response not seen (kundu) ○ only fanning without great toe extension OR ○ asymmetry of flexor response on both sides OR ○ hemiplgic side no response with normal side's flexor response • Minimal plantar: Contraction of Add Magnus is checked in case of equivocal, if +nt: Described as minimal plantar • withdrawal response : initial normal flexor quickly f/b mass extension of toes with withdrawal of entire leg (kundu) • Mute Plantar: Absence of Babinski sign in an otherwise Spastic (Dr. DineshBabu) • Pseudo Extensor reflex: Due to anxiety. Greater toe reacts right at the beginning of the stroke.	1. stroke /paraplegia 2. infancy 3. deep sleep 4. deep anesthesia 5. hep encephalopathy 6. narcotic overdose 7. Alcohol intoxication 8. Coma 9. post ictal state

Other methods of eliciting Plantar response - plantar equivalence

- Oppenheim- Firm stroke with the finger and thumb down the anterior border of the tibia
- Chaddock- Light stroke from just below the lateral malleolus down to the small toe.
- Gordon- Squeezing of the calf muscle
- Schaefer- Deep pressure on achilles tendon
- Strümpell's- Forceful pressure over the anterior tibial region.
- Brissauds : contraction of TFL
 - helpful in :
 - amputated or absent great toe
 - complete paralysis of extensor toes

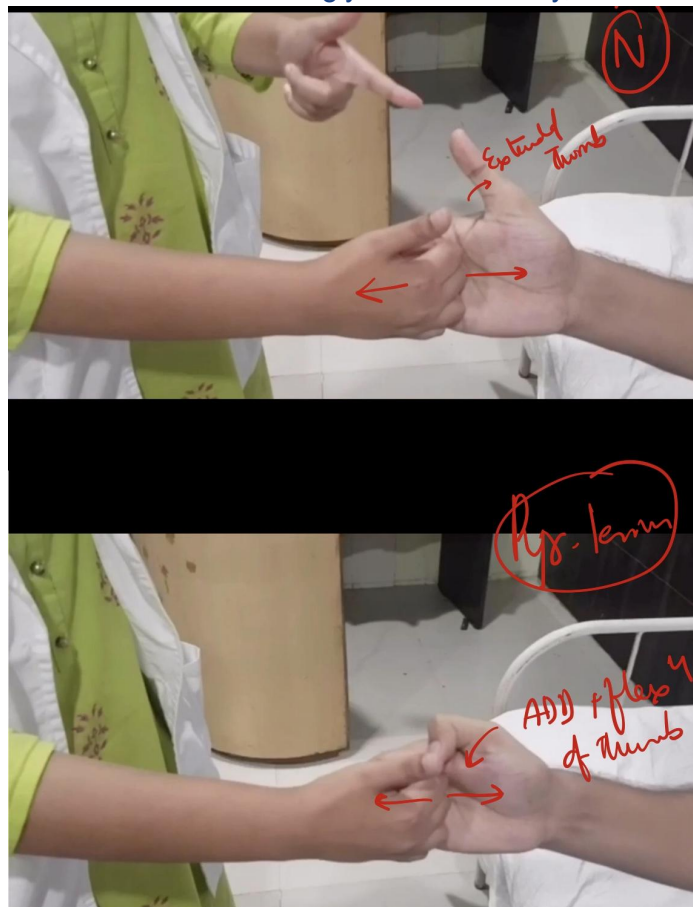
Descriptive Name	Eponym	Maneuver	
A. Plantar toe reflex	Babinski	Move an object along the lateral aspect of the sole.	
B. None	Chaddock	Move an object along the lateral side of the foot.	
C. Achilles-toe reflex	Schaeffer	Squeeze hard on the Achilles tendon.	
D. Shin-toe reflex	Oppenheim	Press your knuckles on the patient's shin and move them down.	
E. Calf-toe reflex	Gordon	Squeeze the calf muscles momentarily.	
F. Pinprick-toe reflex	Bing	Make multiple light pinpricks on the dorsolateral surface of the foot.	
G. Toe-pull reflex	Gonda, Stransky	Pull the fourth toe outward and downward for a brief time and release suddenly.	

Physiology behind it

- Seen in infants up to 1 year. After that, the corticospinal tract becomes completely myelinated and there is descending control over the reflex, so it switches to flexor response to assist in ambulation.
- Why do we start from the lateral side? Because, in initial pyramidal tract lesions, the **receptive focus** is confined to the lateral aspect of the sole. As the lesion progresses, it can be elicited from other parts also
- Why don't we touch the ball of the great toe? If it is stimulated, the **grasp reflex** of the legs may be stimulated causing flexion of the great toe even in the presence of a pyramidal tract lesion

Upper limb equivalents of Babinski***

- **Hoffman reflex**- terminal phalanx of the patient's **middle finger** is flicked downwards between the examiner's finger and thumb. In hypertonia, the tips of the other fingers flex and the thumb flexes and adducts
- **Wartenberg sign**- Patient supinates his hand, slightly flexes his fingers with the thumb in adduction. The examiner pronates his hand and links his flexed fingers with that of the patient's fingers. Both flex their fingers against each other's resistance. In hypertonia, the thumb adducts and flexes strongly. While normally, the thumb extends



Causes of hyperreflexia : UMN , thyrotox , anxiety , tetanus
Slow relaxation of ankle : hypothyroidism

Primitive reflexes :

What is the reflex	How to do it	Where do you see it
palmomental	stroke on thenar eminence with key → puckering of skin over chin	C/L Frontal Lobe disease.
Palmar Grasp reflex		
Avoidance reflex		C/L Parietal lobe disease
Glabellar tap		B/L UMN Facial palsy (pseudobulbar palsy)
Snout reflex	lip tap → puckering	
Sucking reflex		

Deep Tendon Reflexes- Root values from MacLeod

These are monosynaptic i.e Ia (muscle spindle) → Alpha motor neuron

- Biceps jerk- C₅
- Supinator jerk- C₆ (*Bc Brachioradialis previously known as Supinator longus*)
- Triceps jerk- C₇
- Knee jerk- L₃, L₄
- Ankle jerk- S₁

0: absent reflex

1: like a normal ankle reflex

2: Like a normal knee

3: brisk

4: Clonus

→ What is the Inverted Biceps reflex?

Absence of Biceps Contraction/Presence of Triceps contraction/Finger flexion

→ What is the paradoxical Triceps reflex?

Elbow flexion

→ What is inverted supinator Jerk?

No biceps contraction + No Brachioradialis contraction + Presence of Finger flexion.

→ What is the pendular knee Jerk?

>3 to and fro movements of leg d/t hypotonia in cerebellar lesions

Coordination

- Finger-nose test
- Finger-finger nose test: Patient should touch the examiner's finger before touching their nose. Can detect milder degrees of incoordination
- Heel to shin test detects lower limb incoordination
- Dysdiadochokinesis
- Other signs of cerebellar disease- nystagmus, ataxia, dysarthria, pendular reflexes, hypotonia

- Basic anatomy about Cerebellum:
 - In general, the cerebellum controls the ipsilateral half of the body because the cerebellar input reaches the contralateral cortex which goes to the contralateral half of the body
 - Cerebellar nuclei
 - Dentate nucleus- contralateral VA/VL nuclei of thalamus
 - Emboliform and globose nuclei- contralateral red nucleus
 - Fastigial nucleus- vestibular nuclei and reticular formation
 - Lateral lesions of the cerebellum
 - Affect the extremities- distal parts
 - Can't do finger-nose test
 - Intention tremor
 - Fall towards injured side
 - Involve the hemispheres and dentate nucleus
 - Midline lesions of cerebellum
 - Affect the trunk
 - Truncal ataxia
 - Can't stand independently
 - Fall over when sitting
 - Involve the vermis, emboliform, globose, fastigial nuclei
 - Flocculonodular lobe also- nystagmus, vertigo
 - Cerebellar ataxia
 - All cerebellar lesions
 - Loss of balance
 - Wide based gait- will fall if the feet are together
 - Hypotonia
 - Scanning speech
 - Dyssynergia- dysmetria (finger-nose), intention tremor, dysdiadochokinesia
 - Nystagmus- vertical
 - Nausea, vomiting, vertigo
 -

Stance and Gait

- Stance- ask the patient to stand with his feet close together and eyes open.
 - Swaying, lurching or inability to stand with the feet together when the eyes are open suggests cerebellar ataxia.
 - Ask the patient to close his eyes but be ready to catch him . Repeatedly falling is positive Romberg sign.
 - i. Inference :
 1. When vision is taken away, the other two i.e vestibular and proprioception have to maintain balance
 2. If there is a lesion in either of the 2 pathways then Romberg's is positive
- Gait-
 - Time the patient walking a measured 10 metres, turning through 180° and returning.
 - Note stride length, arm swing, steadiness, limping.
 - Ask the patient to walk on the tip toes then on the heels.
 - Ask the patient to **walk heel to toe (tandem gait) in a straight line to test for gait ataxia.**
 - i. sensitive for early ataxia, test of vermis of cerebellum

- o Abnormal gaits
 - i. **Hemiplegic gait/Pyramidal gait/Circumduction gait-** Components
 1. Adduction at shoulder
 2. Flexion at elbow, flexion at wrist
 3. Hand pronated
 4. LL **extended at hip and knee**
 5. Ankle and toes are **plantar flexed**
 6. **Toes** strike the ground while walking. **To prevent this, the gait is of circumduction type.**
 - ii. Festinant gait of Parkinsonism- short, shuffling steps
 - iii. High stepping gait : Foot drop
 - iv. Waddling gait : Myopathy
 - v. Ataxic gait : Cerebellar disease
 - vi. March a' petits pas : shuffling , progressively accelerating gait : Parkinsonism

Involuntary movements (MacLeod)

1. A – Athetosis (slower, writhing movement), Asterixis.
 2. B – Ballismus(violent flinging movements)
 3. C – Chorea (brief, jerky, random, purposeless movements)
 4. D – Dystonia (sustained muscle contractions, leading to twisting, repetitive movements)
 5. E – Essential tremors (involuntary, oscillatory movement about a joint or a group of joints, resulting from alternating contraction and relaxation of muscles)
- Types
- Physiological- fine,fast,postural tremor seen with anxiety
 - Essential- MC Pathological action tremor, AD inheritance
 - Parkinson's - slow,coarse tremor worse at rest but improved with voluntary movement
 - Isolated head tremor- dystonic with abnormal neck posture
 - Intentional tremor- absent at rest, max at movement → cerebellar lesion
6. F – Fasciculations(irregular twitching under the skin overlying the resting muscles)
 7. M - Myoclonus (Rapid, irregular, jerky movements that persist even during sleep)
 8. T-Tics (Rapid, repetitive, coordinated, stereotyped movements that can be mimicked)

Causes of chorea-

Sydenham's chorea (St. Vitus' Dance),
 Huntington's chorea,
 SLE, Wilson's,
 Chorea **gravidarum** (post rheumatic recurrence of chorea in pregnancy),
 Hyperthyroidism

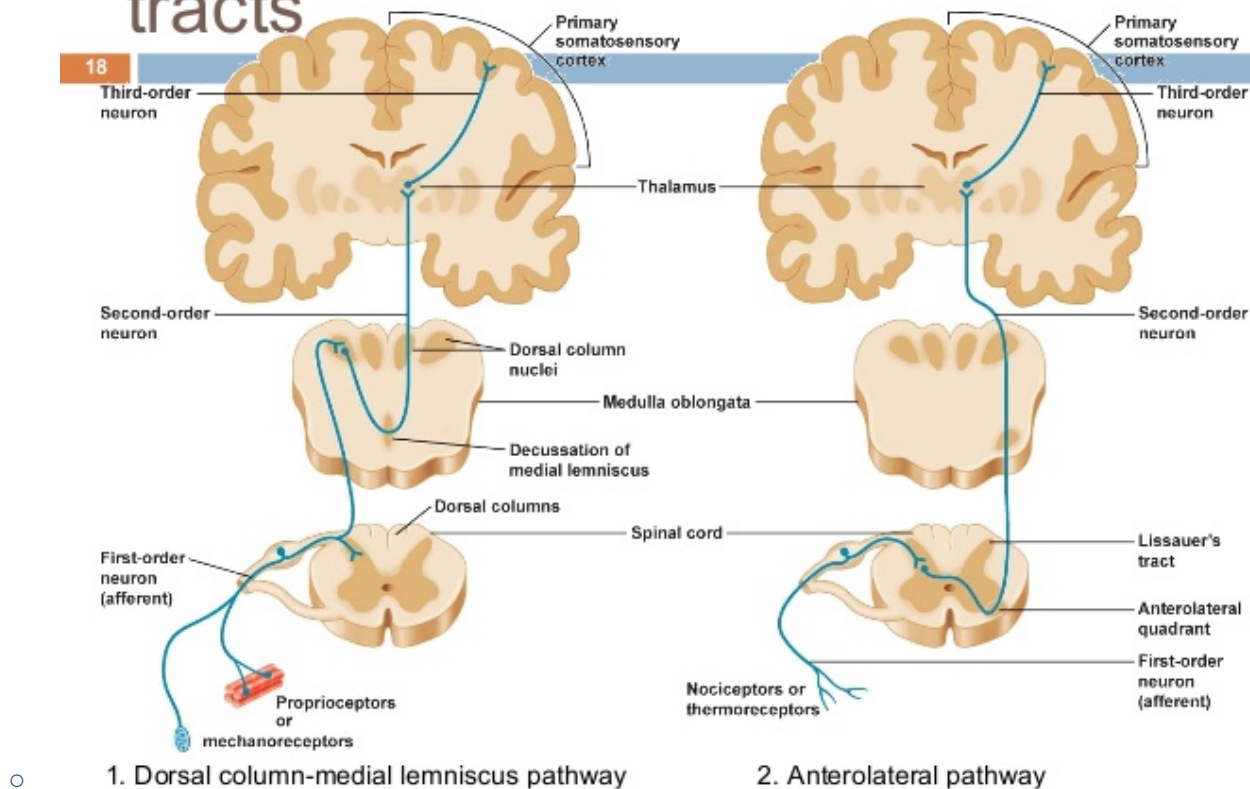
Sensory Examination

- Fine Touch- Ask the patient to look away /close his eyes. Use cotton wool and dab.
- Crude Touch-
- Pain- Use a neurological pin
- Vibration- 128 Hz tuning fork. Put it over bony prominences (big toe and ankle- MacLeod) Touch the fork to stop vibration to see for malingering. If sensation is impaired, keep going proximally to the IP joint, medial malleolus, tibial tuberosity. Do the same for the upper limb.
 - o Why 128 Hz for vibration? Lower frequencies tend to produce more vibratory effect and tend to decay slower?

- Proprioception = Joint sense + Position sense
 - (a) Position sense- Ask the patient to close his eyes. Then, set the given joint in a particular position in space. Ask the patient to place the same joint of the other limb in a similar position. If the other limb is paralysed, ask him to repeat the same action with the joint or ask him to describe the present spatial orientation of the joint.
 - (b) Joint sense- Grasp the proximal interphalangeal joint of the great toe with the thumb and index finger and then use the other hand to move the toe in different directions (Ref Macleod P.269, Fig 11.31)
- Relevant Basics:
 - Proprioceptive sensations- Position, vibration, deep pain. Derived from impulses from the body itself
 - Exteroceptive sensations- those which are derived from sources outside the body. Include pain, temperature and fine touch

Types of Pain	Fast pain	Slow Pain
<u>Example</u>	Pin prick	Burn
<u>Carried by</u>	A δ , Group II and Group III fibres	C fibres
<u>Localisation</u>	Precise	Poor; burning/throbbing type

Somatosensory projections: tracts



	Dorsal Column pathway	Anterolateral/Spinothalamic
Carries	Fine touch Vibration Proprioception Cortical Sensations	Pain Temperature Crude touch
Order of neurons (Follow along with the diagram above)	1. I/L ascent to nucleus gracilis (LL) and nucleus cuneatus (UL) of the medulla. 2. Crossover at medulla and ascent to Thalamus 3. Thalamus to Somato sensory cortex 4. Integration and complex formation.	1. Synapse onto (2) at SpC 2. Cross midline and reach C/L Thalamus 3. Thalamus to cortex 4. Integration and complex formation.

What are the sensory areas of Brain?

1. Primary somatosensory area (S1)- present in the postcentral gyrus. Includes Brodmann's areas 3,1,2.
2. Primary somatosensory area (S2)- located in the wall of the sylvian fissure
3. Somatosensory association cortex- area 5
4. Precentral gyrus

Cortical sensation examination- do it only if sensory examination is normal

1. **Tactile localisation** : Localises accurately where the patient was touched
2. **Two point discrimination** : 2-3 cm on the palm
3. **Stereognosis** : recognition of an object purely from the feel of its shape and size
4. **Graphesthesia** : Ability to recognise numbers/letters/diagrams written on the skin with a blunt point

Localisation

Hemiplegia ± VII N involvement

- Same side- Cortex/corona/capsule
- Opposite side (hemiplegia)- brain stem, spinal cord (C₁-C₄)

Cortex

- Not dense
- Hemiparesis/Monoparesis
- Seizure/altered sensorium
- Lobar dysfunction-
 - o Dominant temporal- aphasia
 - o Non dominant parietal- hemineglect
- Cortical sensations lost- Two point discrimination, tactile localisation, graphaesthesia, stereognosis

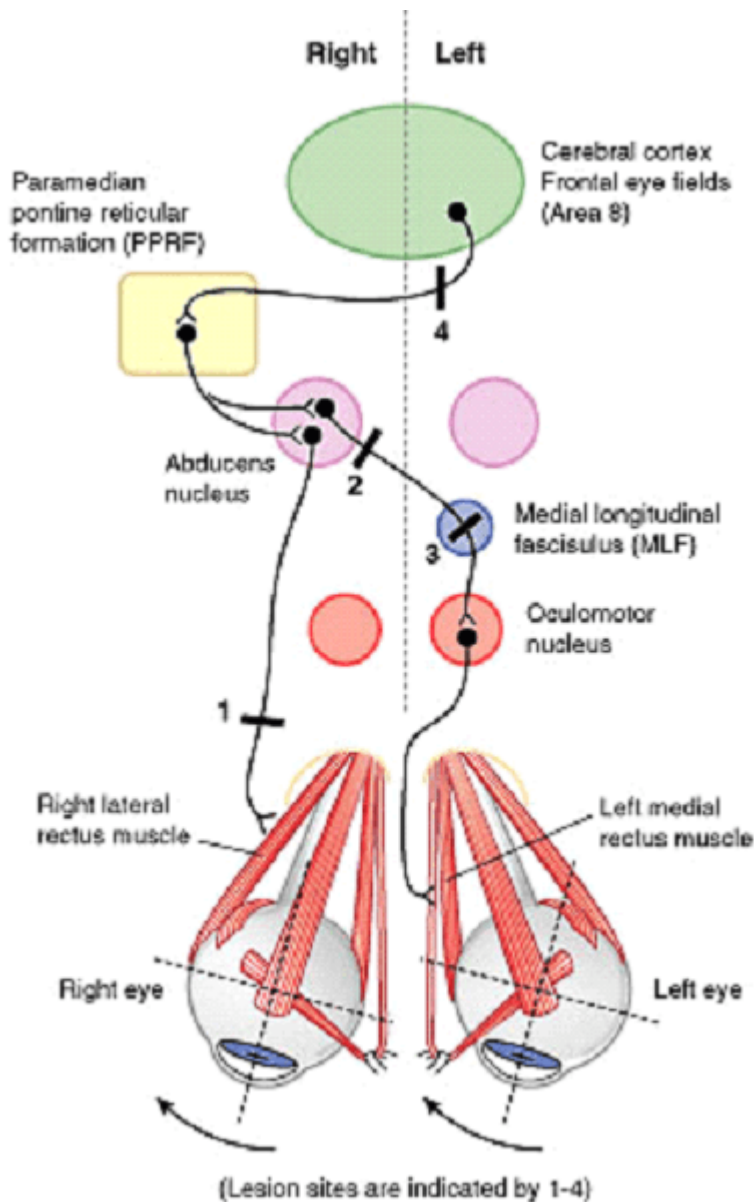
Subcortical

- Same as cortical but the cortical sensations are preserved

Internal capsule

- Dense hemiplegia (Power **equally reduced** to a marked extent UL and LL. Ex: UL and LL power is 0/5)
- Hemianesthesia
- Homonymous hemianopia
- Dysarthria
- Global aphasia (L side)
- Apractognosia

- Conjugate gaze and gaze palsies



- In this diagram, the person wants to look to the right. So the right eye is being ABducted (Lateral rectus, VI Nerve) and the left eye is being ADducted (Medial rectus, III nerve)
- To look to the right, the signal arises from the left cortex from the frontal eye fields (area 8)
- From the FEF, the fibres cross the midline and reach the contralateral PPRF.
- From the PPRF, the fibres go to the abducens nucleus and reach the right lateral rectus muscle.
- To reach the left medial rectus muscle, they go to the oculomotor nucleus by crossing the midline and reaching the MLF.

- Lesions, their causes and their effects
 - Left FEF- Both eyes can't look right. So they slowly drift to the left
 - Cause- frontal lobe seizures/frontal lobe lesions
 - Right PPRF or Right VI nucleus (both have the same effect because they are so close). Both eyes can't look right (lateral gaze palsy)
 - Cause- medial pons lesions
 - Left MLF- left eye can't look right (affected eye cannot adduct) while right eye can look right (unaffected eye can abduct)- this is called Internuclear ophthalmoplegia. Convergence is intact (mediated by another pathway) and the unaffected eye develops nystagmus
 - Commonly occurs in multiple sclerosis because the MLF is highly myelinated
 - One and a half syndrome
 - Damage to the PPRF+MLF
 - INO plus loss of lateral gaze to the affected side.

Theory part of Stroke

***Definitions

1. **Stroke-** A sudden and dramatic onset of focal neurological deficit from a vascular aetiology. (Dr. Vivekanandan). The main deficit occurs over several hours, but not beyond 12 hours and persists for more than 24 hours.
2. **TIA-** Sudden onset of focal neurological deficit from a vascular aetiology where the patient recovers completely within 24 hours.
Most TIA last for 5-20 minutes. Episodes that last for >1hr are generally due to small infarcts. Once there is an infarct, there is no more TIA concept.
80% affect anterior circulation, 20% affect posterior circulation
3. **RIND** (Reversible ischaemic neurological deficit)- Neurological deficit lasting more than 24 hours, but less than 7 days. (<3 weeks- Alagappan)

Classification

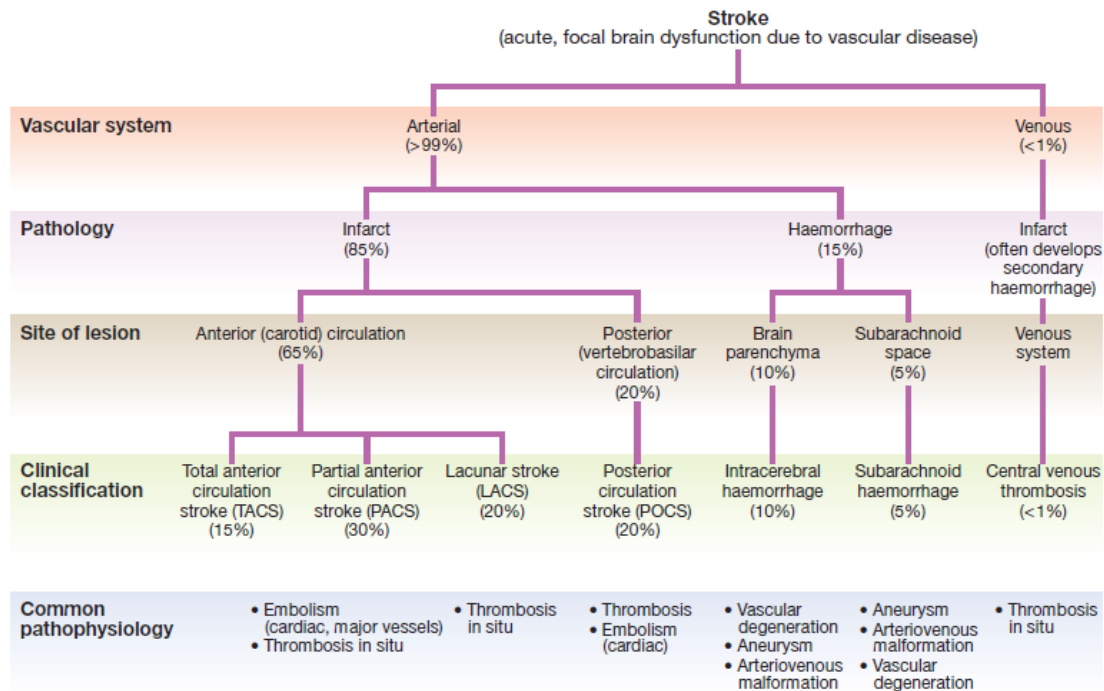


Fig. 27.1 A classification of stroke disease.

Management of Stroke

Investigations

- if anterior circulation stroke is suspected then **CT IS DONE**
- if posterior circulation stroke is suspected then **MRI IS DONE** *cause in posterior region the base of skull comes in the way of visualization so we do an **MRI***

NCCT changes s/o infarct

- Loss of grey-white differentiation
- Loss of gyri-sulci differentiation
- Midline shift
- Ventricular compression
- Insular ribbon sign???



27.5 Investigation of a patient with an acute stroke

Diagnostic question	Investigation
Is it a vascular lesion?	CT/MRI
Is it ischaemic or haemorrhagic?	CT/MRI
Is it a subarachnoid haemorrhage?	CT/lumbar puncture
Is there any cardiac source of embolism?	Electrocardiogram (ECG) 24-hour ECG Echocardiogram
What is the underlying vascular disease?	Duplex ultrasound of carotids Magnetic resonance angiography (MRA) CT angiography (CTA) Contrast angiography
What are the risk factors?	Full blood count Cholesterol Blood glucose
Is there an unusual cause?	Erythrocyte sedimentation rate (ESR) Serum protein electrophoresis Clotting/thrombophilia screen

TIA – risk stratification

- ABCD2 algorithm predicts very early risk of stroke following a TIA
 - **A – Age**
 - >60 = 1
 - **B – Blood pressure**
 - >140/90 mmHg = 1
 - **C – Clinical features of the TIA**
 - Unilateral weakness = 2
 - Speech disturbance without weakness = 1
 - **D1 – Duration of symptoms**
 - > 60 min = 2
 - 10-59 min = 1
 - <10 min = 0
 - **D2 – Diabetes**
 - Diagnosed with diabetes = 1
- The corresponding 2 day risks for a subsequent stroke are
 - **ABCD2 scores**
 - 0-3 = 1% (low risk, Start Aspirin 300mg OD and refer to TIA clinic)
 - 4-5 = 4% (High risk, consider admission. Aspirin 300mg OD, urgent TIA clinic)
 - 6-7 = 8% (As above)

Mx

1. **Airway**- Aspiration is common
2. **Breathing**- Medullary injury dysregulated breathing
3. **Circulation**
 - a. Bradycardia suggests raised ICT which may be due to haemorrhagic stroke. CT scan rules out haemorrhage.
 - b. Recommended antihypertensives: **Labetalol, Nitroglycerine, Hydralazine, Nicardipine, Clevidipine**
 - c. BP targets: (In ischaemic stroke, cerebral perfusion pressure has to be maintained, so BP targets are a bit higher. (umbra penumbra concept- read) But in haemorrhagic stroke, more bleeding can happen so BP has to be brought down much lower.)
 - i. Haemorrhagic stroke- BP should be aggressively brought down. Target **<140-160/90 mmHg**.
 - ii. Ischaemic stroke
 1. If considering rtPA, BP should be **<185/110 mmHg**
 2. If not considering rtPA or it isn't available, BP should be **<220/120 mmHg**(Sometimes the examiner will not accept 220/120. If they don't accept it, then say 180/120. There are two guidelines, which say the target BP should differ depending on the availability of ICU facilities). Don't give antihypertensives in this case unless there is hypertensive emergency.
4. **Thrombolysis- TENECTEPLASE in JIPMER**
 - a. **Indications** (ABCDE)
 - i. Clinical diagnosis of stroke
 - ii. **<4.5 hours** before first onset of symptoms
 - iii. Age >18 years
 - iv. CT scan excludes haemorrhage (Oedema >1/3 of MCA territory- ischaemia cannot be visualised this early)
If the stroke is thrombotic and resolvable, physiological mechanisms would have already taken place
 - b. **Contraindications**
 - i. BP >185/110 mmHg
 - ii. Bleeding propensity- Thrombocytopenia, raised PT-INR, Heparin
 - iii. Previously thrombolysed in the past 24 hours
 - iv. Rapid improvement
 - v. Recent MI- PT INR has to be maintained at a higher level.
 - vi. Coma

Note: Aspirin use is not a C/I for thrombolysis.
 - c. **Method**
 - i. 2 peripheral iv lines
 - ii. 0.9 mg/kg rtPA, maximum 90 mg.
 - iii. 10% bolus dose.
 - iv. Remaining 90% over 1 hour.
 - v. Treat arrhythmias, if any
 - vi. Give inotropes if BP is low.
 - vii. NG tube if required.
 - viii. If there is a high risk of aspiration then NPO.
 - ix. If a k/c/o DM and blood glucose >200mg/dL, then give insulin

5. Anticoagulation

No need to give Heparin. Start on Warfarin directly because the thrombus has already been cleared.

6. Complications of Stroke

- a. Seizures- due to oedema/cortical involvement/hypoglycaemia
- b. DVT- develops 2 to 5 days after hemorrhagic stroke
- c. Pneumonia
 - i. Aspiration- prevent by suction, semi recumbent position, NGT, NPO
 - ii. Ventilator associated pneumonia
 - iii. Hospital acquired pneumonia
- d. Pressure sores- prevented by air mattress
- e. Catheter associated UTI

Stroke in the young- Causes and Work up- (<40 years)

1. Cerebral infarct

- Cardiac embolism (especially in RHD/MS/AF) **tell this first**- Echocardiography (TEE)
- Premature atherosclerosis- serum lipids
- Arterial dissection- MRI, angiography
- Thrombophilia- Protein C, Protein S, Antithrombin III, Factor V Leiden, Prothrombin 20210
- Homocystinuria- Urinary amino acids
- Neurovascular syphilis- VDRL
- Sickle cell anaemia- HMG with PS, HbS
- APLAb syndrome- Anticardiolipin antibodies
- SLE- ANA
- Vasculitis- ESR, CRP, ANCA
- CADASIL- (Cerebral Autosomal Dominant Arteriopathy with Subcortical Infarcts and Leukoencephalopathy)

1. Primary intracerebral haemorrhage

- Warfarin toxicity- esp in RHD patients: PT/INR, aPTT
- Arteriovenous malformation- MRI/Angio
- Drug use- Amphetamines, Cocaine
- Sites of haemorrhage
 - o Putamen
 - o Thalamus
 - o Cerebellum
 - o Pons

UPDATE — WHAT CHANGED IN THE STROKE GUIDELINES

The 2026 AHA/ASA Guideline for the Early Management of Patients with Acute Ischemic Stroke (published Jan 2026) replaces the 2018 guideline and its 2019 update entirely. The points below are the ones that contradict or extend what is written in the Mx section above — learn these as the current answer.

1. THROMBOLYTIC AGENT — TENECTEPLASE IS NOW CO-FIRST-LINE

- **Class I recommendation:** within 4.5 h of onset/last known well, give **either tenecteplase 0.25 mg/kg (max 25 mg) or alteplase 0.9 mg/kg (max 90 mg)**. TNK is no longer 'the JIPMER alternative' — it is guideline-endorsed.
- **Dosing difference is the exam point:**
 - TNK — **single IV bolus over 5 s**. No infusion.
 - Alteplase — 10% as bolus, remaining 90% over 1 h.
- Practical advantage of TNK: greater fibrin specificity, longer half-life, single bolus → faster door-to-needle, easier during inter-hospital transfer.
- **Do not delay IVT** for CTA/MRA or perfusion imaging in the <4.5 h window (Class I). Give it for any *disabling* deficit regardless of NIHSS.

2. EXTENDED THROMBOLYSIS WINDOWS — NEW

Scenario	Window	Selection	Class
Wake-up stroke	within 9 h of the midpoint of sleep	Salvageable penumbra on automated perfusion imaging (or DWI-FLAIR mismatch on MRI)	2a
Known onset	4.5–9 h from last known well	Salvageable penumbra on automated perfusion imaging	2a
LVO who cannot get EVT	4.5–24 h from last known well	LVO + salvageable penumbra; IVT may be beneficial	2b

Trials to quote: EXTEND and WAKE-UP (extended window / DWI-FLAIR), TRACE-3 and HOPE (late TNK in LVO). TIMELESS was negative — know that it is the exception.

3. MECHANICAL THROMBECTOMY (EVT) — MISSING ABOVE, ASK-EVERY-TIME TOPIC

- EVT is the single biggest omission in the Mx notes. Learn the four boxes:

Setting	Criteria	Class
Anterior circulation, early	ICA or M1 occlusion, <6 h , NIHSS ≥6, pre-stroke mRS 0–1, ASPECTS 3–10	I
Anterior circulation, late	ICA or M1, 6–24 h , age <80, NIHSS ≥6, mRS 0–1, ASPECTS 3–5 , no significant mass effect	I
Very large core	ICA or M1, <6 h, age <80, NIHSS ≥6, mRS 0–1, ASPECTS 0–2	2a
Posterior circulation	Basilar artery occlusion , <24 h, baseline mRS 0–1, NIHSS ≥10, PC-ASPECTS ≥6	I

- **Key shift:** a large established core (low ASPECTS) is no longer an automatic contraindication — SELECT2, ANGEL-ASPECT, RESCUE-Japan LIMIT and TENSION pushed the floor down to ASPECTS 3 (and 2a for 0–2).
- **If eligible for both, give both**, rapidly and sequentially. Do not withhold or delay EVT to 'see if the patient improves after lysis' — explicit in the 2026 document.
- Imaging: for suspected LVO within 24 h, emergent CT/CTA or MRI/MRA of cervical + intracranial vessels (Class I). **Do not wait for serum creatinine** before CTA/CTP.
- **No benefit:** pre-procedural tirofiban before EVT.

4. BLOOD PRESSURE — TARGETS REVISED IN BOTH DIRECTIONS

Situation	Target / rule	Comment
Ischaemic, planning IVT	<185/110 before and during; <180/105 for 24 h after	Unchanged — still the safe answer
Ischaemic, no IVT	Permissive hypertension up to <220/120	Treat only for hypertensive emergency or a co-indication
After IVT, mild-moderate stroke	Do NOT target SBP <140 (vs <180)	Class 3 — No Benefit. New.
After successful EVT (mTICI 2b/2c/3)	Intensive SBP <140 for 72 h is HARMFUL	Class 3 — Harm. New; high-yield.
Haemorrhagic (ICH), SBP 150–220	Lower to ~ 140 , keep 130–150 , start within 2 h, maintain ~7 days	Note: SBP <130 is harmful (Class 3). Notes above say '<140–160' — the correct range is 130–150.
ICH, SBP >220	Reduce ~20% in the first hour; avoid a drop >70 mmHg	Excess drop → AKI and neurological deterioration

Agents unchanged: labetalol, nicardipine, clevidipine (hydralazine and NTG are less favoured — unpredictable response, NTG may raise ICP). Add the INTERACT3 care bundle for ICH: SBP <140 + glycaemic control + temperature ≤37.5 °C + rapid reversal of anticoagulation to INR <1.5.

5. ANTIPLATELETS — DAPT WINDOW

- **Minor non-cardioembolic stroke (NIHSS ≤5) or high-risk TIA (ABCD2 ≥4)**, not thrombolysed, presumed atherosclerotic: **aspirin + clopidogrel for 21 days, then single agent** (Class 2a). Trials: CHANCE, POINT.
- Note the 2026 wording widened this to NIHSS ≤5 within 24–72 h (and NIHSS 4–5 within 24 h) with evidence of ≥50% stenosis or new infarcts of large-artery origin.
- If CYP2C19 loss-of-function allele → **ticagrelor + aspirin** instead (CHANCE-2, THALES).
- Aspirin within 24–48 h for everyone else; if thrombolysed, **wait 24 h** and repeat imaging first.

6. ANTICOAGULATION IN AF — THE NOTES ABOVE ARE OUT OF DATE

- The note 'start Warfarin directly' should be corrected: for **non-valvular AF a DOAC is preferred over warfarin**. Warfarin remains the agent only for **valvular AF (moderate-severe mitral stenosis / RHD) and mechanical valves** — which is still the common JIPMER patient, so state the distinction explicitly.
- **Timing (new):** in carefully selected, milder strokes, **early oral anticoagulation is low-risk and reasonable** compared with delayed initiation (Class 2a). Trial: **ELAN**. The old rigid **1-3-6-12 day rule** (TIA-mild-moderate-severe) is now a guide, not a mandate.
- Still no role for **urgent full-dose heparin** in acute ischaemic stroke to prevent early recurrence — the note above is right on that point.

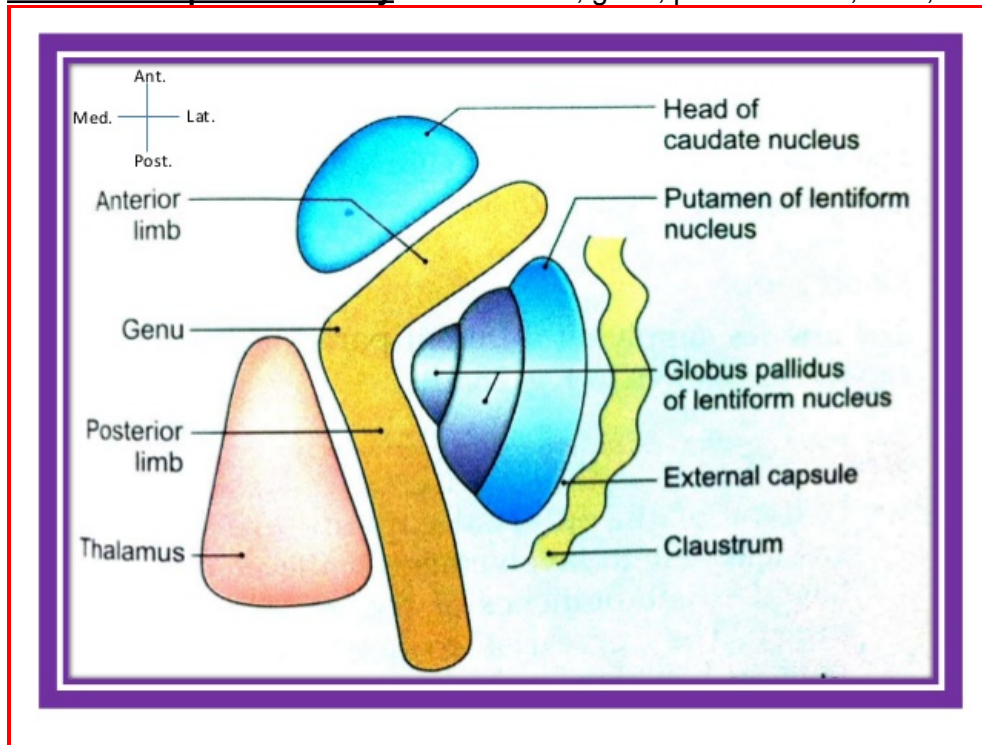
7. SUPPORTIVE CARE — TWO REVERSALS WORTH MARKING

- **Glucose:** IV insulin to a tight target of **80–130 mg/dL is NOT recommended** (Class 3 — No Benefit; trial: SHINE). Treat to a **140–180 mg/dL** range instead. The note above ('insulin if glucose >200') is acceptable but should not become a tight-control protocol.
- **Oxygen:** supplemental O₂ only to keep SpO₂ >94%. Routine supplemental oxygen in non-hypoxic patients has no benefit.
- Baseline ECG and troponin are recommended but **must not delay** IVT or EVT.
- **Decompressive hemicraniectomy** for malignant MCA infarction — within 48 h, greatest benefit at age ≤60, still reduces mortality above 60 with more residual disability. **Suboccipital craniectomy** for space-occupying cerebellar infarct with deteriorating consciousness.
- Mobile stroke units (new section) and direct transport to the nearest **EVT-capable** centre where transfer systems are weak.

ONE-LINE ANSWERS IF PUSHED IN VIVA

If asked...	Say
Thrombolytic of choice	Tenecteplase 0.25 mg/kg single bolus (max 25 mg), Class I alongside alteplase 0.9 mg/kg, within 4.5 h
Window for lysis	4.5 h standard; up to 9 h (or 9 h from mid-sleep in wake-up stroke) with perfusion mismatch; up to 24 h in LVO not eligible for EVT
Window for thrombectomy	6 h routinely; 6–24 h in selected anterior circulation LVO; up to 24 h for basilar occlusion
ASPECTS cut-off for EVT	3–10 (and 2a even for 0–2) — large core is no longer an automatic exclusion
BP after successful EVT	Do not drive SBP below 140 — that is harm
BP in ICH	Target ~140, keep 130–150, start within 2 h; below 130 is harmful
AF with RHD/mitral stenosis	Warfarin — DOACs are for non-valvular AF only
When to start anticoagulation	Early is reasonable in mild stroke (ELAN); 1–3–6–12 rule as a guide

*****Internal capsule anatomy-** Anterior limb, genu, posterior limb, retro, sub lentiform parts



Motor fibres

1. Corticopontine fibres

- Originate from the cerebral cortex and form the largest group of projection fibres in the capsule.
- Named according to the lobe from which they arise
- Frontopontine fibres- pass through anterior limb, genu, posterior limb
- Corticopontine fibres synapse in the ipsilateral pontine nuclei. The fibres arising from the pontine nuclei cross the midline to relay in the cortex of the opposite cerebellar hemisphere to form the cortico-ponto-cerebellar pathway.

2. Pyramidal fibres

- Corticonuclear fibres- synapse with the motor nuclei of the cranial nerves and thus form the supranuclear pathway. Run through the **genu**.
- Corticospinal fibres- Synapse with the anterior horn cells of the opposite half of the spinal cord. Run through the **anterior 2/3 of the posterior limb**. Upper limb fibres are most anterior, then trunk, then LL.

3. Extrapyramidal fibres

- Arise in the cortex and relay into the subcortical grey matter of the extrapyramidal system.
- Named according to their destination- corticorubral, corticostriate, corticonigral etc.
- Occupy the position near the CST fibres.

Sensory fibres

- Mainly thalamocortical fibres which radiate from the thalamus in different directions to reach the cortex.
- Mostly they are 3rd order neurons of sensory pathways conveying sensations to the cortex.
- Anterior, superior, posterior, inferior thalamic radiation
- Posterior thalamic radiation passes through retrolentiform part of the capsule and connects LGB to the primary visual cortex- optic radiation/**geniculocalcarine** tract
- Inferior thalamic radiation passes through the sublenticular part and connects MGB with the primary auditory cortex- auditory radiation.

Arterial supply-

1. Medial and lateral striate branches of MCA-

- One of the striate branches is larger and is more frequently ruptured. It is called **Charcot's artery of cerebral haemorrhage**.
- It enters through the anterior perforated substance and supplies the posterior limb of the capsule.

2. Striate branches of ACA

- One of these branches is larger and takes a recurrent course. It is called **Recurrent artery of Heubner**. It supplies the anterior limb and genu.
3. Central branches of Anterior choroidal artery- sublentiform
 4. Direct branches from internal carotid artery- genu
 5. Central branches of posterior communicating artery

Recurrent artery of Heubner infarct- Faciobrachial monoplegia

- Tracts involved- corticonuclear in the genu, so paralysis of the face
- Paralysis of the upper limb on the opposite side because of involvement of adjacent pyramidal fibres supplying the upper limb (they are the anterior most)

Descending (Motor) Tracts

Lateral system pathways- descend in the lateral column of the spinal cord. Involved in the regulation of skilled voluntary movements because they innervate the distal limb muscles.

- Corticospinal tract
- Rubrospinal tract

Medial system pathways- descend down in the medial and anterior columns of the spinal cord. Involved in the regulation of posture

- Reticulospinal
- Vestibulospinal
- Tectospinal
- Anterior corticospinal

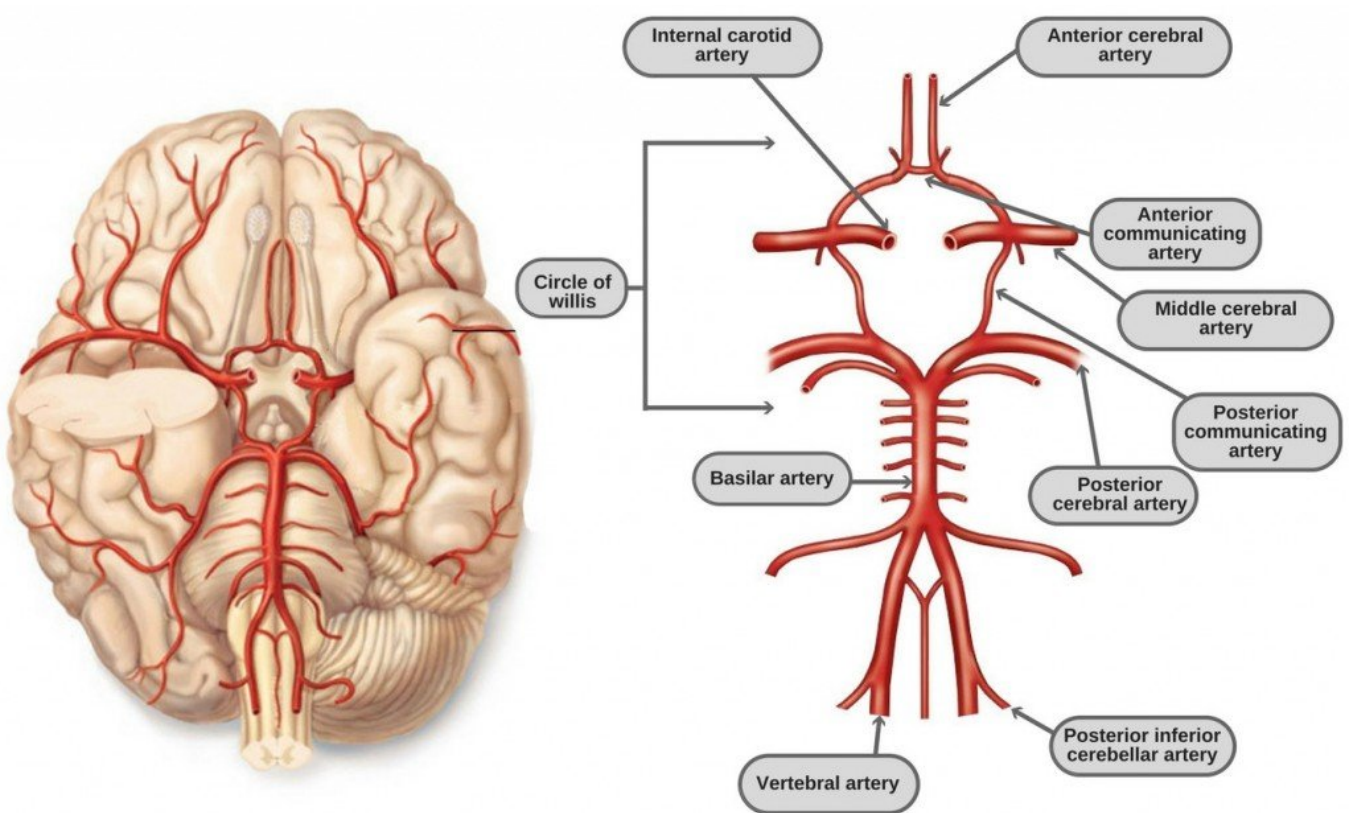
Corticospinal Tract

Origin:

- Primary motor cortex (precentral gyrus, Brodmann area 4)
- Premotor cortex (area 6)
- Supplementary motor cortex (area 6)
- Primary somatosensory cortex (3,1,2)
- Parietal lobe
- Cingulate gyrus

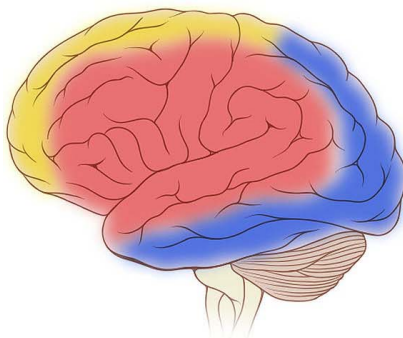
Course

- Fibres converge in the subcortical areas- corona radiata
- Converge through the posterior limb of the internal capsule
- Descend down in the ventral brainstem as the cerebral peduncles
- In the lower medulla, they pass through the pyramid where 80% of the fibres decussate and descend in the lateral funiculus of the spinal cord forming the lateral CST.
- The remaining 20% descend down ipsilaterally (do not decussate) in the anterior funiculus of the spinal cord constituting the anterior CST. They cross over only at the spinal cord segments.
- The lateral CST fibres terminate on the lateral group of motor neurons in the ventral horn of the spinal cord.
- 30% terminate directly on the motor neurons and 70% terminate through interneurons.

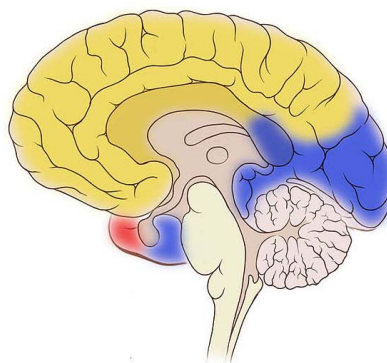


Cortical strokes

Lateral Brain

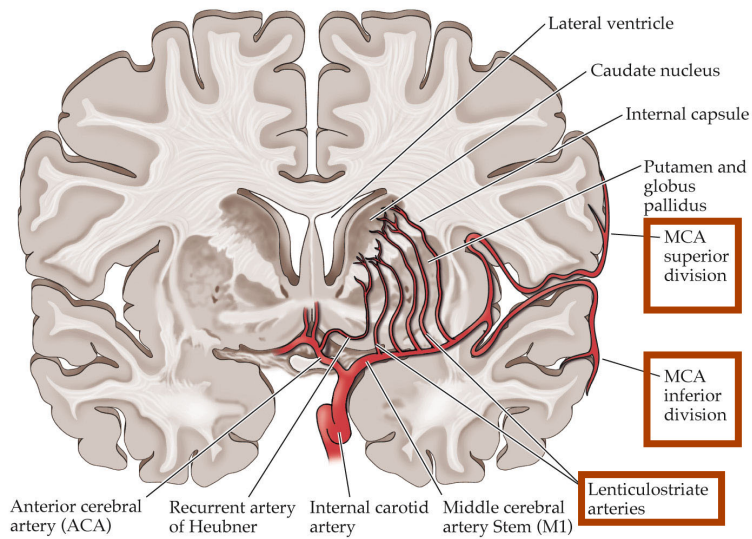


Medial Brain



- Anterior Cerebral Artery
- Middle Cerebral Artery
- Posterior Cerebral Artery

- Middle cerebral artery



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- o Supplies all of the lateral hemisphere except
 - The frontal pole and strip along the superomedial frontal and parietal lobes (ACA)
 - Lower temporal and occipital pole convolutions (PCA)
- o MCA branches
 - Lenticulostriate branches- penetrating branches of proximal MCA (M1 segment)- supply putamen, globus pallidus externus, posterior limb of internal capsule, adjacent corona radiata, most of the caudate nucleus
 - M2 branches- in the sylvian fissure, it divides into superior and inferior branches
- o Manifestations of MCA strokes
 - Complete MCA stroke
 - Contralateral hemiplegia, hemianaesthesia, homonymous hemianopia
 - Dominant hemisphere- global aphasia
 - Non-dominant hemisphere**
 - o Anosognosia- patient is unaware of their disability
 - o Constructional apraxia- inability or difficulty to build/assemble/draw objects
 - o Hemineglect
 - Occur when an embolus occludes the stem of the artery
 - Partial MCA strokes
 - Weakness of hand/arm alone
 - Facial weakness with Broca aphasia
 - Frontal opercular syndrome- Facial weakness+ Aphasia+ arm weakness
- ACA strokes
 - o Anatomy
 - A1 segment- connects the ICA to Anterior communicating artery
 - A2 segment- distal to A.Comm
 - A1 segment gives rise to deep penetrating branches that supply the anterior limb of internal capsule, anterior perforated substance, amygdala, anterior hypothalamus
 - o A1 strokes are rare due to collaterals through A.comm

- o A2 strokes
 - Unpaired A1 with both A2 branches from a single A1
 - Profound abulia- delay in verbal and motor response
 - Spastic quadriparesis
 - Urinary incontinence- paracentral lobule involvement
 - Single ACA
 - Contralateral leg and foot
 - Lesser degree of contralateral arm
 - Urinary incontinence
 - Frontal lobe symptoms
 - o Contralateral grasp reflex, sucking reflex- release signs
 - o Gegenhalten
- PCA strokes
 - o Anatomy
 - 75% of PCAs arise from the basilar artery
 - In 20%, one has its origin from the ipsilateral carotid artery via the P.comm
 - o Proximal PCA strokes- midbrain stroke syndromes mentioned below- Weber/Claude/Benedikt
 - o Distal PCA strokes
 - Causes infarction of the medial temporal and occipital lobes
 - Contralateral homonymous hemianopia with macular sparing
 - Peduncular hallucinosis- hallucinations of brightly coloured scenes and objects
 - Bilateral PCA strokes
 - Cortical blindness (blindness with preserved pupillary reaction) with anosognosia- Anton syndrome
 - Complicated stuff like
 - o Balint syndrome
 - o Palinopsia
 - o Asimultanagnosia

Lacunar strokes

- Infarction following atherothrombotic or lipohyalinotic occlusion of a small artery in the brain
- Called so because the thrombosis of these small vessels forms small lakes of fluid or lacunes in autopsy
- Four characteristic lacunar stroke syndromes
 - o Pure motor hemiparesis- infarct in the posterior limb of internal capsule
 - o Pure sensory stroke- thalamic infarct
 - o Ataxic hemiparesis- ventral pons or internal capsule
 - o Dysarthria and clumsy hand/arm syndrome- ventral pons or genu of internal capsule

Hypertensive IC haemorrhage

- Most common sites in decreasing order of frequency
 - o Putamen
 - o Thalamus
 - o Cerebellum
 - o Pons
- Putamen bleed
 - o Presents with **hemiparesis** because the bleed in the putamen compresses on the internal capsule
- Thalamic bleed- also hemiparesis for the same reason. Prominent sensory deficit
- Pontine haemorrhage- deep coma with quadriplegia over minutes
- Cerebellar haemorrhage- occipital headache, repeated vomiting, gait ataxia

TABLE 446-6 CAUSES OF INTRACRANIAL HEMORRHAGE

Cause	Location	Comments
Head trauma	Intraparenchymal: frontal lobes, anterior temporal lobes; subarachnoid; extra-axial (subdural, epidural)	Coup and contrecoup injury during brain deceleration
Hypertensive hemorrhage	Putamen, globus pallidus, thalamus, cerebellar hemisphere, pons	Chronic hypertension produces hemorrhage from small (~30–100 µm) vessels in these regions
Transformation of prior ischemic infarction	Basal ganglion, subcortical regions, lobar	Occurs in 1–6% of ischemic strokes with predilection for large hemispheric infarctions
Metastatic brain tumor	Lobar	Lung, choriocarcinoma, melanoma, renal cell carcinoma, thyroid, atrial myxoma
Coagulopathy	Any	Risk for ongoing hematoma expansion
Drug	Any, lobar, subarachnoid	Cocaine, amphetamine
Arteriovenous malformation	Lobar, intraventricular, subarachnoid	Risk is ~2–3% per year for bleeding if previously unruptured
Aneurysm	Subarachnoid, intraparenchymal, rarely subdural	Mycotic and nonmycotic forms of aneurysms
Amyloid angiopathy	Lobar	Degenerative disease of intracranial vessels; associated with dementia, rare in patients <60 years
Cavernous angioma	Intraparenchymal	Multiple cavernous angiomas linked to mutations in <i>KRT11</i> , <i>CCM2</i> , and <i>PDCD10</i> genes
Dural arteriovenous fistula	Lobar, subarachnoid	Produces bleeding by venous hypertension
Capillary telangiectasias	Usually brainstem	Rare cause of hemorrhage

Brainstem strokes

Localisation- the rule of 4's

- 4 cranial nerves in Medulla, Pons, Above Pons (2 in midbrain and 2 above brainstem)
- 4 cranial nerves divide into 12 (3, 4, 6, 12)- their motor nuclei are located **medially**
- 4 cranial nerves do not divide into 12 (5, 7, 9, 11)- their motor nuclei are located **laterally**
- 4 midline columns- Motor nucleus (of 3, 4, 6, 12), Motor pathway (Corticospinal tract), MLF, Medial lemniscus (dorsal column pathway)
- 4 lateral (side) columns- Sympathetic, Spinothalamic, Sensory nucleus of V nerve, Spinocerebellar
- Lesions in all the cranial nerve nuclei cause **ipsilateral deficits**
- Midline column lesions
 - Motor pathway- contralateral weakness
 - MLF- ipsilateral INO- Ex: In left INO, when an attempt is made to look right, the left eye will adduct minimally whereas the right eye will abduct with nystagmus
 - Medial lemniscus- contralateral proprioception/vibration loss
- Side structure lesions
 - Spinocerebellar tract- ipsilateral ataxia
 - Spinothalamic- contralateral pain/temperature loss
 - Sensory nucleus of V nerve- ipsilateral loss of pain/temperature in face
 - Sympathetic pathway- ipsilateral Horner's syndrome
- Caveats
 - V nerve involvement should not be used to localise to pons, rather to **lateral side**
 - VIII nerve involvement should only be considered as **deafness** because vestibular signs can be from the pons or medulla.

<u>Site</u>	<u>Eponym</u>	<u>Artery</u>	<u>Structures affected</u>	<u>Differentiating clinical features</u>	<u>Other features</u>
Lateral Medullary Syndrome	Wallenberg	PICA	1. Spinothalamic tract 2. Spinocerebellar tract 3. Sensory nucleus of V nerve 4. Horner's 5. IX and X nerve	IX and X nerve involvement- loss of gag reflex, hoarseness, dysphagia, Ryle's tube in situ, Hiccups	Vertigo and nystagmus due to vestibular nucleus involvement
Medial medullary syndrome	Dejerine	Vertebral artery	1. Corticospinal tract 2. Medial lemniscus 3. MLF 4. XII nerve	XII nerve involvement- tongue deviated to ipsilateral side	
Lateral pontine syndrome		AICA	1. Spinothalamic tract 2. Spinocerebellar tract 3. Sympathetic pathway 4. Spinal nucleus of VN 5. VII N	VII nerve palsy	
Medial pontine syndrome		Paramedian branches of basilar A	1. Corticospinal tract 2. Medial lemniscus 3. MLF 4. VI nerve	VI nerve palsy- Strabismus, LR palsy	
Ventral pontine syndrome	Locked in syndrome	Basilar artery	Bilateral long tract signs Sometimes supplemented by V-VIII nerve dysfunction	Quadriplegia, aphasia Usually no eye paralysis Communicate by eye movements or blinking	

Medial midbrain syndrome	Weber	Penetrating branches of PCA	1.Corticobulbar tract 2.Corticospinal tract 3.III nerve	(Does not exactly follow the rule of 4) Contralateral face weakness due to corticobulbar involvement, III nerve- ipsilateral eye down and out with pupillary dilation	
Lateral midbrain syndrome	Benedikt Claude syndrome	Penetrating branch of PCA	1.III nerve 2.Red nucleus 3.Dentatorubrothalamic tract	Red nucleus- contralateral body Tremor/ataxia- DRT tract	
Dorsal midbrain syndrome	Parinaud	Not usually vascular. Pinealoma, MS etc cause	1.Rostral interstitial nucleus of MLF- upward gaze palsy 2. Collier's sign- eyelid retraction		Most common cause is Pinealoma

Viva points

- Causes of recurrent hemiplegia
 - Embolic stroke
 - Vasculitis
 - Demyelination- Multiple sclerosis
 - Todd's paralysis
 - Infection
 - Hypoglycaemia
- Sudden onset non-vascular aetiology weakness
 - Todd's paralysis- post ictal
 - Trauma
- Strokes which can cause quadriparesis
 - Stroke in unpaired ACA
 - Basilar artery stroke
- Decorticate vs. Decerebrate rigidity
- Hypertensive bleed- Thalamus and Putamen
- Uncal herniation order of weakness
- Lacunar infarcts- read
- Watershed infarcts- caused by hypotension
- Pseudobulbar palsy vs bulbar palsy
- Pseudobulbar palsy causes
 - Recurrent stroke
 - Multiple sclerosis
 - Motor neurone disease
- Inobvious stroke- stroke which is not being recognised as one
 - Homonymous hemianopia
 - Frontal stroke
 - Lateral medullary syndrome- hiccups, vertigo, vomiting

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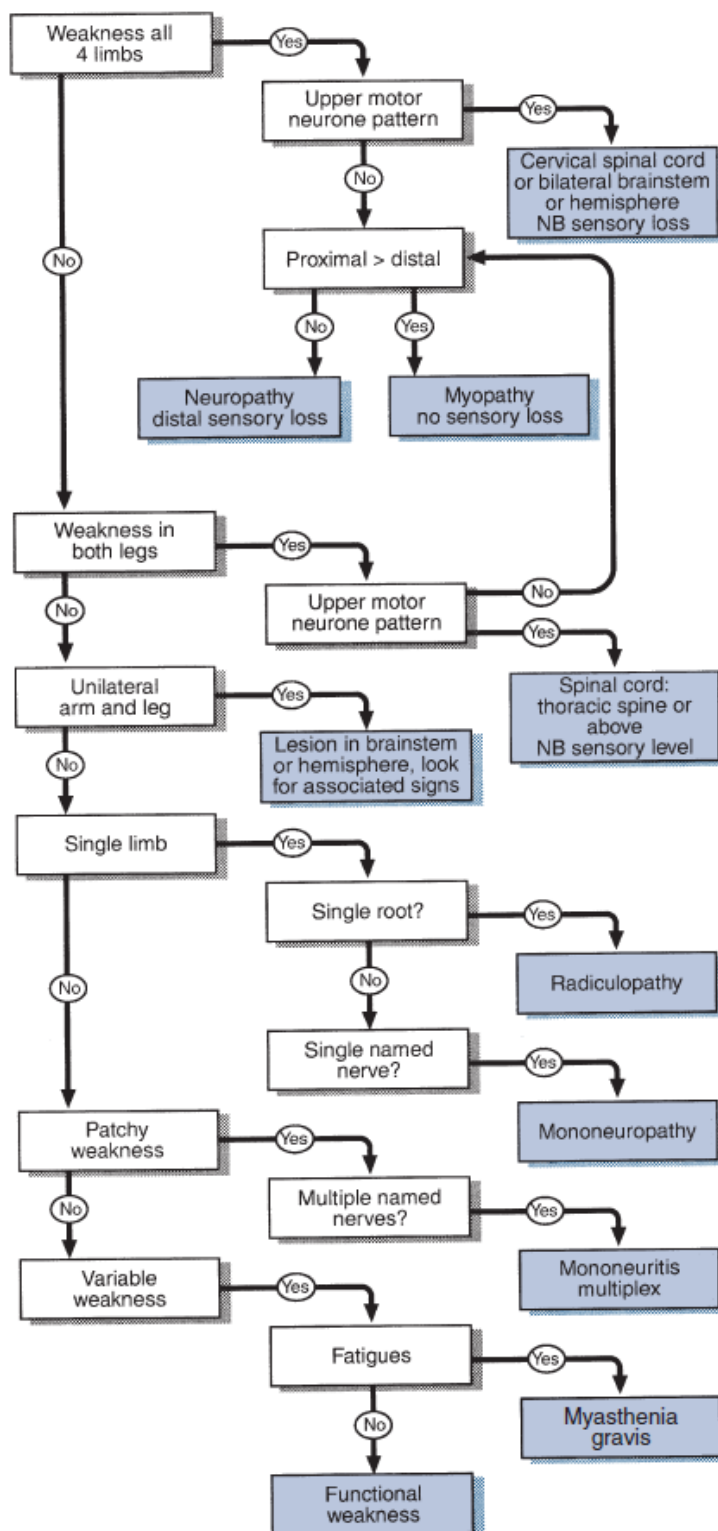


Figure 20.1

HISTORY AND BEDSIDE TECHNIQUE

- **Case sheet header** — record **bed, ward number, floor, place** and **date, month, day or night, time** at the top of every case sheet. [Niharika's notes]
- **Hand grip** is graded as a **percentage** — **40 to 50%** — not on the MRC scale. [Niharika's notes]
- **Triceps power**: ask the patient to extend the elbow and hold it there, then the examiner tries to flex it. [Niharika's notes]
- **Mentalis** — test by asking the patient to **evert the lower lip**. [Niharika's notes]
- **Orbicularis oculi** — “*thirakka vidaadhu*”: the patient should not let you open the eye. Useful Tamil bedside phrases: tone — “*maram maari irkaa, loose aa?*” (stiff like wood, or loose?); progression — “*adhu mosamaayittaa?*” (is it getting worse?). [Niharika's notes]
- **Left-handed patients still have LEFT hemisphere dominance in 95%**. [Niharika's notes]
- **Lambert-Eaton** — **reflexes ABSENT** (they facilitate after exercise). **Myasthenia** — **reflexes PRESERVED**. The quickest bedside discriminator. [Niharika's notes]
- **Renal tubular acidosis causes hypokalaemia** — keep it on the list for acute flaccid weakness with normal sensation. [Niharika's notes]

PUPILS, VISUAL PATHWAY, EYE MOVEMENTS

- **Marcus Gunn pupil = RAPD**. **Argyll Robertson pupil** — responds well to **near**, not to **light** (light-near dissociation). [Niharika's notes]
- **Lateral geniculate nucleus sits in the thalamus**. [Niharika's notes]
- **Meyer's loop** (temporal) → contralateral **superior** quadrantanopia; **Baum's loop** (parietal) → contralateral **inferior** quadrantanopia. [Niharika's notes]
- **The macula is supplied by the MCA** — which is why macular sparing occurs in PCA occlusion. An optic tract lesion should make you think of an **MCA stroke involving the internal capsule**. [Niharika's notes]
- **Pursuit eye movements** are tested in both the **horizontal and vertical** plane. [Niharika's notes]
- **Causes of VERTICAL nystagmus**: demyelination · stroke · **Wernicke's encephalopathy** · Arnold-Chiari malformation · cervicomedullary junction lesions. [Niharika's notes]
- **SO4, LR6** — superior oblique is CN IV, lateral rectus is CN VI (abduction). [Niharika's notes]

BRAINSTEM, CEREBELLUM, SENSORY

- **Millard-Gubler syndrome** = ipsilateral VI nerve palsy (and VII) with contralateral hemiplegia — ventral pons. [Niharika's notes]
- **CN XII involvement ⇒ medial medullary syndrome** (“12 ⇒ MM”). PICA ⇒ lateral medullary. Facial palsy that is **LMN** localises to the pons; **UMN** facial palsy points above it. [Niharika's notes]
- **Rebound phenomenon and past-pointing** — add these to the lateral cerebellar lesion signs. [Niharika's notes]
- **Cutaneous receptors**: Ruffini endings — proprioception; Pacinian — vibration; **Merkel's and Meissner's** — fine touch. [Niharika's notes]

SEIZURES — A BLOCK WORTH LEARNING WHOLE

- **Grey matter initiates seizures; white matter cannot**. Hence **pure demyelinating disease does not cause seizures**. The only grey matter in the brainstem is the cranial nerve nuclei — so **brainstem lesions do not cause seizures unless the reticular activating system is involved**. [Niharika's notes]
- **Non-convulsive status epilepticus** — the seizure is present but muscle fatigue means convulsions are not manifested. Use the **midazolam challenge test**. [Niharika's notes]
- **Seizures are rhythmic; hemiballismus is not** — and in hemiballismus the sensorium is intact. **DKA can cause hemiballismus**. [Niharika's notes]
- **Scar epilepsy** and **focal with secondary generalisation** — name both. [Niharika's notes]
- **Post-episode features to ask for**: palpitations, vomiting, sweating, postural change, **tongue bite**, frothing, clenching of teeth, upward rolling of the eyes, loss of consciousness, incontinence of urine and stool. [Niharika's notes]
- **Top of basilar artery (TOBA) syndrome** with cerebral oedema — seizures will be a manifestation. [Niharika's notes]

STROKE — VIVA POINTS

- **Bilateral extensor plantars with unilateral weakness** ⇒ **think raised ICP or a metabolic cause** — hepatic encephalopathy, uraemic encephalopathy, CO₂ narcosis, hyponatraemia. A high-yield box. [Niharika's notes]
- **Amyloid angiopathy** — lobar/cortical bleed in the elderly. **Thrombotic stroke in old age — think malignancy. Venous** — superior sagittal sinus (**empty delta sign**) and sigmoid sinus. [Niharika's notes]
- **Haemorrhagic stroke by mechanism:** arterial → hypertension; venous → pregnancy, OCP, alcohol, connective tissue disease; plus haemorrhagic transformation of an infarct. [Niharika's notes]
- **Thrombotic — weakness first, then oedema causes seizures. Embolic — weakness improves** (the clot fragments and moves distally). [Niharika's notes]
- **Strokes cause SIADH → hyponatraemia.** Check sodium in every deteriorating stroke patient. [Niharika's notes]
- **CVT — anticoagulation IS given**, even in the presence of haemorrhagic infarction. Note also that **venous thrombosis and iron deficiency anaemia are related**, and **OCP causes DVT**. [Niharika's notes]
- **Provoked DVT — 3 months** of anticoagulation; unprovoked, usually **6 months** or longer. **APLA with low platelets → warfarin/acenocoumarol; DOACs have no role.** [Niharika's notes]
- **Anti-oedema measures run 4-5 days** — mannitol, glycerol, 3% saline. **NTG causes vasodilatation → reduced cerebral perfusion pressure in raised ICP** — avoid it. **CPP = MAP – ICP.** [Niharika's notes]
- **Hypertensive emergency** (haemorrhagic stroke, aortic dissection): **reduce by 25% in the first hour**, then normalise over 24 hours. Note **reactionary hypertension** after a stroke needs no treatment. [Niharika's notes]
- **Haemorrhagic metastases** occur in malignancy. In trauma remember the **lucid interval** — EDH, SDH, SAH. [Niharika's notes]
- **Carotid bruit — auscultate with the DIAPHRAGM.** Examine for an abdominal aneurysm in the same case. [Niharika's notes]
- **PICCLE mnemonic → iron deficiency:** glossitis, cheilitis, knuckle hyperpigmentation, koilonychia. [Niharika's notes]

Case-taking proforma

History

- Chief complaint
- History of presenting illness
 - Weakness- same format as hemiplegia
 - Progressive weakness (increasing in severity and extent)- more likely to be cord compression
 - Improving weakness- inflammatory, acute transverse myelitis,
 - Waxing and waning- MS
 - Sensory involvement
 - Radicular pain- sharp shooting severe dermatomal pain radiating down a limb or around the trunk. Exacerbated by conditions that increase the pressure in the spinal cord like coughing, sneezing, straining etc
 - Spinothalamic tract- Poorly localised aching pain
 - Dorsal column-
 - **Wash basin sign**- unsteadiness on closing eyes to wash face
 - Difficulty in walking at night
 - Constricting band like sensation
 - Autonomic involvement
 - Bladder
 - Urgency, frequency, precipitancy- UMN
 - Hesitancy, poor stream, retention of urine and overflow incontinence- LMN
 - Can patient feel the bladder filling?
 - Catheterisation done?
 - Credes manoeuvre- pressing the lower abdomen to pass urine
 - Bowel
 - Constipation/incontinence
 - Cranial nerves
 - VII- GBS
 - II- Optic neuritis, colour vision disturbance- MS. Optic atrophy in B12 deficiency
 - Ophthalmoplegia- Miller Fischer
 - History for aetiology
 - h/o cough with expectoration, weight loss, night sweats, evening rise of temp- Pott's spine
 - h/o recent URI/diarrhoea- GBS, hypokalaemia
 - h/o trauma
 - lower urinary tract symptoms (in elderly male)- Ca prostate
 - breast lump in elderly female
 - cough, haemoptysis, weight loss- Ca lung
 - h/o dog bite and vaccination- rabies
 - Personal history
 - Smoking for Ca lung- quantify pack years
 - Alcohol for B12 deficiency
 - Veg or non veg for B12 deficiency
 - Past history
 - TB and contact with TB
 - Family history
 - Hereditary spastic paraplegia

Examination

Neurocutaneous markers- be ready for questions on the diseases

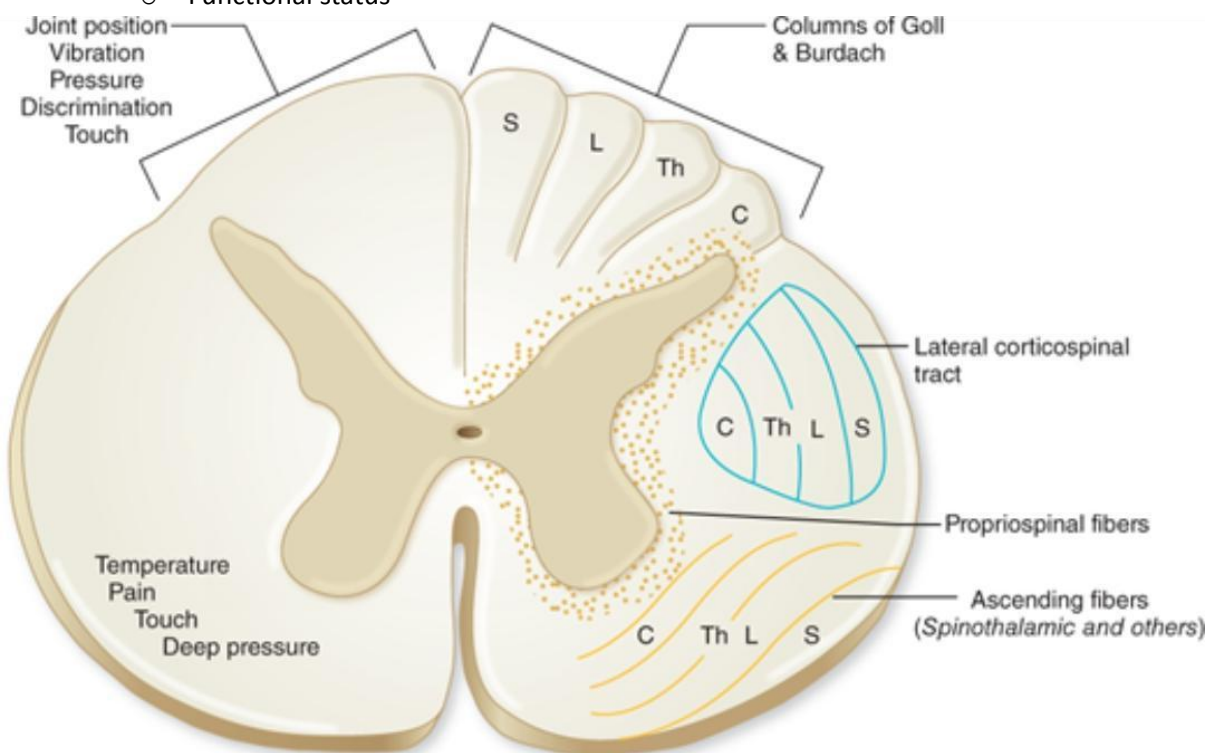
- Pallor- B12 deficiency
- Lymphadenopathy- TB/mets

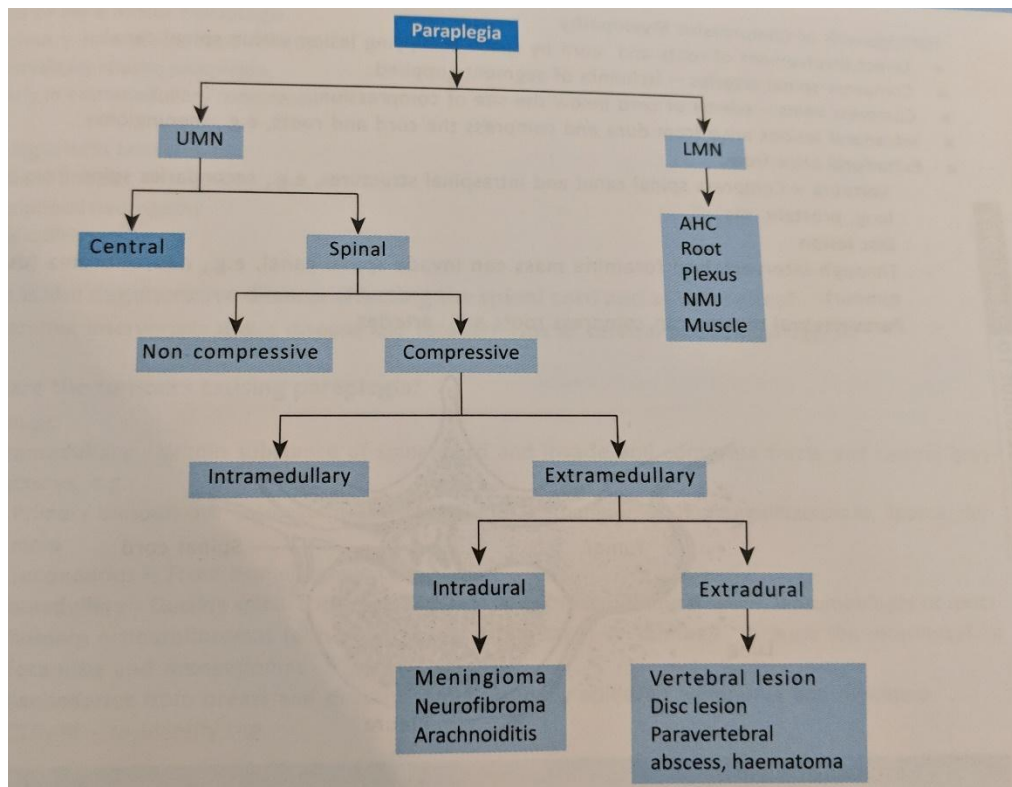
Spine examination

- Gibbus
- Kyphosis
- Scoliosis
- Tuft of hair
- Tenderness over spine

Diagnosis

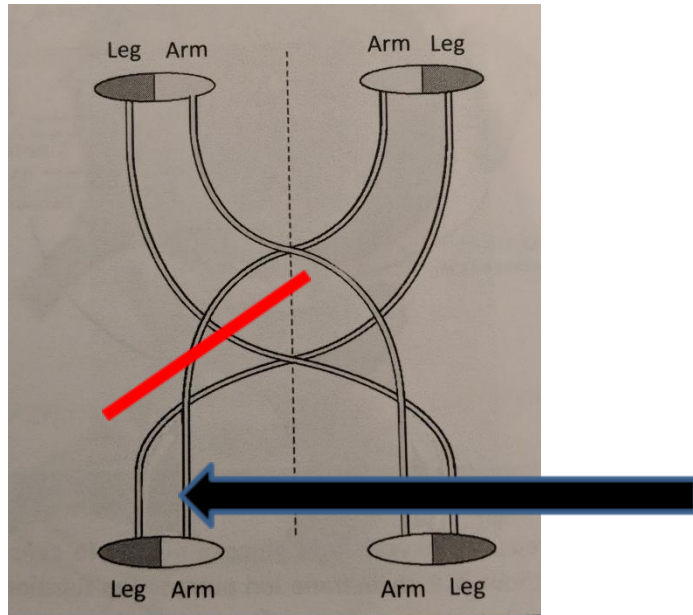
- Paraparesis/paraplegia
- Anatomical level
 - Vertical- Sensory, motor, reflex
 - Horizontal- extradural/extramedullary/intramedullary
 - Pathological- Compressive/Vascular/Infective/Demyelinating
 - Aetiology
 - Functional status





- UMN causes of paraplegia
 - Central
 - Parasagittal meningioma
 - Trauma at parasagittal area
 - Thrombosis
 - Superior sagittal sinus thrombosis
 - Thrombus of unpaired ACA
 - Bilateral ACA thrombosis
 - Spinal
 - Compressive
 - Intramedullary
 - Extramedullary
 - Non compressive
 - Traumatic
 - Post infectious
 - MS, NMO
 - Post vaccinia** - Polio, tetanus, rabies vaccines
 - Familial - Motor neuron disease, Familial spastic paraplegia, Spinocerebellar degeneration
 - Nutritional - B12, Copper deficiency
 - Lathyrism
 - Systemic diseases
 - Paraneoplastic

Cruciate hemiplegia and Elsberg phenomenon



- Pyramidal tract decussates at the cervicomedullary junction
- Arm fibres cross before leg fibres
- Cruciate hemiplegia- controversy
 - <https://www.nature.com/articles/3101861>
 - Multiple journal articles say ipsilateral UL and contralateral LL because the site of the lesion is between the crossing of UL and LL fibres, where the UL fibres have already crossed
 - Alagappan says contralateral UL and ipsilateral LL which cannot be localised anywhere in the diagram
- Elsberg
 - 1. Ipsilateral arm fibres (black arrow)
 - 2. Ipsilateral leg fibres
 - 3. Contralateral leg fibres
 - 4. Contralateral arm fibres

Table I: Various causes of cruciate paralysis described^{1-3,7,8,13-30}

Traumatic injury	Atlanto-occipital dislocation Jefferson fracture Odontoid peg fracture (most common) C2/C3 fractures and dislocations SCIWORA Penetrating injury, i.e. gunshot
Disc herniation	C2/3 disc herniation
Congenital anomalies	Congenital anomalies of C1 and C2 with atlanto-axial instability Chiari malformations ± syringomyelia
Tumour or infection	Spinal metastasis Odontoid tuberculoma
Rheumatoid arthritis	
Complications of surgery to cervicomedullary junction	

Feature	Compressive myelopathy	Non compressive
Bony pain	Present	Absent
Bony tenderness	Present	Absent
Girdle like sensation	Present	Absent
Upper level of sensory loss	Present	Absent
Zone of hyperaesthesia	Present	Absent
Root pain	Present	Absent
Onset and progression	Gradual onset	Can be acute
Symmetry	Asymmetrical	Usually symmetrical
Flexor spasm	Common	Usually absent
Pattern of neurodeficit	U shaped (in cervicomedullary lesions)	Bilaterally symmetrical
Bowel and bladder involvement	Early	Late
Classical example	Pott's spine	MND

Determining the level of the lesion

- The presence of a horizontally defined level below which sensory, motor and autonomic function is impaired is a hallmark of a spinal cord lesion
- Sensory loss below the level of the lesion is due to damage to the spinothalamic tract on the **opposite side, one or two segments higher in case of a unilateral lesion and at the level of a bilateral lesion**. This is because the second order sensory fibres of the STT ascend for one or two levels as they cross anterior to the central canal to join the opposite STT.
- Lesions that transect the descending CST and other motor tracts cause paraplegia or quadriplegia with heightened DTR, Babinski and eventual spasticity
- Transverse damage to the cord also produces autonomic disturbances consisting of absent sweating below the implicated cord level and bowel, bladder and sexual dysfunction

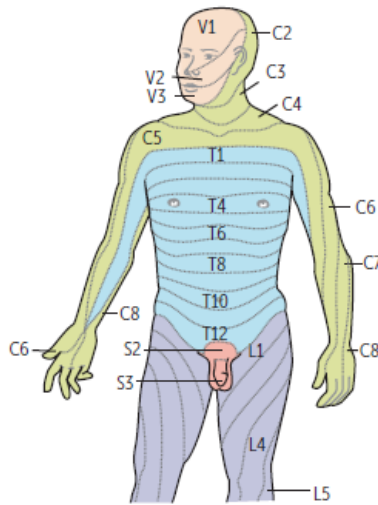
- A band of **altered sensation** (hyperalgesia or hyperpathia) at the upper end of the sensory disturbance, **fasciculations**, **atrophy** of the muscles and muted or absent DTR may be noted
- Cervical cord:
 - Quadriplegia and weakness at the diaphragm
 - Horner's syndrome may accompany a cervical cord lesion at any level
 - Loss of biceps jerk, finger and wrist extensors and triceps jerk
- Thoracic cord
 - Localised by sensory level on the trunk, and if, present, by the site of midline back pain
 - Nipples- T4
 - Umbilicus- T10
 - **Beevor's sign**- T9-T10 lesions paralyse the lower, but not the upper, abdominal muscles resulting in an upward movement of the umbilicus when the abdominal muscles contract
- Lumbar cord
 - Knee jerk, Ankle jerk
 - Hip movements
- Sacral cord/Conus medullaris- refer conus medullaris syndrome

Spastic vs flaccid paraplegia

Legend	Spastic (UMN)	Flaccid (LMN)
History	Feeling of tightness	Looseness
Examination – bulk	Minimal wasting	Wasting present
Tone	Increased	Decreased
Fasciculation	Absent	Present
Superficial reflex	Lost	Lost
Deep tendon reflex	Exaggerated	Lost
Plantar reflex	Extensor plantar response	Flexor or no response

Important dermatomes for localisation

Landmark dermatomes



C2—posterior half of the skull.

C3—high turtleneck shirt.

C4—low-collar shirt.

C6—includes thumbs.

T4—at the nipple.

T7—at the xiphoid process.

T10—at the umbilicus (important for early appendicitis pain referral).

L1—at the inguinal ligament.

L4—includes the kneecaps.

S2, S3, S4—erection and sensation of penile and anal zones.

Causes table

Duration	Cerebral	Non-compressive	Extradural extramedullary	Intradural extramedullary	Intramedullary
Acute	Unpaired ACA thrombosis, SSS thrombosis	Acute transverse myelitis	IV disc prolapse, Epidural abscess	None	Haematomyelia
Chronic	Parasagittal meningioma	B12 deficiency, Hypocupric myelopathy, Frederick Ataxia, Lathyrism	Pott's spine	Neurofibroma	Ependymoma, Astrocytoma, Syringomyelia, Glioma

Approach to spastic paraplegia

- 5 T's- Trauma, TB, Thrombosis, Tumour, Transverse myelitis
- Can be gradual or sudden onset
- Gradual onset
 - Cerebral cause- Cerebral diplegia (Little's disease), Parasagittal meningioma
 - Spinal- Cord compression, non compressive (motor neurone disease, MS, SCD, Lathyrism, syringomyelia)
- Sudden onset
 - Cerebral causes- Superior sagittal sinus thrombosis, Unpaired Anterior Cerebral Artery Thrombosis
 - Spinal causes- Compressive- PIVD, Non compressive- Acute transverse myelitis, Post vaccinal, thrombosis of anterior spinal artery
- Of two types, Paraplegia in flexion and paraplegia in extension

Feature	Paraplegia in extension	Paraplegia in flexion
Definition	Lower limbs take an extension attitude and extensor muscles are spastic	Lower limbs take a flexion attitude due to flexor spasm
Pathology	Only pyramidal tracts are involved	Pyramidal and extrapyramidal involvement
Evolution	Early	Late
Attitude	Hip extended, adducted Knee extended, feet plantarflexed	Thigh and knee flexed Feet dorsiflexed Lateral decubitus position in bed
Tone	Clasp-knife spasticity in extensor muscles	Tone increased in flexor muscles
Jerks	Brisk	Present but diminished
Plantar response	Extensor	Extensor, but associated with flexor spasm
Reflex evacuation of bowel and bladder	Absent	Present
Mass reflex	Absent	May be present

- Cerebral causes of paraplegia
 - h/o Jacksonian fits and cortical sensory loss
 - Spastic paraplegia in extension
 - Features of increased ICT
 - Bladder dysfunction common
 - Abnormalities in higher mental function, speech, cranial nerves

Flaccid paraplegia

- UMN in spinal shock
- Anterior horn cell lesion- Poliomyelitis, Non-polio enterovirus, progressive muscular atrophy
- Nerve root (radiculopathy)- GBS, Tabes dorsalis
- Peripheral neuropathy
- NMJ- Myasthenia, LEMS
- Muscle- myopathy, muscular dystrophy, hypokalaemic periodic paralysis

Compressive vs. Non-compressive

- Compressive lesions
 - Bony deformity
 - Bony tenderness
 - Upper level of sensory loss
 - Root pain
 - Zone of hyperaesthesia
 - Gradual onset
 - Asymmetrical
 - U shaped pattern of neurodeficit (Elsberg phenomenon/Around the clock phenomenon- Harrison's) Ipsilateral UL, Ipsilateral LL, Contralateral LL, Contralateral UL.
 - Early bowel and bladder involvement
- Transverse myelitis is a non-compressive pathology that behaves like compressive

Localisation

- The uppermost level of a spinal cord lesion can also be localised by segmental signs corresponding to disturbed motor or sensory innervation by an individual cord segment
 - Band of altered sensation- hyperalgesia
 - Atrophy or fasciculations of muscles supplied by that root
 - Absent or inverted DTR at that level
 - May be associated with a nerve disorder as well if the nerve shares that root value
- However, **none of this is reliable in spinal shock**
- Foramen magnum- atrophy of SCM, downbeat nystagmus, C2 sensory loss, Horner's syndrome, lower cranial nerve palsy
- **Hemiplegia cruciata**
 - Sign of a cervicomedullary junction lesion
- C2- suboccipital pain/sensory loss. **Descending tract of V nerve affected, so pain and temperature loss over face.** Loss of trapezius reflex
- C3- Loss of trapezius reflex
- C5- inverted biceps jerk, inverted supinator jerk, **sensory loss over deltoid** (regimental badge)
- C6- loss of or diminished biceps and supinator, exaggerated finger flexor
-
- Inverted reflexes
 - Inverted supinator jerk- brisk finger flexion instead of elbow flexion. Associated with an absent biceps jerk and exaggerated triceps jerk. Indicative of a lesion at C5 or C6. Occurs because LMN lesion of C5 is combined with a UMN lesion affecting reflexes below C5.
 - Inverted biceps jerk- extension of elbow instead of flexion. Indicates a lesion at C5
 - Inverted knee jerk- Flexion of the knee instead of extension. Indicates L3 or L4 lesion

Lesions

- Extramedullary (95%)
 - Intradural (15%)- Meningioma, neurofibroma, AV malformations
 - Extradural (80%)- Pott's spine, mets, PIVD, spondylolisthesis, trauma
- Intramedullary (5%)- glioma, ependymoma, chordoma, syringomyelia

Spinal shock

- Flaccid areflexic paralysis below the level of the lesion in the spinal cord after acute insult
- There may be urinary retention and constipation
- Plantar response may be equivocal or no response

Acute transverse myelitis features

- Often follows a viral illness or post-vaccinal

- Flaccid paraplegia. Onset is acute or subacute
- Fever may be present before paralysis develops
- Bladder (urinary retention) and bowel (unable to pass stool) involvement early
- Although it behaves like a compressive variety, there is absence of root pain, spinal tenderness, spinal deformity
- Back pain and progressive paraparesis are presenting complaints
- Plantar response is extensor and there is partial or complete sensory loss with a definite upper level (vs. GBS- flexor response, no demonstrable sensory loss)
- MRI to exclude cord compression
- Majority recover within 3 months

Interpretation of the history

- Root pain definition*- sharp, stabbing, constant shock like pain below the level of the lesion radiating to both limbs. Characteristic of extramedullary pathology
- Funicular pain- poorly localised burning pain. Characteristic of intramedullary pathology
- Intramedullary syndromes- Arise from within the substance of the cord
- Extramedullary syndromes- Lie outside the cord. Radicular pain is prominent, early sacral sensory loss, spastic weakness in the legs, incontinence
- Compressive myelopathies- Warning signs like back, neck pain, bladder disturbances and sensory symptoms precede the paralysis.
- Non-compressive myelopathies
- Extradural vs Intradural
- Causes of ascending paralysis
- Causes of descending paralysis
- **Positive sensations:** girdle like sensation, pins and needles
- **Negative sensations:** numbness
-

Extramedullary	Intramedullary
More common	Less common
Prominent root pain	Funicular pain
Early sacral sensory loss (sacral distribution in spinothalamic tract is close to the outer side .. Hence)	Sacral sparing
Early spastic weakness of legs (same reason)	Less pronounced spasticity and it's a late feature
Reflexes markedly brisk early	Less brisk late
Bowel bladder involvement is late (controversial point – harrisons says early but both kundu and alagappan mention late)	early

Extra dural	Intradural
More common	Less common
Long duration symptoms	Shorter duration
benign	malignant
Both sides root pain (symmetrical symptoms)	One side root pain (asymmetrical)

Conus medullaris syndrome

- Damage to the spinal cord segments S3-S5 (conus medullaris) which are situated at the level of L1 vertebra
- Aetiology
 - Spinal tumours
 - Trauma- vertebral fracture, spondylolisthesis
- Sudden bilateral onset
- Lower back pain
- Less severe radicular pain
- Symmetric, hyperreflexic distal paresis of the LL, but Ankle jerk may be absent (UMN+LMN)
- Symmetric bilateral perianal numbness (saddle block anaesthesia)
- Sensory dissociation
- Early onset of bowel and bladder incontinence
- Erectile dysfunction

Cauda equina syndrome

- Damage to or compression of the cauda equina with nerve fibres of L3-S5 (below L2)
- Aetiology
 - Disc herniation
 - Trauma
 - Tumours
- Gradual unilateral onset
- Severe radicular pain
- Asymmetric, areflexic, flaccid paresis of the legs
- Saddle block anaesthesia (may be asymmetric)
- Asymmetric, unilateral numbness and/or paraesthesia in LL dermatomes
- Late onset of urinary retention
- Change in bowel habits
- Decreased rectal tone or bulbocavernosus reflex
- Erectile dysfunction

Chronic myelopathies

- Spondylotic Myelopathy
- Vascular malformations
- Retroviral associated
- Syringomyelia

- Chronic myelopathy of MS
- SADC
- Hypocupric myelopathy
- Tabes
- Familial spastic paraplegia
- Adrenoleukodystrophy

Non compressive myelopathies

- Spinal cord infarction
- MS, NMO
- SLE, Sjogren associated
- Post infectious/post vaccination
- Acute infectious- EBV, CMV, HSV, Polio, Enteroviridae

Compressive myelopathies

- Neoplastic spinal cord compression
- Spinal epidural abscess
- Pott's spine
- Spinal epidural haematoma
- Haematomyelia

Viva

- Paraesthesia
- Causalgia
- Dysaesthesia
- Hyperaesthesia
- Hypaesthesia
- Anaesthesia dolorosa
- L'hermitte sign
 - MS
 - B12 deficiency
 - Cervical compressive myelopathy
- Uthoff sign- MS symptoms worsening with exercise
- Motor unit- single α motor neuron and the muscle fibres supplied by it
- ******Cranial causes of paraparesis**- any defect in the paracentral lobule
 - Parasagittal meningioma
 - Superior sagittal sinus thrombosis
 - Unpaired ACA thrombosis
 - Cortical sensory loss, Jacksonian fits help to differentiate
- Urine complaints
 - Initiation
 - Hesitancy
 - Fullness of bladder
 - Maintenance of stream of urine
 - Post voidal dribbling of urine
- *****Amaurosis fugax**: Transient unocular loss of vision. Indicates carotid artery atherosclerosis
- Nerve thickening: Leprosy, Amyloidosis, Sarcoidosis
- Cervical myelopathy
 - Onion pattern of facial involvement
 - Check V nerve carefully

Theory part

SACD, ATM, GBS

GBS

- Aetiology
 - About 2/3 of GBS patients experience symptoms of URI or GI infection 1-4 weeks prior to the onset of GBS
 - Pathogens associated with GBS include Campylobacter, CMV, EBV, HIV, Influenza
 - Campylobacter enteritis is the most common disease associated with GBS
- Pathophysiology
 - Postinfectious autoimmune reaction that generates cross-reactive antibodies (molecular mimicry)
 - Autoantibodies against gangliosides or other unknown antigens of peripheral Schwann cells causes immune mediated segmental demyelination
- Clinical features
 - Ascending paralysis- bilateral flaccid paralysis from the LL to UL in a “stocking glove” distribution
 - Sensory involvement usually absent. Sometimes mild sensory symptoms can be present
 - Relatively symmetric weakness
 - Bladder involvement rare
 - Cranial nerve involvement- frequently b/l VII N palsy
 - Lower cranial nerves are also frequently involved, causing bulbar weakness
 - Fever and constitutional symptoms are usually **absent**
 - Landry paralysis- involvement of the respiratory muscles
 - Reduced or absent reflexes
 - Autonomic dysfunction- arrhythmias, voiding dysfunction, intestinal dysfunction, fluctuations in BP
- Atypical GBS
 - Transient sensory level with band like sensation
 - Urinary retention
 - Babinski sign
- Subtypes and variants
 - Acute inflammatory demyelinating polyneuropathy (AIDP)
 - Adults affected more than children
 - Accounts more >90% of cases in the western world
 - Rapid recovery
 - Anti GM1 antibodies in <50%
 - Acute motor axonal neuropathy (AMAN)
 - Children and young adults
 - China and Mexico
 - May be seasonal
 - Rapid recovery
 - Anti GD1a antibodies
 - Acute motor sensory axonal neuropathy (AMSAN)
 - Mostly adults
 - Uncommon
 - Recovery slow, often incomplete
 - Closely related to AMAN

- Miller Fisher Syndrome (MFS)
 - Adults and children
 - Ophthalmoplegia, ataxia, areflexia
 - Anti GQ-1b antibodies
- Diagnosis
 - CSF
 - Albuminocytologic dissociation
 - Elevated protein levels but normal cell counts in the CSF
 - CSF cell counts $>50/\mu\text{L}$ indicate GBS is unlikely
 - Transient rise in CSF cell counts may occur, but prolonged high values make the diagnosis of GBS unlikely
 - Antibodies against gangliosides
 - Electrodiagnostic studies
 - Nerve conduction studies and EMG are useful in confirming the diagnosis and providing some information regarding prognosis
 - Demyelinating forms of GBS are supported by features of demyelination including decreased motor nerve conduction velocity, prolonged distal motor latency etc
 - Axonal forms of GBS- decreased distal motor and/or sensory amplitudes
- Criteria- Brighton criteria. **All of these**
 - Bilateral and flaccid weakness of limbs
 - Decreased and absent DTR in weak limbs
 - Monophasic illness pattern and interval between onset and nadir of weakness between 12h and 28 days
 - Electrophysiologic findings consistent with GBS
 - AC dissociation and CSF WBC $<50/\mu\text{L}$
 - Absence of alternative diagnosis
- Treatment
 - Supportive management
 - Monitor cardiac and respiratory function
 - Prevent decubitus and/or thrombosis
 - High dose IVIG (2g/kg) as five daily infusions for a total of 2g/kg
 - Plasmapheresis
 - In adults- equivalent outcome as IVIG
 - In children- only recommended in cases with rapidly progressing or severe disease
- Prognosis
 - Up to 70% of patients with GBS have a good prognosis. Symptoms recede in reverse order of development
 - 3-7% of patients with GBS die due to acute complications like respiratory paralysis, pulmonary infection, pulmonary embolism, cardiac dysfunction

CIDP

- Similarities with GBS
 - Symmetric
 - Demyelination in nerve conduction studies
- Characteristics
 - Progression over months
 - Impotence, incontinence can be present
 - Relapsing and remitting course
 - Responsive to corticosteroids
 - Usually sensory and motor involvement
 - Nerve enlargement
- Criteria
 - Mandatory clinical criteria
 - Lasting for **2 months or longer**
 - Symmetrical weakness
 - Hypo or areflexia
 - Laboratory
 - NCS suggestive of demyelination
 -

Subacute combined degeneration (SACD)

- Presents with subacute paraesthesias in the hands and feet, loss of vibration and position sensation and progressive spastic and ataxic weakness
- Tracts affected- SCD- Spinocerebellar (ataxia), Corticospinal (spastic weakness), Dorsal column
- Important diagnostic clue- Extensor plantar (because of UMN) with absent ankle jerk (because of peripheral neuropathy)
- Optic atrophy and irritability may be prominent in advanced cases
- Romberg's sign positive
- Diagnosis
 - Macrocytic RBC
 - Low serum B12
 - Increased serum homocysteine and methylmalonic acid
 - Anti parietal cell antibodies
 - Schilling test
- Treatment
 - 1000 µg B12 every day for one week
 - Every week for 1 month
 - Every month for 6 months

Neurogenic bladder

- Anatomy part
 - S2, S3, S4 contracts the detrusor- parasympathetic innervation
 - T11, T12, L1, L2 contract the internal sphincter
 - External sphincter is under voluntary control
 - Periaqueductal gray in the midbrain and Prefrontal cortex inhibit the pontine micturition centre which initiates micturition by contracting the detrusor and relaxing the internal sphincter
- Infantile/Cortical/Uninhibited bladder
 - Loss of cortical control
 - Voiding in socially inappropriate situations
 - Lesions in the frontal cortex
 - Complete voiding occurs though

- Overflow incontinence/LMN bladder/Autonomous bladder
 - S2, S3, S4 lesion
 - Decreased detrusor contraction causes increased bladder volume and decreased bladder pressure
 - Atonic, flaccid bladder
 - Intermittent voiding and hesitancy
- UMN bladder/Automatic bladder/Spastic bladder
 - Lesion anywhere from the Micturition centre to sacral area
 - Detrusor becomes overactive
 - Spastic, hypertonic bladder
 - Small volume, high pressure bladder
 - As soon as it fills, it empties, so it is called automatic bladder
- Sensory denervated bladder
- Motor denervated bladder

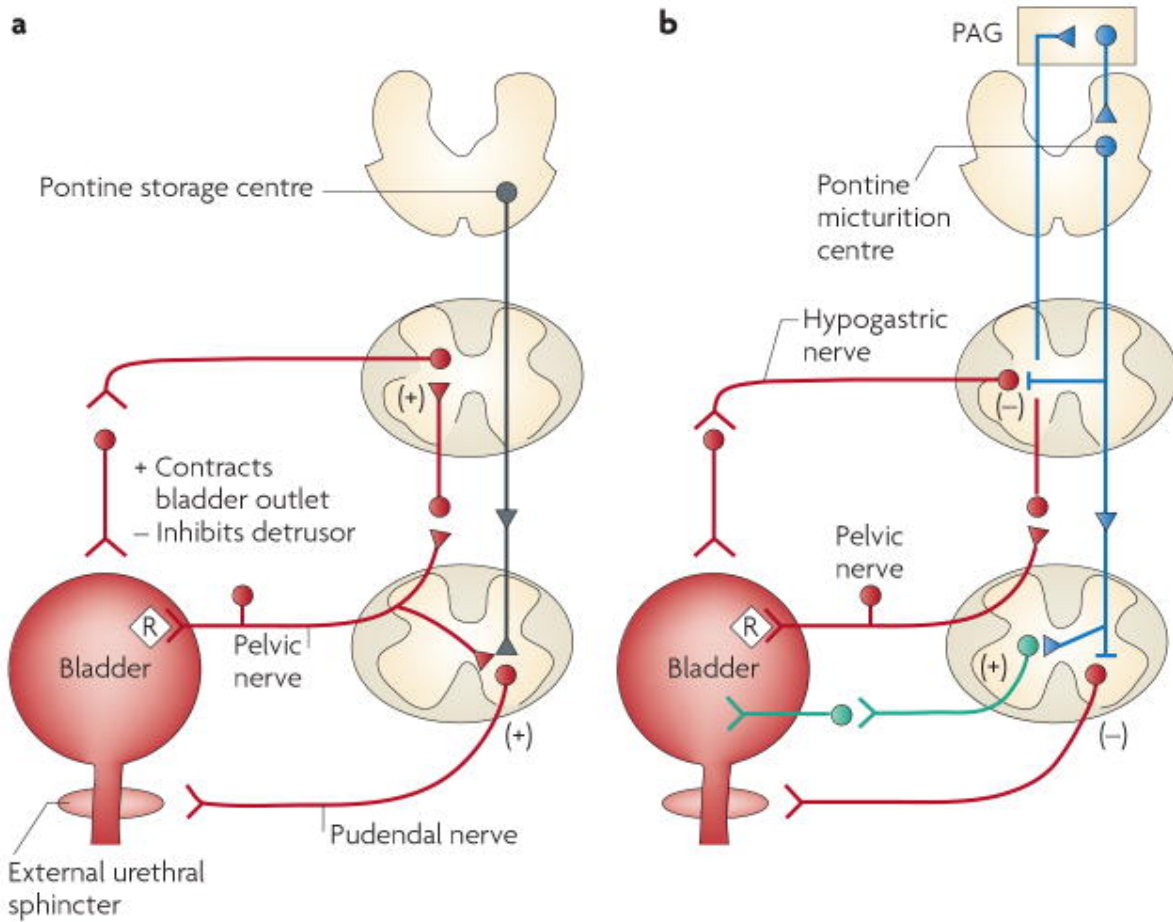
Care of a paraplegic

- Management of the disease
- Positioning in bed- limbs are positioned so contractures do not develop
- Pressure points are adequately padded to prevent pressure sores
- Frequent turning in bed to prevent pressure sores
- Personal hygiene of the patient should be taken care of
- Catheterisation
 - Intermittent catheterisation is better
 - But an indwelling catheter is usually kept
 - Changed once a week
 - Regular urine culture
- Bowel care- periodic enema may be required
- Physiotherapy
- Rehabilitation
 - Physical rehabilitation- Activities of daily living
 - Psychological and social rehabilitation
 - Economic rehabilitation

Acute transverse myelitis

- Aetiologies
 - Demyelinating disease- MS, NMO
 - Systemic immune mediated disorders
 - Post infectious
 - Infectious
- MS
 - Can present with ATM, especially in non-Caucasian populations

NEURAL CONTROL OF MICTURITION — STORAGE vs VOIDING



a — storage (filling) phase · b — voiding phase. PAG = periaqueductal grey.

	STORAGE (panel a)	VOIDING (panel b)
Supraspinal centre	Pontine storage centre (L-region)	Afferents → PAG → Pontine micturition centre (Barrington's nucleus, M-region)
Sympathetic <i>hypogastric n., T11-L2</i>	ON — α_1 contracts bladder neck / internal sphincter; β_3 relaxes detrusor	OFF — inhibited by PMC, so the outlet opens
Parasympathetic <i>pelvic n., S2-S4</i>	OFF — detrusor stays relaxed, bladder fills at low pressure	ON — M_3 muscarinic → detrusor contracts
Somatic <i>pudendal n., Onuf's nucleus S2-S4</i>	ON — external urethral sphincter held closed (the guarding reflex)	OFF — Onuf's nucleus inhibited → external sphincter relaxes

The whole point of the circuit is coordination: the detrusor must contract at the same moment both sphincters relax. When the pontine link is lost, that coordination fails — which is detrusor-sphincter dyssynergia.

THE THREE-LESION RULE — THIS IS WHAT GETS ASKED

Level of lesion	Bladder	Clinical picture
Above the pons stroke, frontal tumour, normal pressure hydrocephalus, Parkinson's	Uninhibited neurogenic bladder	Loss of cortical inhibition, but PAG-PMC loop intact so coordination is preserved → urgency, frequency, urge incontinence. Small-capacity bladder, no residual urine
Between pons and sacral cord spinal cord injury, MS, compressive myelopathy	Reflex / automatic (UMN) bladder	Sacral arc intact but disconnected → involuntary reflex voiding with detrusor-sphincter dyssynergia . High-pressure bladder, incomplete emptying → vesicoureteric reflux and renal failure — the reason these patients need urodynamics
Sacral cord, conus, cauda equina S2-S4 lesion, diabetic autonomic neuropathy	Autonomous / atonic (LMN) bladder	Reflex arc destroyed → flaccid, large-capacity, insensate bladder → overflow incontinence with a large residual volume. Saddle anaesthesia and lax anal tone accompany it

- **Spinal shock** — immediately after a cord lesion the bladder is **areflexic and atonic** whatever the level; the reflex bladder only emerges after weeks. Do not describe an acute cord injury as having an automatic bladder.
- **Sensation:** bladder fullness travels with the **spinothalamic** tract; this is why sensation of filling is lost early in anterior cord syndromes.

Chronic myelopathies

- Spondylotic myelopathy
 - Neck pain and shoulder pain with stiffness
 - Radicular arm pain, often with a C5, C6 distribution
 - Decreased tendon reflexes in UL, often biceps
 - Slowly progressive spastic paraparesis
 - Urinary incontinence in advanced cases
- Vascular malformations
 - Typical presentation- middle aged man with progressive myelopathy that worsens slowly or may be waxing and waning mimicking MS
 - Incomplete sensory, motor, bowel and bladder disturbances
 - Motor deficits may predominate and produce a mixed UMN+LMN involvement like ALS
 - Intermittent claudication may also be seen
- Retroviral myelopathies
 - Tropical spastic paraparesis
 - HTLV-1
 - Progressive syndrome with variable sensory and bladder disturbance
 - May resemble primary progressive MS
 - HIV myelopathy- resembles SACS
- Lathyrism
 - Toxin- BOAA- Beta oxalyl amino alanine
 - Latent stage- altered gait
 - No stick stage- short, jerky steps without the aid of a stick
 - One stick stage
 - Two stick stage
 - Crawler stage
 - Vitamin C prophylaxis
 - Banning the crop prevents it
 - Removal of the toxin by parboiling
- Syringomyelia
 - More than half of all cases are associated with a chiari 1 malformation where cerebellar tonsils herniate through foramen magnum
 - Many young patients acquire a cervical-thoracic scoliosis
 - Regional dissociated sensory loss- loss of pain and temperature
 - Suspended anaesthesia- above and below is normal
 - Extends to involve the CST

HISTORY AND DIAGNOSTIC FORMULATION

- **Progressive weakness suggests cord compression; improving weakness suggests an inflammatory cause.** The shorthand written is “*pure, distal, contrast, motor*” — i.e. is the deficit purely motor, is it distal-predominant, and is it progressing or improving? [Niharika's notes]
- **State the diagnosis in full form:** “*Chronic flaccid/spastic paraparesis due to (aetiology)*” — duration, tone, then cause. [Niharika's notes]
- **Three horizontal levels to localise:** extradural · extramedullary · intramedullary. Extramedullary further divides into intradural and extradural. Extradural — **Pott's spine**; intradural — **neurofibroma, meningioma**. [Niharika's notes]
- **Bladder:** UMN → urgency and frequency; LMN → hesitancy, retention, overflow. (A shorthand ‘Ru / Rc / Ve / TRC’ appears here in the original and was not fully legible — worth checking against the source.) [Niharika's notes]
- **Optic atrophy occurs in both B12 deficiency and MS** — a point of overlap when working up ophthalmoplegia. [Niharika's notes]
- **Family history — hereditary spastic paraplegia.** Remember to ask. [Niharika's notes]

LOCALISING THE LEVEL

- **The rule:** at the level of the lesion there are **LMN signs**; below it, **UMN signs**. Segmental LMN findings are what give you the level. [Niharika's notes]
- **Reflexes are the best way of determining the level.** [Niharika's notes]

Reflex / test	Root value
Biceps jerk	C5
Inverted supinator / inverted biceps jerk	C5-C6 level lesion
Finger flexion test	C8-T1
Cremasteric reflex	Afferent — ilioinguinal nerve; efferent — genital branch of the genitofemoral nerve

- **Vertebral level → cord segment, thoracic spine: T1-T6 → add 2; T7-T9 → add 3.** (The same page also carries “T6 to T12 → add 2” — use the T1-T6/T7-T9 version, which is the standard rule.)
Thoracic-level lesions are relatively uncommonly seen. [Niharika's notes]

CONUS MEDULLARIS AND OTHER SIGNS

- **The conus medullaris comprises segments S3, S4, S5.** [Niharika's notes]
- **Why the ankle jerk can be absent in a conus lesion:** although conus syndrome is classically UMN-predominant, the **S1 segment weakness gives an LMN component distally**, so the ankle jerk is lost. [Niharika's notes]
- **L'hermitte's sign → posterior column involvement.** Seen in MS, B12 deficiency and cervical compressive myelopathy. [Niharika's notes]
- **Onion-skin facial sensory loss in cervical myelopathy — the ‘balaclava helmet’ pattern,** produced by the somatotopy of the **spinal trigeminal nucleus**. [Niharika's notes]

CAUSES — ADDITIONS TO THE TABLE

- **Intramedullary tumours — add oligodendroglioma and medulloblastoma** to ependymoma, astrocytoma, syringomyelia and glioma. [Niharika's notes]
- **Non-compressive causes to remember:** MS and NMO · post-vaccinal and post-traumatic · spinocerebellar degeneration · motor neuron disease. [Niharika's notes]
- **Flaccid paraplegia by level:** anterior horn cell → **root** → **plexus** → **NMJ** → **muscle**. At the NMJ: myasthenia and **LEMS = Lambert-Eaton Myasthenic Syndrome**. [Niharika's notes]
- **Revise the cortical homunculus** — relevant to parasagittal and central causes of paraplegia. [Niharika's notes]

APPENDIX — HARRISON'S 21e

Jugular Venous Pressure · Arterial Pulse · Blood Pressure

Chapter 239 — Physical Examination of the Cardiovascular System



1816

PART 6

Disorders of the Cardiovascular System

tongue, and mucous membranes, as part of the Osler-Weber-Rendu syndrome (hereditary hemorrhagic telangiectasia), resemble spider nevi and can be a source of right-to-left shunting when also present in the lung. Malar telangiectasias also are seen in patients with advanced mitral stenosis (MS) or scleroderma. An unusually tan or bronze discoloration of the skin may suggest hemochromatosis as the cause of the associated systolic heart failure. Jaundice, which may be visible first in the sclerae, has a broad differential diagnosis but, in the appropriate setting, can be consistent with advanced right heart failure and congestive hepatomegaly. Various hereditary lipid disorders sometimes are associated with subcutaneous xanthomas, particularly along the tendon sheaths or over the extensor surfaces of the extremities. Severe hypertriglyceridemia can be associated with eruptive xanthomatosis and lipemia retinalis. Palmar crease xanthomas are specific for type III hyperlipoproteinemia. Pseudoxanthoma elasticum, a disease associated with premature atherosclerosis, is manifested by a leathery, cobblestoned appearance of the skin in the axilla and neck creases and by angiod streaks on funduscopic examination. Extensive lentiginoses have been described in a variety of development delay–cardiovascular syndromes, including Carney's syndrome, which includes multiple atrial myxomas. Cutaneous manifestations of sarcoidosis such as lupus pernio and erythema nodosum may suggest this disease as a cause of an associated dilated cardiomyopathy, especially with heart block, intraventricular conduction delay, or ventricular tachycardia.

Head and Neck Dentition and oral hygiene should be assessed in every patient both as a source of potential infection and as an index of general health. A high-arched palate is a feature of Marfan syndrome and other connective tissue disease syndromes. Bifid uvula has been described in patients with Loeys-Dietz syndrome, and orange tonsils are characteristic of Tangier disease. The ocular manifestations of hyperthyroidism have been well described. Many patients with congenital heart disease have associated hypertelorism, low-set ears, or micrognathia. Blue sclerae are a feature of osteogenesis imperfecta. An arcus senilis pattern lacks specificity as an index of coronary heart disease risk. The funduscopic examination is an often-underused method by which to assess the microvasculature, especially among patients with established atherosclerosis, hypertension, or diabetes mellitus. A mydriatic agent may be necessary for optimal visualization. A funduscopic examination should be performed routinely in the assessment of patients with suspected endocarditis and those with a history of acute visual change. Branch retinal artery occlusion or visualization of a Hollenhorst plaque can narrow the differential diagnosis rapidly in the appropriate setting. Relapsing polychondritis may manifest as an inflamed pinna or, in its later stages, as a saddle-nose deformity because of destruction of nasal cartilage; granulomatosis with polyangiitis (Wegener's) can also lead to a saddle-nose deformity.

Chest Midline sternotomy, left posterolateral thoracotomy, or infraclavicular scars at the site of pacemaker/defibrillator generator implantation should not be overlooked and may provide the first clue regarding an underlying cardiovascular disorder in patients unable to provide a relevant history. A prominent venous collateral pattern may suggest subclavian or vena caval obstruction. If the head and neck appear dusky and slightly cyanotic and the venous pressure is grossly elevated without visible pulsations, a diagnosis of superior vena cava syndrome should be entertained. Thoracic cage abnormalities have been well described among patients with connective tissue disease syndromes. They include pectus carinatum ("pigeon chest") and pectus excavatum ("funnel chest"). Obstructive lung disease is suggested by a barrel chest deformity, especially with tachypnea, pursed-lip breathing, and use of accessory muscles. The characteristically severe kyphosis and compensatory lumbar, pelvic, and knee flexion of ankylosing spondylitis should prompt careful auscultation for a murmur of aortic regurgitation (AR). Straight back syndrome refers to the loss of the normal kyphosis of the thoracic spine and has been described in patients with mitral valve prolapse (MVP) and its variants. In some patients with cyanotic congenital heart disease, the chest wall appears to be asymmetric, with anterior displacement of the left hemithorax.

The respiratory rate and pattern should be noted during spontaneous breathing, with additional attention to depth, audible wheezing, and stridor. Lung examination can reveal adventitious sounds indicative of pulmonary edema, pneumonia, or pleuritis.

Abdomen In some patients with advanced obstructive lung disease, the point of maximal cardiac impulse may be in the epigastrium. The liver is frequently enlarged and tender in patients with chronic heart failure. Systolic pulsations over the liver signify severe tricuspid regurgitation (TR). Splenomegaly may be a feature of infective endocarditis, particularly when symptoms have persisted for weeks or months. Ascites is a nonspecific finding but may be present with advanced chronic right heart failure, constrictive pericarditis, hepatic cirrhosis, or an intraperitoneal malignancy. The finding of an elevated JVP implies a cardiovascular etiology. In nonobese patients, the aorta typically is palpated between the epigastrium and the umbilicus. The sensitivity of palpation for the detection of an abdominal aortic aneurysm (pulsatile and expansile mass) decreases as a function of body size. Because palpation alone is not sufficiently accurate to establish this diagnosis, a screening ultrasound examination is advised when appropriate. The presence of an arterial bruit over the abdomen suggests high-grade atherosclerotic disease, although precise localization is difficult.

Extremities The temperature and color of the extremities, the presence of clubbing, arachnodactyly, and pertinent nail findings can be surmised quickly during the examination. Clubbing implies the presence of central right-to-left shunting, although it has also been described in patients with endocarditis. Its appearance can range from cyanosis and softening of the root of the nail bed, to the classic loss of the normal angle between the base of the nail and the skin, to the skeletal and periosteal bony changes of hypertrophic osteoarthropathy, which is seen rarely in patients with advanced lung or liver disease. Patients with the Holt-Oram syndrome have an unopposable, "fingerized" thumb, whereas patients with Marfan syndrome may have arachnodactyly and a positive "wrist" (overlapping of the thumb and fifth finger around the wrist) or "thumb" (protrusion of the thumb beyond the ulnar aspect of the hand when the fingers are clenched over the thumb in a fist) sign. The Janeway lesions of endocarditis are non-tender, slightly raised hemorrhages on the palms and soles, whereas Osler's nodes are tender, raised nodules on the pads of the fingers or toes. Splinter hemorrhages are classically identified as linear petechiae in the midposition of the nail bed and should be distinguished from the more common traumatic petechiae, which are seen closer to the distal edge.

Lower extremity or presacral edema in the setting of an elevated JVP defines volume overload and may be a feature of chronic heart failure or constrictive pericarditis. Lower extremity edema in the absence of jugular venous hypertension may be due to profound hypoalbuminemia as seen in nephrotic syndrome or liver failure. Other causes include lymphatic or venous obstruction or, more commonly, venous insufficiency, as would be further suggested by the appearance of varicosities, venous ulcers (typically medial in location), and brownish cutaneous discoloration from hemosiderin deposition (eburnation). Pitting edema can also be seen in patients who use dihydropyridine calcium channel blockers. A Homan's sign (posterior calf pain on active dorsiflexion of the foot against resistance) is neither specific nor sensitive for deep venous thrombosis. Muscular atrophy or the absence of hair along an extremity is consistent with severe arterial insufficiency or a primary neuromuscular disorder.

■ CARDIOVASCULAR EXAMINATION

Jugular Venous Pressure and Waveform The JVP is the single most important bedside measurement from which to estimate the volume status. The internal jugular vein is preferred because the external jugular vein is valved and not directly in line with the superior vena cava and right atrium. Nevertheless, the external jugular vein has been used to discriminate between high and low central venous pressure (CVP) when tested among medical students, residents, and attending

physicians. Precise estimation of the central venous or right atrial pressure from bedside assessment of the jugular venous waveform has proved difficult. Venous pressure traditionally has been measured as the vertical distance between the top of the jugular venous pulsation and the sternal inflection point (angle of Louis). A distance >4.5 cm at 30° elevation is considered abnormal. However, the actual distance between the mid-right atrium and the angle of Louis varies considerably as a function of both body size and the patient angle at which the assessment is made (30° , 45° , or 60°). The use of the sternal angle as a reference point leads to systematic underestimation of CVP, and this method should be used less for semiquantification than to distinguish a normal from an abnormally elevated CVP. The use of the clavicle may provide an easier reference for standardization. Venous pulsations above this level in the sitting position are clearly abnormal, as the distance between the clavicle and the right atrium is at least 10 cm. The patient should always be placed in the sitting position, with the legs dangling below the bedside, when an elevated pressure is suspected in the semisupine position. It should also be noted that bedside estimates of CVP are made in centimeters of water, but must be converted to millimeters of mercury to provide correlation with accepted hemodynamic norms ($1.36 \text{ cmH}_2\text{O} = 1.0 \text{ mmHg}$).

The venous waveform sometimes can be difficult to distinguish from the carotid pulse, especially during casual inspection. Nevertheless, the venous waveform has several characteristic features, and its individual components can be appreciated in most patients (Fig. 239-1). The arterial pulsation is not easily obliterated with palpation; the venous waveform in patients with sinus rhythm is usually biphasic, while the carotid pulse is monophasic; and the jugular venous pulsation should change with changes in posture or inspiration (unless the venous pressure is quite elevated).

The venous waveform is divided into several distinct peaks. The *a* wave reflects right atrial presystolic contraction and occurs just after the electrocardiographic P wave, preceding the first heart sound (S_1). A prominent *a* wave is seen in patients with reduced right ventricular compliance; a cannon *a* wave occurs with atrioventricular (AV) dissociation and right atrial contraction against a closed tricuspid valve. In a patient with a wide complex tachycardia, the appreciation of cannon *a* waves in the jugular venous waveform identifies the rhythm as ventricular in origin. The *a* wave is not present with atrial fibrillation. The *x* descent defines the fall in right atrial pressure after inscription of the *a* wave. The *c* wave, which occurs as the closed tricuspid valve is pushed into the right atrium during early ventricular systole, interrupts this *x* descent and is followed by a further descent. The *v* wave represents atrial filling (atrial diastole) and occurs during ventricular systole. The height of the *v* wave is determined by right atrial compliance as well as the volume of blood returning to the right atrium either antegrade from the cavae or retrograde through an incompetent tricuspid valve. In patients with TR, the *v* wave is accentuated and the subsequent fall in pressure (*y* descent) is rapid. With progressive degrees of TR, the *v* wave merges with the *c* wave, and the right atrial and jugular vein waveforms become "ventricularized." The *y* descent, which follows the peak of the *v* wave, can become prolonged or blunted with obstruction to right ventricular inflow, as may occur with tricuspid stenosis or pericardial tamponade. Normally, the venous pressure should fall by at least 3 mmHg with inspiration. Kussmaul's sign is defined by either a rise or a lack of fall of the JVP with inspiration and is classically associated with constrictive pericarditis, although it has been reported in patients with restrictive cardiomyopathy, massive pulmonary embolism, right ventricular infarction, and advanced left ventricular (LV) systolic heart failure. It is also a common, isolated finding in patients after cardiac surgery without other hemodynamic abnormalities.

Venous hypertension sometimes can be elicited by passive leg elevation or performance of the abdominojugular reflux maneuver. When these signs are positive, a volume-overloaded state with limited compliance of an overly distended or constricted venous system is present. Abdominojugular reflux is produced with firm and consistent pressure over the upper portion of the abdomen, preferably over the right upper quadrant, for >15 s. A positive response is defined by a sustained rise of >3 cm in the JVP during the application of firm abdominal pressure.

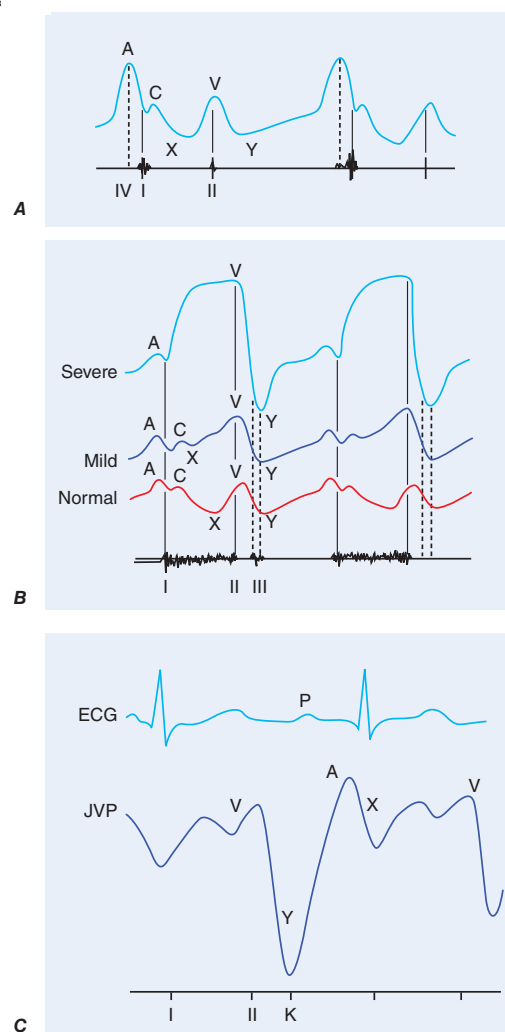


FIGURE 239-1 A. Jugular venous pulse wave tracing (top) with heart sounds (bottom) in a patient with reduced right ventricular compliance. The A wave represents right atrial presystolic contraction and occurs just after the electrocardiographic P wave and just before the first heart sound (I). In this example, the A wave is accentuated and larger than normal due to decreased right ventricular compliance, as also suggested by the right-sided S_3 (IV). The C wave may reflect the carotid pulsation in the neck and/or an early systolic increase in right atrial pressure as the right ventricle pushes the closed tricuspid valve into the right atrium. The *x* descent follows the A wave just as atrial pressure continues to fall. The V wave represents atrial filling during ventricular systole and peaks at the second heart sound (II). The *y* descent corresponds to the fall in right atrial pressure after tricuspid valve opening. B. Jugular venous wave forms in mild (middle) and severe (top) tricuspid regurgitation, compared with normal, with phonocardiographic representation of the corresponding heart sounds below. With increasing degrees of tricuspid regurgitation, the waveform becomes "ventricularized." C. Electrocardiogram (ECG) (top), jugular venous waveform (JVP) (middle), and heart sounds (bottom) in pericardial constriction. Note the prominent and rapid *y* descent, corresponding in timing to the pericardial knock (K). (Reproduced with permission from J Abrams: *Synopsis of Cardiac Physical Diagnosis*, 2nd ed. Boston, Butterworth Heinemann, 2001.)

The response should be assessed after 10 s of continuous pressure to allow for respiratory artifacts and tensing of the abdominal muscles to subside. Patients must be coached to refrain from breath holding or a Valsalva-like maneuver during the procedure. Performance of the abdominojugular reflux maneuver is useful in predicting a pulmonary artery wedge pressure >15 mmHg in patients with heart failure.

Although the JVP estimates right ventricular filling pressure, it has a predictable relationship with the pulmonary artery wedge pressure.



1818 In a large study of patients with advanced heart failure, the presence of a right atrial pressure >10 mmHg (as predicted on bedside examination) had a positive value of 88% for the prediction of a pulmonary artery wedge pressure of >22 mmHg. In addition, an elevated JVP has prognostic significance in patients with both symptomatic heart failure and asymptomatic LV systolic dysfunction. The presence of an elevated JVP is associated with a higher risk of subsequent hospitalization for heart failure, death from heart failure, or both.

PART 6

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Assessment of Blood Pressure Measurement of blood pressure usually is delegated to a medical assistant but should be repeated by the examining clinician. Accurate measurement depends on body position, arm size, time of measurement, place of measurement, device, device size, technique, and examiner. In general, physician-recorded blood pressures are higher than both nurse-recorded pressures and self-recorded pressures at home. Blood pressure is best measured in the seated position with the arm at the level of the heart and the feet on the floor with the back supported, using an appropriately sized cuff, after 5–10 min of relaxation. When it is measured in the supine position, the arm should be raised to bring it to the level of the mid-right atrium. The length and width of the blood pressure cuff bladder should be 80 and 40% of the arm's circumference, respectively. A common source of error in practice is to use an inappropriately small cuff, resulting in marked overestimation of true blood pressure, or an inappropriately large cuff, resulting in underestimation of true blood pressure. The cuff should be inflated to 30 mmHg above the expected systolic pressure and the pressure released at a rate of 2–3 mmHg/s. Systolic and diastolic pressures are defined by the first and fifth Korotkoff sounds, respectively. Very low (even 0 mmHg) diastolic blood pressures may be recorded in patients with chronic, severe AR or a large arteriovenous fistula because of enhanced diastolic "run-off." In these instances, both the phase IV and phase V Korotkoff sounds should be recorded. Blood pressure is best assessed at the brachial artery level, though it can be measured at the radial, popliteal, or pedal pulse level. In general, systolic pressure increases and diastolic pressure decreases when measured in more distal arteries. Blood pressure should be measured in both arms, and the difference should be <10 mmHg. A blood pressure differential that exceeds this threshold may be associated with atherosclerotic or inflammatory subclavian artery disease, supraaortic stenosis, aortic coarctation, or aortic dissection. Systolic leg pressures are usually as much as 20 mmHg higher than systolic arm pressures. Greater leg–arm pressure differences are seen in patients with chronic severe AR as well as patients with extensive and calcified lower extremity peripheral arterial disease. The ankle-brachial index (systolic pressure in the dorsalis pedis and/or posterior tibial artery divided by the higher of the two brachial artery pressures) is a powerful predictor of long-term cardiovascular mortality.

The blood pressure measured in an office or hospital setting may not accurately reflect the pressure in other venues. "White coat hypertension" (elevated clinic blood pressure and normal out of clinic blood pressure) is defined by at least three separate clinic-based measurements >130/80 mmHg and at least two non-clinic-based measurements <130/80 mmHg in the absence of any evidence of target organ damage. Individuals with white coat hypertension may not benefit from drug therapy, although they may be more likely to develop sustained hypertension over time. Masked hypertension (normal or low clinic blood pressure but elevated out of clinic blood pressure) should be suspected when normal or even low blood pressures are recorded in the office in patients with advanced atherosclerotic disease, especially when evidence of target organ damage is present or bruits are audible. Higher systolic blood pressures measured with a 24-h ambulatory blood pressure device are associated with a higher risk of cardiovascular disease and all-cause death independent of blood pressures measured in the outpatient setting.

Orthostatic hypotension is defined by a fall in systolic pressure >20 mmHg or in diastolic pressure >10 mmHg in response to assumption of the upright posture from a supine position within 3 min. There may also be a lack of a compensatory tachycardia, an abnormal response that suggests autonomic insufficiency, as may be seen in patients with

diabetes mellitus or Parkinson's disease. Orthostatic hypotension is a common cause of postural lightheadedness/syncope and should be assessed routinely in patients for whom this diagnosis might pertain. It can be exacerbated by advanced age, dehydration, certain medications, food, deconditioning, and ambient temperature/humidity.

Arterial Pulse The carotid artery pulse occurs just after the ascending aortic pulse. The aortic pulse is best appreciated in the epigastrium, just above the level of the umbilicus. Peripheral arterial pulses that should be assessed routinely include the subclavian, brachial, radial, ulnar, femoral, popliteal, dorsalis pedis, and posterior tibial. In patients in whom the diagnosis of either temporal arteritis or polymyalgia rheumatica is suspected, the temporal arteries also should be examined. Although one of the two pedal pulses may not be palpable in up to 10% of normal subjects, the pair should be symmetric. The integrity of the arcuate system of the hand is assessed by Allen's test, which is performed routinely before instrumentation of the radial artery. The pulses should be examined for their symmetry, volume, timing, contour, amplitude, and duration. If necessary, simultaneous auscultation of the heart can help identify a delay in the arrival of an arterial pulse. Simultaneous palpation of the radial and femoral pulses may reveal a femoral delay in a patient with upper extremity hypertension and suspected aortic coarctation. The carotid upstrokes should never be examined simultaneously or before listening for a bruit. Light pressure should always be used to avoid precipitation of carotid hypersensitivity syndrome and syncope in a susceptible elderly individual. The arterial pulse usually becomes more rapid and spiking as a function of its distance from the heart, a phenomenon that reflects the muscular status of the more peripheral arteries and the summation of the incident and reflected waves. In general, the character and contour of the arterial pulse depend on the stroke volume, ejection velocity, vascular compliance, and systemic vascular resistance. The pulse examination can be misleading in patients with reduced cardiac output and in those with stiffened arteries from aging, chronic hypertension, or peripheral arterial disease.

The character of the pulse is best appreciated at the carotid level (**Fig. 239-2**). A weak and delayed pulse (*pulsus parvus et tardus*) defines severe aortic stenosis (AS). Some patients with AS may also have a slow, notched, or interrupted upstroke (anacrotic pulse) with a thrill or shudder. With chronic severe AR, by contrast, the carotid upstroke has a sharp rise and rapid fall-off (Corrigan's or water-hammer pulse). Some patients with advanced AR may have a bifid or bisferiens pulse, in which two systolic peaks can be appreciated. A bifid pulse is also described in patients with hypertrophic obstructive cardiomyopathy (HOCM), with inscription of percussion and tidal waves. A bifid pulse is easily appreciated in patients on intraaortic balloon counterpulsation (IABP), in whom the second pulse is diastolic in timing.

Pulsus paradoxus refers to a fall in systolic pressure >10 mmHg with inspiration that is seen in patients with pericardial tamponade but also is described in those with massive pulmonary embolism, hemorrhagic shock, severe obstructive lung disease, and tension pneumothorax. Pulsus paradoxus is measured by noting the difference between the systolic pressure at which the Korotkoff sounds are first heard (during expiration) and the systolic pressure at which the Korotkoff sounds are heard with each heartbeat, independent of the respiratory phase. Between these two pressures, the Korotkoff sounds are heard only intermittently and during expiration. The cuff pressure must be decreased slowly to appreciate the finding. It can be difficult to measure pulsus paradoxus in patients with tachycardia, atrial fibrillation, or tachypnea. A pulsus paradoxus may be palpable at the brachial artery or femoral artery level when the pressure difference exceeds 15 mmHg. This inspiratory fall in systolic pressure is an exaggerated consequence of interventricular dependence.

Pulsus alternans, in contrast, is defined by beat-to-beat variability of pulse amplitude. It is present only when every other phase I Korotkoff sound is audible as the cuff pressure is lowered slowly, typically in a patient with a regular heart rhythm and independent of the respiratory cycle. Pulsus alternans is seen in patients with severe LV systolic dysfunction and is thought to be due to cyclic changes in intracellular

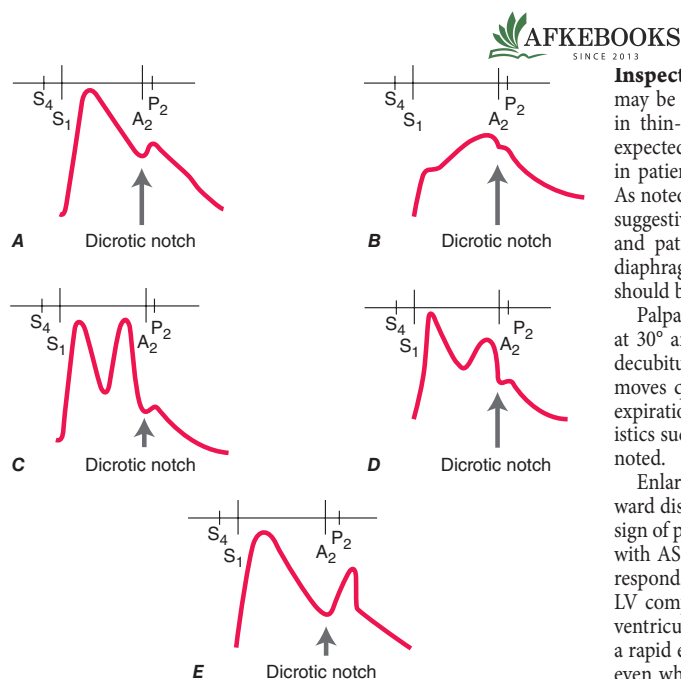


FIGURE 239-2 Schematic diagrams of the configurational changes in carotid pulse and their differential diagnoses. Heart sounds are also illustrated. **A.** Normal. S_4 , fourth heart sound; S_1 , first heart sound; A_2 , aortic component of second heart sound; P_2 , pulmonic component of second heart sound. **B.** Aortic stenosis. Anacrotic pulse with slow upstroke to a reduced peak. **C.** Bisferiens pulse with two peaks in systole. This pulse is rarely appreciated in patients with severe aortic regurgitation. **D.** Bisferiens pulse in hypertrophic obstructive cardiomyopathy. There is a rapid upstroke to the first peak (percussion wave) and a slower rise to the second peak (tidal wave). **E.** Dicrotic pulse with peaks in systole and diastole. This waveform may be seen in patients with sepsis or during intraaortic balloon counterpulsation with inflation just after the dicrotic notch. (Reproduced with permission from K Chatterjee, W Parmley [eds]: *Cardiology: An Illustrated Text/Reference*. Philadelphia, Gower Medical Publishers, 1991.)

calcium and action potential duration. When pulsus alternans is associated with electrocardiographic T-wave alternans, the risk for an arrhythmic event appears to be increased.

Ascending aortic aneurysms can rarely be appreciated as a pulsatile mass in the right parasternal area. Appreciation of a prominent abdominal aortic pulse should prompt noninvasive imaging with ultrasound or computed tomography for better characterization. Femoral and/or popliteal artery aneurysms should be sought in patients with abdominal aortic aneurysm disease.

The level of a claudication-producing arterial obstruction can often be identified on physical examination (Fig. 239-3). For example, in a patient with calf claudication, a decrease in pulse amplitude between the common femoral and popliteal arteries will localize the obstruction to the level of the superficial femoral artery, although inflow obstruction above the level of the common femoral artery may coexist. Auscultation for carotid, subclavian, abdominal aortic, and femoral artery bruits should be routine. However, the correlation between the presence of a bruit and the degree of vascular obstruction is poor. A cervical bruit is a weak indicator of the degree of carotid artery stenosis; the absence of a bruit does not exclude the presence of significant luminal obstruction. If a bruit extends into diastole or if a thrill is present, the obstruction is usually severe. Another cause of an arterial bruit is an arteriovenous fistula with enhanced flow.

The likelihood of significant lower extremity peripheral arterial disease increases with typical symptoms of claudication, cool skin, abnormalities on pulse examination, or the presence of a vascular bruit. Abnormal pulse oximetry (a >2% difference between finger and toe oxygen saturation) can be used to detect lower extremity peripheral arterial disease and is comparable in its performance characteristics to the ankle-brachial index.

Inspection and Palpation of the Heart

The LV apex beat may be visible in the midclavicular line at the fifth intercostal space in thin-chested adults. Visible pulsations anywhere other than this expected location are abnormal. The left anterior chest wall may heave in patients with an enlarged or hyperdynamic left or right ventricle. As noted previously, a visible right upper parasternal pulsation may be suggestive of ascending aortic aneurysm disease. In thin, tall patients and patients with advanced obstructive lung disease and flattened diaphragms, the cardiac impulse may be visible in the epigastrium and should be distinguished from a pulsatile liver edge.

Palpation of the heart begins with the patient in the supine position at 30° and can be enhanced by placing the patient in the left lateral decubitus position. The normal LV impulse is <2 cm in diameter and moves quickly away from the fingers; it is better appreciated at end expiration, with the heart closer to the anterior chest wall. Characteristics such as size, amplitude, and rate of force development should be noted.

Enlargement of the LV cavity is manifested by a leftward and downward displacement of an enlarged apex beat. A sustained apex beat is a sign of pressure overload, such as that which may be present in patients with AS or chronic hypertension. A palpable presystolic impulse corresponds to the fourth heart sound (S_4) and is indicative of reduced LV compliance and the forceful contribution of atrial contraction to ventricular filling. A palpable third sound (S_3), which is indicative of a rapid early filling wave in patients with heart failure, may be present even when the gallop itself is not audible. A large LV aneurysm may sometimes be palpable as an ectopic impulse, discrete from the apex beat. HOCM may very rarely cause a triple cadence beat at the apex with contributions from a palpable S_4 and the two components of the bisferiens systolic pulse.

Right ventricular pressure or volume overload may create a sternal lift. Signs of either TR (cv waves in the jugular venous pulse) and/or pulmonary arterial hypertension (a loud single or palpable P_2) would be confirmatory. The right ventricle can enlarge to the extent that left-sided events are obscured. A zone of retraction between the right ventricular and LV impulses sometimes can be appreciated in patients with right ventricle pressure or volume overload when they are placed in the left lateral decubitus position. Systolic and diastolic thrills signify turbulent and high-velocity blood flow. Their locations help identify the origin of heart murmurs.

CARDIAC AUSCULTATION

Heart Sounds Ventricular systole is defined by the interval between the first (S_1) and second (S_2) heart sounds (Fig. 239-4). The first heart sound (S_1) includes mitral and tricuspid valve closure. Normal splitting can be appreciated in young patients and those with right bundle branch block, in whom tricuspid valve closure is relatively delayed. The intensity of S_1 is determined by the distance over which the anterior leaflet of the mitral valve must travel to return to its annular plane, leaflet mobility, LV contractility, and the PR interval. S_1 is classically loud in the early phases of rheumatic MS and in patients with hyperkinetic circulatory states or short PR intervals. S_1 becomes softer in the later stages of MS when the leaflets are rigid and calcified, after exposure to β -adrenergic receptor blockers, with long PR intervals, and with LV contractile dysfunction. The intensity of heart sounds, however, can be reduced by any process that increases the distance between the stethoscope and the responsible cardiac event, including mechanical ventilation, obstructive lung disease, obesity, pneumothorax, and a pericardial effusion.

Aortic and pulmonic valve closure constitutes the second heart sound (S_2). With normal or physiologic splitting, the A_2 - P_2 interval increases with inspiration and narrows during expiration. This physiologic interval will widen with right bundle branch block because of the further delay in pulmonic valve closure and in patients with severe MR because of the premature closure of the aortic valve. An unusually narrowly split or even a singular S_2 is a feature of pulmonary arterial hypertension. Fixed splitting of S_2 , in which the A_2 - P_2 interval is wide and does not change during the respiratory cycle, occurs in patients with a

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